Sweep tables at $\alpha_0 = 0.30$ (compact and full)

paper style; main grid $(\alpha_0, \alpha, \delta, K)$, Corridor and Maze

Anytime-Valid Certification of Robustness Margins for Safe Control via E-Values: N200 data ($N_{\mathrm{MC}} = 200$ trajectories per method and setting)

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Every caption names the file, the data it was generated from and the exact line that restores the display in an edition of the paper (the arXiv version, the ablation report or the unpublished extended version). File paths are relative to the root of the paper's LaTeX source; its `tools/` and `catalog/` folders are published as `code/tools/` and `code/catalog/` of the code and data repository, while the other source folders (`sections/`, `appendices/`, `figures/`, `report/`) are not included there. Numbers with the prefix A (for example Table A.I or Appendix A.E-A) are those of the arXiv version of the paper. The settings T1–T5 of the captions (New1–New5 in file names) are the settings S1–S5 of the paper. Table fragments of the catalog are restored with `\ntable{../catalog/<style>/<name>}` (the macro `\ntable` of `appendices/n200_assets.tex` reads `appendices/generated_n200/<edition style>/`); figures of the catalog with `\nfigure{../catalog/<style>/<path>}`, inside the float and caption of the edition. Table values are generated by `tools/n200_tables/build_tables.py` (`--catalog` for the catalog fragments) and re-derived from the exports by `tools/n200_tables/check_tables.py`. The level at $K = \infty$ used by the check marks is

$$\delta_1 = \frac{\delta}{2b \ln^2 2}. \tag{1}$$

Abbreviations in the captions ([F], [A] and [T] are files of the N200 summary exports, which are not included in the repository):

[F] `N200_numbers/final/common/metrics.json` (main grid, baselines, default)
[A] `N200_numbers/new_anchors/common/metrics.json` (sweeps at T1–T5)
[T] `N200_numbers/final/temporal/temporal_metrics.json`
[V] `tools/n200_tables/tables_values.json.gz` (every printed cell of the editions, with its view id)
[C] `appendices/generated_n200/catalog/CATALOG_TABLES.json` (catalog tables: view ids, restore lines)

The key below holds for every table.

Key to all tables. $\pm$, $^{+}$, $\downarrow/\uparrow$, counts, ✓, parentheses and N-A as in Table A.I; bold: best mean of the four certified variants at one setting (ties all bold; never SDs, $\widehat{m}_{\mathrm{cert}}$, $\widehat{m}_{\tau^{\mathrm{use}}}$, $\widehat{m}_{p^{\max}}$); a/m: a trajectories with an excess above $\varepsilon_{\mathrm{chk}}$ among the m with a checked decision (a alone if $m = N_{\mathrm{MC}}$), bound over all N_{MC}.

I. Compact sweep tables at $\alpha_0 = 0.30$

A. *Corridor,* $\alpha_0 = 0.30$

TABLE I

COMPACT SWEEP (BEST VARIANT), CORRIDOR, $\alpha_0 = 0.30$. *File:* APPENDICES/GENERATED_N200/CATALOG/PAPER/COMPACT_CORRIDOR_A030.TEX. *Data:* [F]; FAMILIES MAIN_GRID_200, MATCHED_BASELINES_200; 130 VIEWS (IDS IN [C]). *Restore:* \NTABLE{../CATALOG/PAPER/COMPACT_CORRIDOR_A030}; NOT USED IN THE CURRENT EDITIONS.

Compact sweep. One row per setting $(\alpha_0, \alpha, \delta, K)$ of the main grid. Variant: the certified variant with the most best means in the four columns (best: the bold rule of the key; ties of a column all count); (k): the number k of the four columns in which it has the best mean; ties in k: smaller printed $\widehat{m}_c/c_{\max}$, then $\widehat{q}^{+}_{\text{unsafe}}$, $\widehat{q}^{+}_{\text{CBF}}$, $\widehat{T}_{\text{mis}}$, then the order V-U, V-Δ, V-L, V-C. ±: one sample SD over the trajectories that enter the mean; $^{+}$: decision fraction; ↓: lower is better.

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$

Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A	30.69±13.51	22.32±10.04	70.00±0.00
$c = 0$	0.00±0.00	28.00±2.99	15.27±4.17	32.45±0.77
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	35.91±0.92
ACI	0.22±0.04	17.68±1.59	2.74±1.43	33.13±0.81
QT	0.30±0.03	15.19±1.31	0.61±0.52	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$

α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
$\alpha = 0.05$	$\delta = 0.025$	10	V-U (4)	0.96±0.03	0.33±0.38	0.00±0.00	35.72±0.92
		50	V-U (4)	0.96±0.03	0.32±0.37	0.00±0.00	35.72±0.92
		100	V-U (3)	0.96±0.03	0.32±0.37	0.00±0.00	35.73±0.92
		200	V-U (3)	0.96±0.03	0.31±0.37	0.00±0.00	35.73±0.92
		∞	V-U (3)	0.96±0.03	0.30±0.37	0.00±0.00	35.73±0.92
	$\delta = 0.05$	10	V-L (3)	0.94±0.04	0.47±0.54	0.00±0.00	35.65±0.92
		50	V-L (3)	0.94±0.04	0.47±0.53	0.00±0.00	35.66±0.92
		100	V-L (3)	0.94±0.04	0.47±0.55	0.00±0.00	35.65±0.92
		200	V-L (3)	0.94±0.04	0.46±0.53	0.00±0.00	35.66±0.92
		∞	V-L (3)	0.94±0.04	0.46±0.53	0.00±0.00	35.66±0.92
	$\delta = 0.10$	10	V-U (3)	0.93±0.04	0.62±0.63	0.00±0.00	35.59±0.92
		50	V-U (4)	0.93±0.04	0.63±0.63	0.00±0.00	35.58±0.92
		100	V-L (4)	0.93±0.04	0.63±0.63	0.00±0.00	35.58±0.92
		200	V-L (3)	0.93±0.04	0.62±0.63	0.00±0.00	35.59±0.92
		∞	V-L (3)	0.93±0.04	0.62±0.63	0.00±0.00	35.59±0.92
	$\delta = 0.15$	10	V-U (4)	0.92±0.04	0.73±0.70	0.00±0.00	35.54±0.94
		50	V-U (4)	0.92±0.04	0.72±0.70	0.00±0.00	35.54±0.93
		100	V-U (4)	0.92±0.04	0.72±0.68	0.00±0.00	35.54±0.92
		200	V-L (3)	0.92±0.05	0.72±0.68	0.00±0.00	35.54±0.93
		∞	V-L (3)	0.92±0.05	0.72±0.68	0.00±0.00	35.54±0.93
	$\delta = 0.20$	10	V-U (4)	0.91±0.05	0.87±0.86	0.00±0.00	35.49±0.95
		50	V-L (4)	0.91±0.05	0.88±0.89	0.00±0.00	35.48±0.94
		100	V-U (4)	0.91±0.05	0.86±0.84	0.00±0.00	35.49±0.94
		200	V-L (3)	0.91±0.05	0.86±0.87	0.00±0.00	35.49±0.94
		∞	V-L (3)	0.91±0.05	0.86±0.87	0.00±0.00	35.49±0.94
$\alpha = 0.10$	$\delta = 0.025$	10	V-L (3)	0.84±0.05	1.87±1.07	0.00±0.00	35.20±0.91
		50	V-L (3)	0.84±0.05	1.88±1.07	0.00±0.00	35.21±0.91
		100	V-L (3)	0.84±0.05	1.88±1.09	0.00±0.00	35.21±0.91
		200	V-L (3)	0.84±0.05	1.87±1.07	0.00±0.00	35.21±0.91
		∞	V-L (3)	0.84±0.05	1.87±1.07	0.00±0.00	35.21±0.91

Table I (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A	30.69±13.51	22.32±10.04	70.00±0.00
$c = 0$	0.00±0.00	28.00±2.99	15.27±4.17	32.45±0.77
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	35.91±0.92
ACI	0.22±0.04	17.68±1.59	2.74±1.43	33.13±0.81
QT	0.30±0.03	15.19±1.31	0.61±0.52	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.10$	$\delta = 0.05$	10	V-L (3)	0.83±0.05	2.09±1.14	0.00±0.00	35.14±0.91
		50	V-L (3)	0.83±0.05	2.12±1.18	0.00±0.00	35.14±0.91
		100	V-L (3)	0.83±0.05	2.10±1.17	0.00±0.00	35.14±0.91
		200	V-L (3)	0.83±0.05	2.09±1.14	0.00±0.00	35.14±0.91
		∞	V-L (3)	0.83±0.05	2.08±1.14	0.00±0.00	35.14±0.91
	$\delta = 0.10$	10	V-L (3)	0.81±0.06	2.35±1.21	0.00±0.00	35.07±0.91
		50	V-L (3)	0.81±0.06	2.37±1.24	0.00±0.00	35.06±0.91
		100	V-L (3)	0.81±0.06	2.36±1.22	0.00±0.00	35.06±0.91
		200	V-L (3)	0.81±0.06	2.34±1.20	0.00±0.00	35.07±0.91
		∞	V-L (3)	0.81±0.06	2.33±1.20	0.00±0.00	35.07±0.91
	$\delta = 0.15$	10	V-L (3)	0.79±0.06	2.61±1.39	0.00±0.00	35.01±0.92
		50	V-L (3)	0.79±0.06	2.62±1.41	0.00±0.00	35.01±0.91
		100	V-L (3)	0.79±0.06	2.62±1.39	0.00±0.00	35.01±0.91
		200	V-L (3)	0.80±0.06	2.61±1.40	0.00±0.00	35.01±0.91
		∞	V-L (3)	0.80±0.06	2.60±1.40	0.00±0.00	35.01±0.91
	$\delta = 0.20$	10	V-L (3)	0.78±0.07	2.86±1.71	0.00±0.02	34.96±0.93
		50	V-L (3)	0.78±0.06	2.87±1.55	0.00±0.00	34.96±0.92
		100	V-L (3)	0.78±0.06	2.86±1.53	0.00±0.00	34.95±0.92
		200	V-L (3)	0.78±0.06	2.83±1.53	0.00±0.00	34.96±0.92
		∞	V-L (3)	0.78±0.06	2.82±1.54	0.00±0.00	34.96±0.92
$\alpha = 0.15$	$\delta = 0.025$	10	V-C (3)	0.74±0.06	3.71±1.63	0.00±0.02	34.80±0.89
		50	V-C (3)	0.74±0.05	3.69±1.49	0.00±0.00	34.80±0.89
		100	V-C (3)	0.74±0.05	3.66±1.51	0.00±0.00	34.81±0.89
		200	V-C (3)	0.75±0.06	3.64±1.51	0.00±0.00	34.81±0.89
		∞	V-C (3)	0.75±0.06	3.63±1.51	0.00±0.00	34.81±0.89
	$\delta = 0.05$	10	V-C (3)	0.73±0.06	4.08±1.67	0.00±0.02	34.73±0.90
		50	V-C (3)	0.72±0.06	4.08±1.65	0.00±0.02	34.73±0.90
		100	V-C (3)	0.73±0.06	4.04±1.62	0.00±0.02	34.74±0.90
		200	V-C (3)	0.73±0.06	4.02±1.60	0.00±0.02	34.74±0.90
		∞	V-C (3)	0.73±0.06	4.02±1.60	0.00±0.02	34.74±0.90
	$\delta = 0.10$	10	V-C (3)	0.70±0.07	4.52±1.83	0.00±0.02	34.63±0.91
		50	V-C (3)	0.70±0.06	4.52±1.80	0.00±0.02	34.63±0.91
		100	V-C (3)	0.70±0.06	4.49±1.77	0.00±0.02	34.64±0.91
		200	V-C (3)	0.70±0.06	4.47±1.76	0.00±0.02	34.64±0.90
		∞	V-C (3)	0.70±0.06	4.46±1.76	0.00±0.02	34.64±0.90
	$\delta = 0.15$	10	V-C (3)	0.69±0.07	4.84±1.88	0.00±0.03	34.57±0.91
		50	V-C (3)	0.69±0.07	4.85±1.86	0.00±0.03	34.57±0.92
		100	V-C (3)	0.69±0.07	4.82±1.87	0.00±0.03	34.57±0.91
		200	V-C (3)	0.69±0.07	4.79±1.86	0.00±0.03	34.58±0.91
		∞	V-C (3)	0.69±0.07	4.79±1.86	0.00±0.03	34.58±0.91
	$\delta = 0.20$	10	V-C (3)	0.67±0.08	5.14±2.11	0.00±0.03	34.52±0.94
		50	V-C (3)	0.67±0.07	5.18±2.07	0.00±0.03	34.51±0.93
		100	V-C (3)	0.67±0.07	5.14±2.08	0.00±0.03	34.52±0.93
		200	V-C (3)	0.67±0.07	5.10±2.08	0.00±0.03	34.53±0.94
		∞	V-C (3)	0.67±0.08	5.10±2.08	0.00±0.03	34.53±0.94

Table I (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A	30.69±13.51	22.32±10.04	70.00±0.00
$c = 0$	0.00±0.00	28.00±2.99	15.27±4.17	32.45±0.77
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	35.91±0.92
ACI	0.22±0.04	17.68±1.59	2.74±1.43	33.13±0.81
QT	0.30±0.03	15.19±1.31	0.61±0.52	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.20$	$\delta = 0.025$	10	V-C (2)	0.65±0.06	5.83±1.86	0.01±0.06	34.42±0.87
		50	V-C (2)	0.65±0.06	5.86±1.81	0.01±0.06	34.41±0.87
		100	V-C (2)	0.65±0.06	5.83±1.81	0.01±0.06	34.42±0.87
		200	V-C (2)	0.65±0.06	5.80±1.85	0.01±0.06	34.42±0.87
		∞	V-C (2)	0.65±0.06	5.80±1.85	0.01±0.06	34.42±0.87
	$\delta = 0.05$	10	V-C (2)	0.63±0.06	6.13±1.85	0.01±0.11	34.36±0.89
		50	V-C (2)	0.63±0.06	6.13±1.84	0.01±0.11	34.36±0.88
		100	V-C (2)	0.63±0.06	6.09±1.84	0.01±0.11	34.37±0.88
		200	V-C (2)	0.63±0.06	6.07±1.85	0.01±0.11	34.37±0.88
		∞	V-C (2)	0.63±0.06	6.07±1.86	0.01±0.11	34.37±0.89
	$\delta = 0.10$	10	V-C (2)	0.61±0.07	6.70±2.06	0.03±0.14	34.26±0.91
		50	V-C (2)	0.61±0.07	6.71±2.03	0.03±0.15	34.27±0.91
		100	V-C (2)	0.61±0.07	6.65±2.04	0.02±0.14	34.27±0.91
		200	V-C (2)	0.61±0.07	6.65±2.04	0.02±0.14	34.27±0.91
		∞	V-C (2)	0.61±0.07	6.65±2.04	0.02±0.14	34.27±0.91
	$\delta = 0.15$	10	V-C (2)	0.59±0.07	7.07±2.13	0.05±0.25	34.22±0.91
		50	V-C (2)	0.59±0.07	7.11±2.10	0.05±0.25	34.22±0.92
		100	V-C (2)	0.59±0.07	7.06±2.12	0.04±0.25	34.22±0.91
		200	V-C (2)	0.59±0.07	7.05±2.12	0.05±0.26	34.23±0.91
		∞	V-C (2)	0.59±0.07	7.04±2.11	0.04±0.25	34.23±0.91
	$\delta = 0.20$	10	V-C (2)	0.58±0.07	7.36±2.23	0.06±0.26	34.18±0.91
		50	V-C (2)	0.58±0.07	7.44±2.16	0.07±0.27	34.18±0.91
		100	V-C (2)	0.58±0.07	7.39±2.20	0.06±0.26	34.18±0.92
		200	V-C (2)	0.58±0.07	7.35±2.21	0.06±0.26	34.19±0.92
		∞	V-C (2)	0.58±0.07	7.33±2.21	0.06±0.26	34.19±0.92
$\alpha = 0.25$	$\delta = 0.025$	10	V-C (2)	0.57±0.05	7.83±1.77	0.05±0.19	34.12±0.89
		50	V-C (2)	0.57±0.05	7.83±1.74	0.05±0.19	34.13±0.89
		100	V-C (2)	0.58±0.05	7.74±1.71	0.05±0.19	34.14±0.88
		200	V-C (2)	0.59±0.05	7.52±1.72	0.05±0.17	34.16±0.89
		∞	V-C (2)	0.59±0.05	7.47±1.73	0.05±0.17	34.17±0.89
	$\delta = 0.05$	10	V-C (2)	0.55±0.06	8.33±1.84	0.08±0.26	34.04±0.90
		50	V-C (2)	0.55±0.05	8.41±1.80	0.08±0.27	34.03±0.90
		100	V-C (2)	0.56±0.05	8.30±1.79	0.08±0.26	34.05±0.90
		200	V-C (2)	0.57±0.05	8.03±1.80	0.07±0.25	34.08±0.90
		∞	V-C (2)	0.57±0.05	7.95±1.79	0.06±0.23	34.08±0.90
	$\delta = 0.10$	10	V-C (2)	0.53±0.06	8.94±1.88	0.12±0.36	33.97±0.88
		50	V-C (2)	0.53±0.06	9.00±1.93	0.13±0.37	33.96±0.88
		100	V-C (2)	0.53±0.06	8.90±1.88	0.12±0.36	33.98±0.87
		200	V-C (2)	0.54±0.06	8.60±1.87	0.11±0.35	34.00±0.88
		∞	V-C (2)	0.55±0.06	8.49±1.87	0.10±0.34	34.01±0.89
	$\delta = 0.15$	10	V-C (2)	0.52±0.06	9.31±1.97	0.14±0.36	33.93±0.88
		50	V-C (2)	0.52±0.06	9.35±1.96	0.14±0.36	33.92±0.88
		100	V-C (2)	0.52±0.06	9.23±1.94	0.14±0.37	33.94±0.87
		200	V-C (2)	0.53±0.06	8.94±1.93	0.12±0.35	33.96±0.88
		∞	V-C (2)	0.54±0.06	8.80±1.93	0.11±0.33	33.98±0.89

Table I (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A	30.69±13.51	22.32±10.04	70.00±0.00
$c = 0$	0.00±0.00	28.00±2.99	15.27±4.17	32.45±0.77
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	35.91±0.92
ACI	0.22±0.04	17.68±1.59	2.74±1.43	33.13±0.81
QT	0.30±0.03	15.19±1.31	0.61±0.52	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.25$	$\delta = 0.20$	10	V-C (2)	0.50±0.06	9.67±2.09	0.18±0.41	33.88±0.89
		50	V-C (2)	0.50±0.06	9.74±2.12	0.18±0.41	33.86±0.89
		100	V-C (2)	0.50±0.06	9.61±2.08	0.19±0.41	33.88±0.87
		200	V-C (2)	0.52±0.06	9.32±2.05	0.17±0.40	33.91±0.89
		∞	V-C (2)	0.52±0.06	9.15±2.05	0.15±0.39	33.93±0.90

B. Maze, $\alpha_0 = 0.30$

TABLE II

COMPACT SWEEP (BEST VARIANT), MAZE, $\alpha_0 = 0.30$. *File:* APPENDICES/GENERATED_N200/CATALOG/PAPER/COMPACT_MAZE_A030.TEX. *Data:* [F]; FAMILIES MAIN_GRID_200, MATCHED_BASELINES_200; 130 VIEWS (IDS IN [C]). *Restore:* \NTABLE{../CATALOG/PAPER/COMPACT_MAZE_A030}; NOT USED IN THE CURRENT EDITIONS.

Compact sweep. One row per setting $(\alpha_0, \alpha, \delta, K)$ of the main grid. Variant: the certified variant with the most best means in the four columns (best: the bold rule of the key; ties of a column all count); (k): the number k of the four columns in which it has the best mean; ties in k: smaller printed $\widehat{m}_c/c_{\max}$, then $\widehat{q}^{+}_{\text{unsafe}}$, $\widehat{q}^{+}_{\text{CBF}}$, $\widehat{T}_{\text{mis}}$, then the order V-U, V-Δ, V-L, V-C. ±: one sample SD over the trajectories that enter the mean; $^{+}$: decision fraction; ↓: lower is better.

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$

Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A	55.68±1.58	53.27±0.95	42.16±0.14
$c = 0$	0.00±0.00	24.93±2.57	3.86±2.16	48.67±0.27
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	54.44±0.32
ACI	0.22±0.04	15.84±1.93	0.48±0.65	49.71±0.32
QT	0.50±0.03	8.04±1.31	0.05±0.17	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$

α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
$\alpha = 0.05$	$\delta = 0.025$	10	V-Δ (4)	0.97±0.03	0.08±0.21	0.00±0.00	54.21±0.38
		50	V-Δ (4)	0.97±0.03	0.08±0.21	0.00±0.00	54.21±0.39
		100	V-Δ (4)	0.97±0.03	0.08±0.21	0.00±0.00	54.21±0.39
		200	V-Δ (4)	0.97±0.03	0.08±0.21	0.00±0.00	54.22±0.38
		∞	V-Δ (4)	0.97±0.03	0.08±0.21	0.00±0.00	54.22±0.38
	$\delta = 0.05$	10	V-U (4)	0.96±0.03	0.10±0.25	0.00±0.00	54.16±0.40
		50	V-U (4)	0.96±0.03	0.10±0.26	0.00±0.00	54.15±0.40
		100	V-U (4)	0.96±0.03	0.10±0.25	0.00±0.00	54.16±0.40
		200	V-U (4)	0.96±0.03	0.10±0.25	0.00±0.00	54.16±0.40
		∞	V-U (4)	0.96±0.03	0.10±0.25	0.00±0.00	54.16±0.40
	$\delta = 0.10$	10	V-C (4)	0.95±0.04	0.18±0.35	0.00±0.00	54.04±0.43
		50	V-U (4)	0.95±0.04	0.19±0.37	0.00±0.00	54.05±0.42
		100	V-C (4)	0.95±0.04	0.17±0.34	0.00±0.00	54.05±0.43
		200	V-U (4)	0.95±0.04	0.17±0.35	0.00±0.00	54.06±0.42
		∞	V-U (4)	0.95±0.04	0.17±0.35	0.00±0.00	54.06±0.42
	$\delta = 0.15$	10	V-U (4)	0.94±0.04	0.21±0.41	0.00±0.00	53.96±0.44
		50	V-U (4)	0.94±0.04	0.22±0.41	0.00±0.00	53.96±0.44
		100	V-U (4)	0.94±0.04	0.21±0.41	0.00±0.00	53.97±0.44
		200	V-U (4)	0.94±0.04	0.21±0.41	0.00±0.00	53.97±0.44
		∞	V-U (4)	0.94±0.04	0.21±0.41	0.00±0.00	53.97±0.44
	$\delta = 0.20$	10	V-U (4)	0.93±0.04	0.25±0.47	0.00±0.00	53.91±0.45
		50	V-U (4)	0.93±0.04	0.26±0.48	0.00±0.00	53.91±0.45
		100	V-U (4)	0.93±0.04	0.25±0.47	0.00±0.00	53.91±0.45
		200	V-U (4)	0.93±0.04	0.25±0.47	0.00±0.00	53.92±0.45
		∞	V-U (4)	0.93±0.04	0.25±0.47	0.00±0.00	53.92±0.45
$\alpha = 0.10$	$\delta = 0.025$	10	V-Δ (3)	0.88±0.04	0.55±0.55	0.00±0.00	53.56±0.46
		50	V-Δ (3)	0.88±0.04	0.55±0.55	0.00±0.00	53.57±0.46
		100	V-L (3)	0.88±0.04	0.62±0.61	0.00±0.00	53.52±0.47
		200	V-L (3)	0.88±0.04	0.61±0.61	0.00±0.00	53.53±0.47
		∞	V-L (3)	0.88±0.04	0.61±0.61	0.00±0.00	53.53±0.47

Table II (continued).

Maze	**Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$			
Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A	55.68±1.58	53.27±0.95	42.16±0.14
$c = 0$	0.00±0.00	24.93±2.57	3.86±2.16	48.67±0.27
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	54.44±0.32
ACI	0.22±0.04	15.84±1.93	0.48±0.65	49.71±0.32
QT	0.50±0.03	8.04±1.31	0.05±0.17	51.18±0.32

Maze	$\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$						
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.10$	$\delta = 0.05$	10	V-C (3)	0.86±0.05	0.75±0.73	0.00±0.00	53.41±0.45
		50	V-C (3)	0.86±0.05	0.77±0.73	0.00±0.00	53.40±0.44
		100	V-C (3)	0.86±0.05	0.76±0.71	0.00±0.00	53.41±0.44
		200	V-C (3)	0.86±0.05	0.75±0.72	0.00±0.00	53.41±0.44
		∞	V-C (3)	0.86±0.05	0.75±0.72	0.00±0.00	53.41±0.44
	$\delta = 0.10$	10	V-L (3)	0.84±0.05	1.08±0.90	0.00±0.00	53.25±0.49
		50	V-U (3)	0.84±0.05	0.95±0.80	0.00±0.00	53.30±0.49
		100	V-C (3)	0.84±0.05	1.07±0.91	0.00±0.00	53.25±0.50
		200	V-L (3)	0.84±0.05	1.04±0.90	0.00±0.00	53.27±0.50
		∞	V-L (3)	0.84±0.05	1.04±0.90	0.00±0.00	53.27±0.50
	$\delta = 0.15$	10	V-L (3)	0.82±0.06	1.21±1.02	0.00±0.00	53.14±0.52
		50	V-C (3)	0.82±0.06	1.21±1.03	0.00±0.00	53.13±0.53
		100	V-L (3)	0.82±0.06	1.19±1.01	0.00±0.00	53.15±0.51
		200	V-C (3)	0.82±0.06	1.18±1.01	0.00±0.00	53.15±0.52
		∞	V-C (3)	0.82±0.06	1.18±1.01	0.00±0.00	53.15±0.52
	$\delta = 0.20$	10	V-C (3)	0.81±0.07	1.49±1.31	0.00±0.00	53.02±0.58
		50	V-U (3)	0.81±0.06	1.37±1.15	0.00±0.00	53.08±0.54
		100	V-L (3)	0.81±0.07	1.49±1.30	0.00±0.00	53.03±0.57
		200	V-L (3)	0.81±0.07	1.47±1.30	0.00±0.00	53.04±0.57
		∞	V-L (3)	0.81±0.07	1.47±1.30	0.00±0.00	53.04±0.58
$\alpha = 0.15$	$\delta = 0.025$	10	V-C (3)	0.78±0.05	1.85±1.10	0.00±0.00	52.83±0.46
		50	V-L (3)	0.78±0.05	1.84±1.12	0.00±0.00	52.83±0.46
		100	V-L (3)	0.78±0.05	1.84±1.11	0.00±0.00	52.83±0.46
		200	V-L (3)	0.78±0.05	1.83±1.11	0.00±0.00	52.84±0.46
		∞	V-L (3)	0.78±0.05	1.82±1.12	0.00±0.00	52.84±0.46
	$\delta = 0.05$	10	V-L (3)	0.76±0.06	2.16±1.27	0.00±0.00	52.72±0.52
		50	V-L (3)	0.76±0.06	2.16±1.28	0.00±0.00	52.72±0.52
		100	V-L (3)	0.76±0.06	2.14±1.28	0.00±0.00	52.72±0.52
		200	V-L (3)	0.76±0.06	2.13±1.28	0.00±0.00	52.73±0.52
		∞	V-L (3)	0.76±0.06	2.13±1.28	0.00±0.00	52.73±0.52
	$\delta = 0.10$	10	V-C (3)	0.73±0.06	2.67±1.53	0.00±0.00	52.53±0.52
		50	V-C (3)	0.73±0.06	2.68±1.51	0.00±0.00	52.55±0.53
		100	V-C (3)	0.73±0.06	2.65±1.53	0.00±0.00	52.55±0.53
		200	V-C (3)	0.73±0.06	2.64±1.53	0.00±0.00	52.56±0.53
		∞	V-C (3)	0.73±0.06	2.64±1.53	0.00±0.00	52.56±0.53
	$\delta = 0.15$	10	V-C (3)	0.71±0.07	3.04±1.67	0.00±0.00	52.43±0.52
		50	V-C (3)	0.71±0.07	3.01±1.62	0.00±0.00	52.43±0.52
		100	V-L (3)	0.72±0.07	2.98±1.62	0.00±0.00	52.44±0.52
		200	V-C (3)	0.72±0.07	2.97±1.62	0.00±0.00	52.44±0.53
		∞	V-L (3)	0.72±0.07	2.97±1.62	0.00±0.00	52.45±0.53
	$\delta = 0.20$	10	V-C (3)	0.70±0.08	3.38±1.87	0.00±0.06	52.33±0.59
		50	V-C (3)	0.70±0.08	3.35±1.85	0.00±0.06	52.33±0.58
		100	V-C (3)	0.70±0.08	3.36±1.88	0.00±0.06	52.32±0.58
		200	V-L (3)	0.70±0.08	3.31±1.89	0.00±0.06	52.34±0.58
		∞	V-C (3)	0.70±0.08	3.30±1.87	0.00±0.06	52.34±0.58

Table II (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$					
Method			$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered			N-A	55.68±1.58	53.27±0.95	42.16±0.14
$c = 0$			0.00±0.00	24.93±2.57	3.86±2.16	48.67±0.27
$c_{\max}$			1.00±0.00	0.00±0.00	0.00±0.00	54.44±0.32
ACI			0.22±0.04	15.84±1.93	0.48±0.65	49.71±0.32
QT			0.50±0.03	8.04±1.31	0.05±0.17	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.20$	$\delta = 0.025$	10	V-C (3)	0.69±0.06	3.67±1.58	0.00±0.00	52.20±0.47
		50	V-C (3)	0.69±0.06	3.60±1.57	0.00±0.00	52.21±0.47
		100	V-C (3)	0.69±0.06	3.60±1.57	0.00±0.00	52.21±0.48
		200	V-C (3)	0.69±0.06	3.59±1.57	0.00±0.00	52.22±0.48
		∞	V-C (3)	0.69±0.06	3.59±1.57	0.00±0.00	52.22±0.48
	$\delta = 0.05$	10	V-C (3)	0.66±0.06	4.06±1.74	0.00±0.00	52.09±0.49
		50	V-C (3)	0.67±0.06	4.04±1.74	0.00±0.00	52.08±0.49
		100	V-C (3)	0.67±0.06	4.02±1.72	0.00±0.00	52.08±0.49
		200	V-C (3)	0.67±0.06	4.01±1.73	0.00±0.00	52.09±0.49
		∞	V-C (3)	0.67±0.06	4.01±1.73	0.00±0.00	52.09±0.49
	$\delta = 0.10$	10	V-C (3)	0.64±0.07	4.60±1.93	0.00±0.06	51.94±0.54
		50	V-C (3)	0.64±0.07	4.57±1.91	0.00±0.06	51.94±0.53
		100	V-C (3)	0.64±0.07	4.56±1.89	0.00±0.06	51.96±0.54
		200	V-C (3)	0.64±0.07	4.52±1.91	0.00±0.06	51.96±0.54
		∞	V-C (3)	0.64±0.07	4.52±1.91	0.00±0.06	51.96±0.54
	$\delta = 0.15$	10	V-C (3)	0.62±0.07	5.12±2.18	0.00±0.06	51.82±0.57
		50	V-C (3)	0.62±0.08	5.05±2.14	0.00±0.06	51.83±0.56
		100	V-C (3)	0.62±0.08	5.01±2.11	0.00±0.06	51.84±0.55
		200	V-C (3)	0.62±0.08	4.99±2.12	0.00±0.06	51.85±0.56
		∞	V-C (3)	0.62±0.08	4.99±2.12	0.00±0.06	51.85±0.56
	$\delta = 0.20$	10	V-C (3)	0.61±0.08	5.31±2.18	0.00±0.06	51.79±0.57
		50	V-C (3)	0.61±0.08	5.38±2.19	0.00±0.06	51.79±0.58
		100	V-C (3)	0.61±0.08	5.31±2.17	0.00±0.06	51.79±0.58
		200	V-C (3)	0.61±0.08	5.26±2.17	0.00±0.06	51.81±0.58
		∞	V-C (3)	0.61±0.08	5.26±2.17	0.00±0.06	51.81±0.58
$\alpha = 0.25$	$\delta = 0.025$	10	V-C (3)	0.60±0.06	5.73±1.88	0.00±0.04	51.67±0.47
		50	V-C (3)	0.60±0.06	5.77±1.86	0.00±0.04	51.67±0.48
		100	V-C (2)	0.60±0.06	5.75±1.88	0.01±0.05	51.67±0.47
		200	V-C (3)	0.60±0.06	5.75±1.88	0.00±0.04	51.67±0.47
		∞	V-C (3)	0.60±0.06	5.74±1.89	0.00±0.04	51.67±0.47
	$\delta = 0.05$	10	V-C (2)	0.58±0.06	6.27±1.93	0.01±0.06	51.58±0.50
		50	V-C (2)	0.58±0.06	6.26±1.92	0.01±0.06	51.57±0.49
		100	V-C (2)	0.58±0.06	6.24±1.93	0.01±0.06	51.58±0.49
		200	V-C (2)	0.58±0.06	6.22±1.94	0.01±0.06	51.58±0.49
		∞	V-C (2)	0.58±0.06	6.22±1.94	0.01±0.06	51.58±0.49
	$\delta = 0.10$	10	V-C (2)	0.55±0.07	6.76±2.17	0.01±0.09	51.43±0.52
		50	V-C (2)	0.55±0.07	6.78±2.18	0.01±0.09	51.44±0.54
		100	V-C (2)	0.55±0.07	6.74±2.17	0.01±0.09	51.46±0.54
		200	V-C (2)	0.56±0.07	6.71±2.18	0.01±0.09	51.47±0.54
		∞	V-C (2)	0.56±0.07	6.71±2.18	0.01±0.09	51.47±0.54
	$\delta = 0.15$	10	V-C (2)	0.53±0.07	7.22±2.34	0.01±0.07	51.34±0.54
		50	V-C (2)	0.53±0.08	7.28±2.37	0.01±0.07	51.33±0.53
		100	V-C (2)	0.53±0.08	7.21±2.33	0.01±0.08	51.33±0.54
		200	V-C (2)	0.53±0.08	7.15±2.36	0.01±0.07	51.35±0.53
		∞	V-C (2)	0.53±0.08	7.15±2.36	0.01±0.07	51.35±0.53

Table II (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$				
Method	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A	55.68±1.58	53.27±0.95	42.16±0.14
$c = 0$	0.00±0.00	24.93±2.57	3.86±2.16	48.67±0.27
$c_{\max}$	1.00±0.00	0.00±0.00	0.00±0.00	54.44±0.32
ACI	0.22±0.04	15.84±1.93	0.48±0.65	49.71±0.32
QT	0.50±0.03	8.04±1.31	0.05±0.17	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
α	δ	K	Variant (k)	$\widehat{m}_c/c_{\max}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.25$	$\delta = 0.20$	10	V-C (2)	0.52±0.07	7.38±2.26	0.01±0.06	51.26±0.52
		50	V-C (2)	0.52±0.07	7.40±2.27	0.01±0.06	51.25±0.52
		100	V-C (2)	0.52±0.07	7.42±2.27	0.01±0.06	51.25±0.52
		200	V-C (2)	0.52±0.08	7.32±2.30	0.01±0.06	51.27±0.52
		∞	V-C (2)	0.52±0.08	7.32±2.30	0.01±0.06	51.27±0.52

II. Full sweep tables at $\alpha_0 = 0.30$

A. *Corridor,* $\alpha_0 = 0.30$

TABLE III

Full sweep, Corridor, $\alpha_0 = 0.30$. *File:* appendices/generated_n200/catalog/paper/sweep_corridor_a030.tex. *Data:* [F]; families main_grid_200, matched_baselines_200; 505 views (ids in [C]). *Restore:* \ntable{../catalog/paper/sweep_corridor_a030}; not used in the current editions.

$\pm$: one sample SD over the trajectories that enter the mean; $^+$: decision fraction; $\downarrow/\uparrow$: lower/higher is better; bold: lowest ($\downarrow$) or highest ($\uparrow$) mean of the four certified variants at one setting; N-A: not applicable.

Corridor Baselines $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$

Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$	$\widehat{q}^+_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^+_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$

α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^+_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^+_{Y}\downarrow(\%)$	$\widehat{q}^+_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^+_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.025	10	V-U	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 4.43±4.00	**0.33**±0.38 **0.54**±0.55	38.0±24.1 **0.00**±0.00	**59.3**±24.1 2.7±2.6	**0.04**±0.01 **0.17**±0.01	18.94±6.98 1.5±0.8	**35.72**±0.92
			V-Δ	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 4.43±4.00	**0.33**±0.38 **0.54**±0.55	38.0±24.1 **0.00**±0.00	**59.3**±24.1 2.7±2.6	**0.04**±0.01 **0.17**±0.01	18.94±6.98 1.5±0.8	**35.72**±0.92
			V-L	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 4.43±4.00	**0.33**±0.38 0.59±0.61	**39.7**±23.4 **0.00**±0.00	**59.3**±24.1 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.31±7.33 1.5±0.8	**35.72**±0.92
			V-C	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 4.43±4.00	**0.33**±0.38 0.59±0.61	**39.7**±23.4 **0.00**±0.00	**59.3**±24.1 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.31±7.33 1.5±0.8	**35.72**±0.92
		50	V-U	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 2.04±1.27	**0.32**±0.37 **0.54**±0.54	38.2±23.5 **0.00**±0.00	**59.0**±23.6 2.8±2.9	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	**35.72**±0.92
			V-Δ	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 2.04±1.27	**0.32**±0.37 **0.54**±0.54	38.2±23.5 **0.00**±0.00	**59.0**±23.6 2.8±2.9	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	**35.72**±0.92
			V-L	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 2.04±1.27	0.33±0.38 0.59±0.60	**40.0**±23.0 **0.00**±0.00	**59.0**±23.6 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.38±7.27 1.5±0.8	**35.72**±0.92
			V-C	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 2.04±1.27	0.33±0.38 0.59±0.60	**40.0**±23.0 **0.00**±0.00	**59.0**±23.6 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.38±7.27 1.5±0.8	**35.72**±0.92
		100	V-U	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.47±0.86	**0.32**±0.37 **0.52**±0.53	36.0±22.6 **0.00**±0.00	**59.2**±23.6 4.9±3.6	**0.04**±0.01 **0.17**±0.01	19.11±6.96 1.5±0.8	35.73±0.92
			V-Δ	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.47±0.86	**0.32**±0.37 **0.52**±0.53	36.0±22.6 **0.00**±0.00	**59.2**±23.6 4.9±3.6	**0.04**±0.01 **0.17**±0.01	19.11±6.96 1.5±0.8	35.73±0.92
			V-L	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.47±0.86	0.33±0.38 0.58±0.60	**39.8**±23.0 **0.00**±0.00	**59.2**±23.6 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.47±7.29 1.5±0.8	**35.72**±0.92
			V-C	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.47±0.86	0.33±0.38 0.58±0.60	**39.8**±23.0 **0.00**±0.00	**59.2**±23.6 **1.1**±0.8	**0.04**±0.01 **0.17**±0.01	19.47±7.29 1.5±0.8	**35.72**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.05$	$\delta = 0.025$	200	V-U	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.17±0.63	**0.31**±0.37 **0.48**±0.52	34.1±22.4 **0.00**±0.00	**59.7**±24.1 6.3±3.5	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	35.73±0.92
			V-Δ	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.17±0.63	**0.31**±0.37 **0.48**±0.52	34.1±22.4 **0.00**±0.00	**59.7**±24.1 6.3±3.5	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	35.73±0.92
			V-L	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.17±0.63	0.33±0.38 0.58±0.60	**39.3**±23.4 **0.00**±0.00	**59.7**±24.1 **1.0**±0.8	**0.04**±0.01 **0.17**±0.01	19.48±7.36 1.5±0.8	**35.72**±0.92
			V-C	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 1.17±0.63	0.33±0.38 0.58±0.60	**39.3**±23.4 **0.00**±0.00	**59.7**±24.1 **1.0**±0.8	**0.04**±0.01 **0.17**±0.01	19.48±7.36 1.5±0.8	**35.72**±0.92
		∞	V-U	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 0.97±0.44	**0.30**±0.37 **0.46**±0.52	33.1±22.1 **0.00**±0.00	**59.7**±24.1 7.3±3.3	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	35.73±0.92
			V-Δ	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 0.97±0.44	**0.30**±0.37 **0.46**±0.52	33.1±22.1 **0.00**±0.00	**59.7**±24.1 7.3±3.3	**0.04**±0.01 **0.17**±0.01	19.02±6.94 1.5±0.8	35.73±0.92
			V-L	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 0.97±0.44	0.33±0.38 0.58±0.60	**39.3**±23.4 **0.00**±0.00	**59.7**±24.1 **1.0**±0.8	**0.04**±0.01 **0.17**±0.01	19.48±7.36 1.5±0.8	**35.72**±0.92
			V-C	**0.96**±0.03 **0.17**±0.01	**0.98**±0.08 0.97±0.44	0.33±0.38 0.58±0.60	**39.3**±23.4 **0.00**±0.00	**59.7**±24.1 **1.0**±0.8	**0.04**±0.01 **0.17**±0.01	19.48±7.36 1.5±0.8	**35.72**±0.92
	$\delta = 0.05$	10	V-U	0.95±0.04 **0.17**±0.01	**0.98**±0.08 5.67±4.47	**0.46**±0.54 **0.72**±0.72	47.2±24.1 **0.00**±0.00	**50.1**±23.8 2.7±2.3	**0.04**±0.01 **0.17**±0.01	16.11±6.78 1.7±1.0	35.66±0.92
			V-Δ	0.95±0.04 **0.17**±0.01	**0.98**±0.08 5.67±4.47	**0.46**±0.54 **0.72**±0.72	47.2±24.1 **0.00**±0.00	**50.1**±23.8 2.7±2.3	**0.04**±0.01 **0.17**±0.01	16.11±6.78 1.7±1.0	35.66±0.92
			V-L	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 5.67±4.47	0.47±0.54 0.77±0.78	**48.5**±23.0 **0.00**±0.00	**50.1**±23.8 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.78±7.48 1.7±1.0	**35.65**±0.92
			V-C	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 5.67±4.47	0.47±0.54 0.77±0.78	**48.5**±23.0 **0.00**±0.00	**50.1**±23.8 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.78±7.48 1.7±1.0	**35.65**±0.92
		50	V-U	0.95±0.04 **0.17**±0.01	**0.98**±0.08 2.46±1.33	**0.46**±0.52 **0.73**±0.71	47.0±23.6 **0.00**±0.00	**50.1**±23.4 2.9±2.8	**0.04**±0.01 **0.17**±0.01	16.30±6.86 1.7±1.0	**35.66**±0.92
			V-Δ	0.95±0.04 **0.17**±0.01	**0.98**±0.08 2.46±1.33	**0.46**±0.52 **0.73**±0.71	47.0±23.6 **0.00**±0.00	**50.1**±23.4 2.9±2.8	**0.04**±0.01 **0.17**±0.01	16.30±6.86 1.7±1.0	**35.66**±0.92
			V-L	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 2.46±1.33	0.47±0.53 0.77±0.77	**48.6**±22.7 **0.00**±0.00	**50.1**±23.4 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.85±7.42 1.7±1.0	**35.66**±0.92
			V-C	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 2.46±1.33	0.47±0.53 0.77±0.77	**48.6**±22.7 **0.00**±0.00	**50.1**±23.4 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.85±7.42 1.7±1.0	**35.66**±0.92
		100	V-U	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.84±0.97	**0.45**±0.53 **0.70**±0.71	45.5±22.8 **0.00**±0.00	**50.3**±23.4 4.2±3.4	**0.04**±0.01 **0.17**±0.01	16.67±7.23 1.7±1.0	35.66±0.92
			V-Δ	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.84±0.97	**0.45**±0.53 **0.70**±0.71	45.5±22.8 **0.00**±0.00	**50.3**±23.4 4.2±3.4	**0.04**±0.01 **0.17**±0.01	16.67±7.23 1.7±1.0	35.66±0.92
			V-L	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.84±0.97	0.47±0.55 0.77±0.78	**48.4**±22.7 **0.00**±0.00	**50.3**±23.4 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.93±7.44 1.7±1.0	**35.65**±0.92
			V-C	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.84±0.97	0.47±0.55 0.77±0.78	**48.4**±22.7 **0.00**±0.00	**50.3**±23.4 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	16.93±7.44 1.7±1.0	**35.65**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.05	0.05	200	V-U	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.30±0.66	**0.43**±0.52 **0.65**±0.69	42.9±23.1 **0.00**±0.00	**50.7**±23.9 6.4±3.6	**0.04**±0.01 **0.17**±0.01	16.25±6.84 1.7±1.0	**35.66**±0.92
			V-Δ	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.30±0.66	**0.43**±0.52 **0.65**±0.69	42.9±23.1 **0.00**±0.00	**50.7**±23.9 6.4±3.6	**0.04**±0.01 **0.17**±0.01	16.25±6.84 1.7±1.0	**35.66**±0.92
			V-L	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.30±0.66	0.46±0.53 0.76±0.77	**47.9**±23.2 **0.00**±0.00	**50.7**±24.0 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	17.01±7.60 1.7±1.0	**35.66**±0.92
			V-C	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.30±0.66	0.46±0.53 0.76±0.77	**47.9**±23.2 **0.00**±0.00	**50.7**±24.0 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	17.01±7.60 1.7±1.0	**35.66**±0.92
		∞	V-U	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.08±0.46	**0.42**±0.52 **0.63**±0.70	41.3±22.5 **0.00**±0.00	**50.7**±23.9 8.0±2.8	**0.04**±0.01 **0.17**±0.01	16.25±6.84 1.6±1.0	35.67±0.92
			V-Δ	0.95±0.04 **0.17**±0.01	**0.98**±0.08 1.08±0.46	**0.42**±0.52 **0.63**±0.70	41.3±22.5 **0.00**±0.00	**50.7**±23.9 8.0±2.8	**0.04**±0.01 **0.17**±0.01	16.25±6.84 1.6±1.0	35.67±0.92
			V-L	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.08±0.46	0.46±0.53 0.76±0.77	**47.9**±23.2 **0.00**±0.00	**50.7**±24.0 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	17.01±7.60 1.7±1.0	**35.66**±0.92
			V-C	**0.94**±0.04 **0.17**±0.01	**0.98**±0.08 1.08±0.46	0.46±0.53 0.76±0.77	**47.9**±23.2 **0.00**±0.00	**50.7**±24.0 **1.4**±0.9	**0.04**±0.01 **0.17**±0.01	17.01±7.60 1.7±1.0	**35.66**±0.92
0.05	0.10	10	V-U	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 7.02±4.74	**0.62**±0.63 **1.00**±0.88	56.2±22.1 **0.00**±0.00	**41.1**±21.6 2.7±2.2	**0.04**±0.01 **0.17**±0.01	13.94±6.86 1.9±1.3	35.59±0.92
			V-Δ	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 7.02±4.74	**0.62**±0.63 **1.00**±0.88	56.2±22.1 **0.00**±0.00	**41.1**±21.6 2.7±2.2	**0.04**±0.01 **0.17**±0.01	13.94±6.86 1.9±1.3	35.59±0.92
			V-L	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 7.02±4.74	0.63±0.63 1.03±0.92	**57.2**±20.9 **0.00**±0.00	**41.1**±21.6 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.34±7.36 2.0±1.3	**35.58**±0.92
			V-C	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 7.02±4.74	0.63±0.63 1.03±0.92	**57.2**±20.9 **0.00**±0.00	**41.1**±21.6 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.34±7.36 2.0±1.3	**35.58**±0.92
		50	V-U	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 3.09±1.53	**0.63**±0.63 **1.00**±0.88	56.3±21.2 **0.00**±0.00	**41.0**±21.1 2.6±2.3	**0.04**±0.01 **0.17**±0.01	14.22±7.10 2.0±1.3	**35.58**±0.92
			V-Δ	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 3.09±1.53	**0.63**±0.63 **1.00**±0.88	56.3±21.2 **0.00**±0.00	**41.0**±21.1 2.6±2.3	**0.04**±0.01 **0.17**±0.01	14.22±7.10 2.0±1.3	**35.58**±0.92
			V-L	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 3.09±1.53	0.64±0.64 1.04±0.92	**57.4**±20.4 **0.00**±0.00	**41.0**±21.1 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.31±7.17 2.0±1.3	**35.58**±0.92
			V-C	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 3.09±1.53	0.64±0.64 1.04±0.92	**57.4**±20.4 **0.00**±0.00	**41.0**±21.1 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.31±7.17 2.0±1.3	**35.58**±0.92
		100	V-U	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 2.20±0.99	**0.63**±0.63 **0.98**±0.87	55.6±20.5 **0.00**±0.00	**40.8**±20.6 3.6±2.9	**0.04**±0.01 **0.17**±0.01	14.12±6.88 1.9±1.3	35.59±0.92
			V-Δ	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 2.20±0.99	**0.63**±0.63 **0.98**±0.87	55.6±20.5 **0.00**±0.00	**40.8**±20.6 3.6±2.9	**0.04**±0.01 **0.17**±0.01	14.12±6.88 1.9±1.3	35.59±0.92
			V-L	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 2.20±0.99	**0.63**±0.63 1.04±0.92	**57.6**±20.0 **0.00**±0.00	**40.8**±20.6 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.21±6.98 2.0±1.3	**35.58**±0.92
			V-C	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 2.20±0.99	**0.63**±0.63 1.04±0.92	**57.6**±20.0 **0.00**±0.00	**40.8**±20.6 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.21±6.98 2.0±1.3	**35.58**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.10	200	V-U	0.94±0.04 **0.17**±0.01	**0.97**±0.08 1.53±0.71	**0.60**±0.63 **0.92**±0.85	52.1±21.5 **0.00**±0.00	**41.4**±21.7 6.4±3.6	**0.04**±0.01 **0.17**±0.01	14.18±7.09 1.9±1.3	**35.59**±0.93
			V-Δ	0.94±0.04 **0.17**±0.01	**0.97**±0.08 1.53±0.71	**0.60**±0.63 **0.92**±0.85	52.1±21.5 **0.00**±0.00	**41.4**±21.7 6.4±3.6	**0.04**±0.01 **0.17**±0.01	14.18±7.09 1.9±1.3	**35.59**±0.93
			V-L	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 1.53±0.71	0.62±0.63 1.02±0.92	**56.9**±21.0 **0.00**±0.00	41.5±21.7 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.46±7.38 2.0±1.3	**35.59**±0.92
			V-C	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 1.53±0.71	0.62±0.63 1.02±0.92	**56.9**±21.0 **0.00**±0.00	41.5±21.7 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.46±7.38 2.0±1.3	**35.59**±0.92
		∞	V-U	0.94±0.04 **0.17**±0.01	**0.97**±0.08 1.24±0.51	**0.59**±0.63 **0.89**±0.86	50.1±20.9 **0.00**±0.00	**41.4**±21.7 8.4±2.3	**0.04**±0.01 **0.17**±0.01	14.06±6.90 1.9±1.3	35.60±0.93
			V-Δ	0.94±0.04 **0.17**±0.01	**0.97**±0.08 1.24±0.51	**0.59**±0.63 **0.89**±0.86	50.1±20.9 **0.00**±0.00	**41.4**±21.7 8.4±2.3	**0.04**±0.01 **0.17**±0.01	14.06±6.90 1.9±1.3	35.60±0.93
			V-L	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 1.24±0.51	0.62±0.63 1.02±0.92	**56.9**±21.0 **0.00**±0.00	41.5±21.7 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.46±7.38 2.0±1.3	**35.59**±0.92
			V-C	**0.93**±0.04 **0.17**±0.01	**0.97**±0.08 1.24±0.51	0.62±0.63 1.02±0.92	**56.9**±21.0 **0.00**±0.00	41.5±21.7 **1.6**±0.8	**0.04**±0.01 **0.17**±0.01	14.46±7.38 2.0±1.3	**35.59**±0.92
	0.15	10	V-U	**0.92**±0.04 **0.17**±0.01	**0.96**±0.09 7.82±4.95	**0.73**±0.70 **1.20**±0.99	61.7±20.5 **0.00**±0.00	**35.5**±20.2 2.7±2.1	**0.04**±0.01 **0.17**±0.01	12.44±6.94 2.1±1.3	**35.54**±0.94
			V-Δ	**0.92**±0.04 **0.17**±0.01	**0.96**±0.09 7.82±4.95	**0.73**±0.70 **1.20**±0.99	61.7±20.5 **0.00**±0.00	**35.5**±20.2 2.7±2.1	**0.04**±0.01 **0.17**±0.01	12.44±6.94 2.1±1.3	**35.54**±0.94
			V-L	**0.92**±0.05 **0.17**±0.01	**0.96**±0.09 7.82±4.95	0.74±0.70 1.24±1.05	**62.7**±19.6 **0.00**±0.00	**35.5**±20.2 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.56±7.11 2.2±1.4	**35.54**±0.93
			V-C	**0.92**±0.05 **0.17**±0.01	**0.96**±0.09 7.82±4.95	0.74±0.70 1.24±1.05	**62.7**±19.6 **0.00**±0.00	**35.5**±20.2 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.56±7.11 2.2±1.4	**35.54**±0.93
		50	V-U	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 3.37±1.53	**0.72**±0.70 **1.17**±0.98	61.7±19.9 **0.00**±0.00	**35.6**±19.9 2.7±2.3	**0.04**±0.01 **0.17**±0.01	12.47±6.84 2.2±1.4	**35.54**±0.93
			V-Δ	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 3.37±1.53	**0.72**±0.70 **1.17**±0.98	61.7±19.9 **0.00**±0.00	**35.6**±19.9 2.7±2.3	**0.04**±0.01 **0.17**±0.01	12.47±6.84 2.2±1.4	**35.54**±0.93
			V-L	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 3.37±1.53	0.73±0.70 1.23±1.04	**62.7**±19.3 **0.00**±0.00	**35.6**±19.9 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.58±7.01 2.2±1.4	**35.54**±0.93
			V-C	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 3.37±1.53	0.73±0.70 1.23±1.04	**62.7**±19.3 **0.00**±0.00	**35.6**±19.9 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.58±7.01 2.2±1.4	**35.54**±0.93
		100	V-U	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 2.37±1.01	**0.72**±0.68 **1.18**±0.96	61.3±19.1 **0.00**±0.00	**35.2**±19.2 3.5±2.9	**0.04**±0.01 **0.17**±0.01	12.32±6.54 2.1±1.4	**35.54**±0.92
			V-Δ	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 2.37±1.01	**0.72**±0.68 **1.18**±0.96	61.3±19.1 **0.00**±0.00	**35.2**±19.2 3.5±2.9	**0.04**±0.01 **0.17**±0.01	12.32±6.54 2.1±1.4	**35.54**±0.92
			V-L	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 2.37±1.01	0.73±0.69 1.24±1.03	**63.0**±18.6 **0.00**±0.00	**35.2**±19.2 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.44±6.73 2.2±1.4	**35.54**±0.92
			V-C	**0.92**±0.04 **0.17**±0.01	**0.96**±0.08 2.37±1.01	0.73±0.69 1.24±1.03	**63.0**±18.6 **0.00**±0.00	**35.2**±19.2 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.44±6.73 2.2±1.4	**35.54**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.15	200	V-U	0.93±0.04 **0.17**±0.01	0.97±0.08 1.68±0.78	**0.69**±0.68 **1.10**±0.95	57.7±20.3 **0.00**±0.00	**35.9**±20.3 6.4±3.5	**0.04**±0.01 **0.17**±0.01	12.57±6.97 2.1±1.3	**35.54**±0.93
			V-Δ	0.93±0.04 **0.17**±0.01	0.97±0.08 1.68±0.78	**0.69**±0.68 **1.10**±0.95	57.7±20.3 **0.00**±0.00	**35.9**±20.3 6.4±3.5	**0.04**±0.01 **0.17**±0.01	12.57±6.97 2.1±1.3	**35.54**±0.93
			V-L	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 1.68±0.78	0.72±0.68 1.22±1.04	**62.4**±19.6 **0.00**±0.00	**35.9**±20.3 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.68±7.14 2.2±1.4	**35.54**±0.93
			V-C	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 1.68±0.78	0.72±0.68 1.22±1.04	**62.4**±19.6 **0.00**±0.00	**35.9**±20.3 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.68±7.14 2.2±1.4	**35.54**±0.93
		∞	V-U	0.93±0.04 **0.17**±0.01	0.97±0.09 1.34±0.53	**0.69**±0.68 **1.07**±0.95	55.6±19.8 **0.00**±0.00	**35.9**±20.3 8.5±2.1	**0.04**±0.01 **0.17**±0.01	12.57±6.97 2.1±1.3	35.55±0.94
			V-Δ	0.93±0.04 **0.17**±0.01	0.97±0.09 1.34±0.53	**0.69**±0.68 **1.07**±0.95	55.6±19.8 **0.00**±0.00	**35.9**±20.3 8.5±2.1	**0.04**±0.01 **0.17**±0.01	12.57±6.97 2.1±1.3	35.55±0.94
			V-L	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 1.34±0.53	0.72±0.68 1.22±1.04	**62.4**±19.6 **0.00**±0.00	**35.9**±20.3 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.68±7.14 2.2±1.4	**35.54**±0.93
			V-C	**0.92**±0.05 **0.17**±0.01	**0.96**±0.08 1.34±0.53	0.72±0.68 1.22±1.04	**62.4**±19.6 **0.00**±0.00	**35.9**±20.3 **1.7**±0.8	**0.04**±0.01 **0.17**±0.01	12.68±7.14 2.2±1.4	**35.54**±0.93
	0.20	10	V-U	**0.91**±0.05 **0.17**±0.01	**0.96**±0.09 8.70±5.26	**0.87**±0.86 **1.34**±1.14	65.3±18.8 **0.00**±0.00	**32.0**±18.6 2.7±1.9	**0.04**±0.01 **0.17**±0.01	11.27±6.52 2.3±1.6	**35.49**±0.95
			V-Δ	**0.91**±0.05 **0.17**±0.01	**0.96**±0.09 8.70±5.26	**0.87**±0.86 **1.34**±1.14	65.3±18.8 **0.00**±0.00	**32.0**±18.6 2.7±1.9	**0.04**±0.01 **0.17**±0.01	11.27±6.52 2.3±1.6	**35.49**±0.95
			V-L	**0.91**±0.05 **0.17**±0.01	**0.96**±0.09 8.70±5.26	**0.87**±0.88 1.39±1.21	**66.2**±18.0 **0.00**±0.00	**32.0**±18.6 **1.9**±0.8	**0.04**±0.01 **0.17**±0.01	11.38±6.71 2.4±1.6	**35.49**±0.95
			V-C	**0.91**±0.05 **0.17**±0.01	**0.96**±0.09 8.70±5.26	**0.87**±0.88 1.39±1.21	**66.2**±18.0 **0.00**±0.00	**32.0**±18.6 **1.9**±0.8	**0.04**±0.01 **0.17**±0.01	11.38±6.71 2.4±1.6	**35.49**±0.95
		50	V-U	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 3.67±1.57	**0.88**±0.87 **1.34**±1.13	65.6±17.8 **0.00**±0.00	**31.7**±17.8 2.7±2.0	**0.04**±0.01 **0.17**±0.01	11.18±6.25 2.4±1.6	35.49±0.94
			V-Δ	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 3.67±1.57	**0.88**±0.87 **1.34**±1.13	65.6±17.8 **0.00**±0.00	**31.7**±17.8 2.7±2.0	**0.04**±0.01 **0.17**±0.01	11.18±6.25 2.4±1.6	35.49±0.94
			V-L	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 3.67±1.57	**0.88**±0.89 1.41±1.21	**66.4**±17.3 **0.00**±0.00	**31.7**±17.8 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.30±6.44 2.4±1.6	**35.48**±0.94
			V-C	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 3.67±1.57	**0.88**±0.89 1.41±1.21	**66.4**±17.3 **0.00**±0.00	**31.7**±17.8 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.30±6.44 2.4±1.6	**35.48**±0.94
		100	V-U	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 2.53±1.05	**0.86**±0.84 **1.33**±1.11	65.1±17.4 **0.00**±0.00	**31.6**±17.5 3.3±2.9	**0.04**±0.01 **0.17**±0.01	11.13±6.11 2.3±1.6	**35.49**±0.94
			V-Δ	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 2.53±1.05	**0.86**±0.84 **1.33**±1.11	65.1±17.4 **0.00**±0.00	**31.6**±17.5 3.3±2.9	**0.04**±0.01 **0.17**±0.01	11.13±6.11 2.3±1.6	**35.49**±0.94
			V-L	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 2.53±1.05	0.87±0.87 1.40±1.19	**66.6**±17.0 **0.00**±0.00	**31.6**±17.5 **1.9**±0.8	**0.04**±0.01 **0.17**±0.01	11.25±6.31 2.4±1.6	**35.49**±0.94
			V-C	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 2.53±1.05	0.87±0.87 1.40±1.19	**66.6**±17.0 **0.00**±0.00	**31.6**±17.5 **1.9**±0.8	**0.04**±0.01 **0.17**±0.01	11.25±6.31 2.4±1.6	**35.49**±0.94

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.05$ $\delta = 0.20$	200 V-U	0.92±0.05 **0.17**±0.01	**0.96**±0.08 1.87±0.81	**0.84**±0.85 **1.26**±1.10	61.6±19.0 **0.00**±0.00	**32.2**±18.7 6.1±3.4	**0.04**±0.01 **0.17**±0.01	11.36±6.55 2.3±1.6	**35.49**±0.95
	200 V-Δ	0.92±0.05 **0.17**±0.01	**0.96**±0.08 1.87±0.81	**0.84**±0.85 **1.26**±1.10	61.6±19.0 **0.00**±0.00	**32.2**±18.7 6.1±3.4	**0.04**±0.01 **0.17**±0.01	11.36±6.55 2.3±1.6	**35.49**±0.95
	200 V-L	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 1.87±0.81	0.86±0.87 1.38±1.20	**65.9**±18.1 **0.00**±0.00	**32.2**±18.7 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.47±6.73 2.4±1.6	**35.49**±0.94
	200 V-C	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 1.87±0.81	0.86±0.87 1.38±1.20	**65.9**±18.1 **0.00**±0.00	**32.2**±18.7 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.47±6.73 2.4±1.6	**35.49**±0.94
	∞ V-U	0.92±0.05 **0.17**±0.01	**0.96**±0.09 1.42±0.59	**0.83**±0.85 **1.22**±1.11	59.2±18.4 **0.00**±0.00	**32.2**±18.7 8.6±2.0	**0.04**±0.01 **0.17**±0.01	11.36±6.55 2.3±1.6	35.50±0.95
	∞ V-Δ	0.92±0.05 **0.17**±0.01	**0.96**±0.09 1.42±0.59	**0.83**±0.85 **1.22**±1.11	59.2±18.4 **0.00**±0.00	**32.2**±18.7 8.6±2.0	**0.04**±0.01 **0.17**±0.01	11.36±6.55 2.3±1.6	35.50±0.95
	∞ V-L	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 1.42±0.59	0.86±0.87 1.38±1.20	**65.9**±18.1 **0.00**±0.00	**32.2**±18.7 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.47±6.73 2.4±1.6	**35.49**±0.94
	∞ V-C	**0.91**±0.05 **0.17**±0.01	**0.96**±0.08 1.42±0.59	0.86±0.87 1.38±1.20	**65.9**±18.1 **0.00**±0.00	**32.2**±18.7 **1.8**±0.8	**0.04**±0.01 **0.17**±0.01	11.47±6.73 2.4±1.6	**35.49**±0.94
$\alpha = 0.10$ $\delta = 0.025$	10 V-U	0.86±0.04 **0.17**±0.01	0.94±0.08 29.45±8.62	**1.60**±0.91 **2.46**±1.13	**76.5**±6.8 **0.00**±0.00	**21.2**±7.1 **2.2**±0.8	**0.03**±0.01 **0.16**±0.01	7.51±2.57 4.6±1.9	35.25±0.91
	10 V-Δ	0.86±0.04 **0.17**±0.01	0.94±0.08 29.45±8.62	**1.60**±0.91 **2.46**±1.13	**76.5**±6.8 **0.00**±0.00	**21.2**±7.1 **2.2**±0.8	**0.03**±0.01 **0.16**±0.01	7.51±2.57 4.6±1.9	35.25±0.91
	10 V-L	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 29.43±8.53	1.87±1.07 2.98±1.45	**76.5**±6.8 **0.00**±0.00	**21.2**±7.1 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.51±2.57 5.3±2.0	**35.20**±0.91
	10 V-C	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 29.43±8.53	1.87±1.07 2.98±1.45	**76.5**±6.8 **0.00**±0.00	**21.2**±7.1 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.51±2.57 5.3±2.0	**35.20**±0.91
	50 V-U	0.86±0.04 **0.17**±0.01	0.94±0.08 8.70±2.14	**1.63**±0.88 **2.53**±1.10	**76.5**±6.9 **0.00**±0.00	**21.3**±7.2 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.58 4.7±1.9	35.25±0.91
	50 V-Δ	0.86±0.04 **0.17**±0.01	0.94±0.08 8.70±2.14	**1.63**±0.88 **2.53**±1.10	**76.5**±6.9 **0.00**±0.00	**21.3**±7.2 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.58 4.7±1.9	35.25±0.91
	50 V-L	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 8.66±2.12	1.88±1.07 2.99±1.45	**76.5**±6.9 **0.00**±0.00	**21.3**±7.1 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.58 5.3±2.0	**35.21**±0.91
	50 V-C	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 8.66±2.12	1.88±1.07 2.99±1.45	**76.5**±6.9 **0.00**±0.00	**21.3**±7.1 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.58 5.3±2.0	**35.21**±0.91
	100 V-U	0.86±0.04 **0.17**±0.01	0.94±0.08 5.38±1.30	**1.58**±0.92 **2.48**±1.16	76.4±6.9 **0.00**±0.00	**21.3**±7.2 2.4±1.0	**0.03**±0.01 **0.16**±0.01	7.52±2.59 4.8±2.0	35.26±0.91
	100 V-Δ	0.86±0.04 **0.17**±0.01	0.94±0.08 5.38±1.30	**1.58**±0.92 **2.48**±1.16	76.4±6.9 **0.00**±0.00	**21.3**±7.2 2.4±1.0	**0.03**±0.01 **0.16**±0.01	7.52±2.59 4.8±2.0	35.26±0.91
	100 V-L	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 5.35±1.31	1.88±1.09 2.99±1.48	**76.5**±6.9 **0.00**±0.00	**21.3**±7.2 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.59 5.3±2.1	**35.21**±0.91
	100 V-C	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 5.35±1.31	1.88±1.09 2.99±1.48	**76.5**±6.9 **0.00**±0.00	**21.3**±7.2 **2.2**±0.7	**0.03**±0.01 **0.16**±0.01	7.52±2.59 5.3±2.1	**35.21**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.025	200	V-U	0.88±0.04 **0.17**±0.01	0.95±0.08 3.70±0.92	**1.31**±0.85 **2.02**±1.06	67.0±10.9 **0.00**±0.00	**21.2**±7.2 11.7±7.5	**0.04**±0.01 **0.16**±0.01	7.52±2.59 4.6±1.9	35.33±0.90
			V-Δ	0.88±0.04 **0.17**±0.01	0.95±0.08 3.70±0.92	**1.31**±0.85 **2.02**±1.06	67.0±10.9 **0.00**±0.00	**21.2**±7.2 11.7±7.5	**0.04**±0.01 **0.16**±0.01	7.52±2.59 4.6±1.9	35.33±0.90
			V-L	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 3.70±0.94	1.87±1.07 2.98±1.45	**76.5**±6.9 **0.00**±0.00	21.3±7.2 **2.2**±0.7	0.03±0.01 **0.16**±0.01	7.52±2.59 5.3±2.0	**35.21**±0.91
			V-C	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 3.70±0.94	1.87±1.07 2.98±1.45	**76.5**±6.9 **0.00**±0.00	21.3±7.2 **2.2**±0.7	0.03±0.01 **0.16**±0.01	7.52±2.59 5.3±2.0	**35.21**±0.91
		∞	V-U	0.90±0.04 **0.17**±0.01	0.95±0.08 2.44±0.59	**1.14**±0.84 **1.70**±1.03	51.6±7.1 **0.00**±0.00	**21.2**±7.2 27.3±1.4	**0.04**±0.01 **0.16**±0.01	7.52±2.59 4.4±2.0	35.39±0.91
			V-Δ	0.90±0.04 **0.17**±0.01	0.95±0.08 2.44±0.59	**1.14**±0.84 **1.70**±1.03	51.6±7.1 **0.00**±0.00	**21.2**±7.2 27.3±1.4	**0.04**±0.01 **0.16**±0.01	7.52±2.59 4.4±2.0	35.39±0.91
			V-L	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 2.44±0.58	1.87±1.07 2.97±1.45	**76.5**±6.9 **0.00**±0.00	21.3±7.2 **2.2**±0.7	0.03±0.01 **0.16**±0.01	7.52±2.59 5.3±2.0	**35.21**±0.91
			V-C	**0.84**±0.05 **0.17**±0.01	**0.93**±0.09 2.44±0.58	1.87±1.07 2.97±1.45	**76.5**±6.9 **0.00**±0.00	21.3±7.2 **2.2**±0.7	0.03±0.01 **0.16**±0.01	7.52±2.59 5.3±2.0	**35.21**±0.91
	0.05	10	V-U	0.84±0.05 **0.17**±0.01	**0.93**±0.09 31.76±8.94	**1.81**±0.93 **2.80**±1.22	**79.3**±6.2 **0.00**±0.00	**18.2**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.36 5.0±2.0	35.19±0.91
			V-Δ	0.84±0.05 **0.17**±0.01	**0.93**±0.09 31.76±8.94	**1.81**±0.93 **2.80**±1.22	**79.3**±6.2 **0.00**±0.00	**18.2**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.36 5.0±2.0	35.19±0.91
			V-L	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 31.73±8.83	2.09±1.14 3.33±1.59	79.2±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.36 5.6±2.1	**35.14**±0.91
			V-C	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 31.73±8.83	2.09±1.14 3.33±1.59	79.2±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.36 5.6±2.1	**35.14**±0.91
		50	V-U	0.84±0.05 **0.17**±0.01	**0.93**±0.09 9.41±2.38	**1.87**±0.96 **2.95**±1.26	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.37 5.3±2.1	35.18±0.91
			V-Δ	0.84±0.05 **0.17**±0.01	**0.93**±0.09 9.41±2.38	**1.87**±0.96 **2.95**±1.26	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.37 5.3±2.1	35.18±0.91
			V-L	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 9.39±2.38	2.12±1.18 3.38±1.64	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.37 5.7±2.2	**35.14**±0.91
			V-C	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 9.39±2.38	2.12±1.18 3.38±1.64	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.45±2.37 5.7±2.2	**35.14**±0.91
		100	V-U	0.84±0.05 **0.17**±0.01	**0.93**±0.09 5.73±1.36	**1.78**±0.94 **2.83**±1.26	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.8	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.2±2.1	35.19±0.91
			V-Δ	0.84±0.05 **0.17**±0.01	**0.93**±0.09 5.73±1.36	**1.78**±0.94 **2.83**±1.26	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.8	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.2±2.1	35.19±0.91
			V-L	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 5.73±1.36	2.10±1.17 3.36±1.64	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.7±2.2	**35.14**±0.91
			V-C	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 5.73±1.36	2.10±1.17 3.36±1.64	**79.2**±6.2 **0.00**±0.00	**18.3**±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.7±2.2	**35.14**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.05	200	V-U	0.86±0.04 **0.17**±0.01	0.94±0.08 3.95±0.96	**1.52**±0.87 **2.40**±1.13	72.6±10.3 **0.00**±0.00	**18.2**±6.6 9.1±7.6	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.0±2.0	35.26±0.90
			V-Δ	0.86±0.04 **0.17**±0.01	0.94±0.08 3.95±0.96	**1.52**±0.87 **2.40**±1.13	72.6±10.3 **0.00**±0.00	**18.2**±6.6 9.1±7.6	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.0±2.0	35.26±0.90
			V-L	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 3.95±0.97	2.09±1.14 3.33±1.59	**79.2**±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.7±2.1	**35.14**±0.91
			V-C	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 3.95±0.97	2.09±1.14 3.33±1.59	**79.2**±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	**0.03**±0.01 **0.16**±0.01	6.46±2.38 5.7±2.1	**35.14**±0.91
		∞	V-U	0.89±0.04 **0.17**±0.01	0.95±0.08 2.54±0.59	**1.29**±0.88 **1.97**±1.12	54.3±6.5 **0.00**±0.00	**18.2**±6.6 27.5±1.4	**0.04**±0.01 **0.16**±0.01	6.46±2.38 4.8±2.1	35.33±0.91
			V-Δ	0.89±0.04 **0.17**±0.01	0.95±0.08 2.54±0.59	**1.29**±0.88 **1.97**±1.12	54.3±6.5 **0.00**±0.00	**18.2**±6.6 27.5±1.4	**0.04**±0.01 **0.16**±0.01	6.46±2.38 4.8±2.1	35.33±0.91
			V-L	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 2.54±0.59	2.08±1.14 3.32±1.59	**79.2**±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	0.03±0.01 **0.16**±0.01	6.46±2.38 5.6±2.1	**35.14**±0.91
			V-C	**0.83**±0.05 **0.17**±0.01	**0.93**±0.09 2.54±0.59	2.08±1.14 3.32±1.59	**79.2**±6.2 **0.00**±0.00	18.3±6.6 **2.5**±0.7	0.03±0.01 **0.16**±0.01	6.46±2.38 5.6±2.1	**35.14**±0.91
0.10	0.10	10	V-U	0.83±0.05 **0.17**±0.01	0.93±0.09 34.68±9.35	**2.07**±1.00 **3.23**±1.30	**82.1**±5.7 **0.00**±0.00	**15.2**±6.1 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.36±2.21 5.6±2.1	35.12±0.91
			V-Δ	0.83±0.05 **0.17**±0.01	0.93±0.09 34.68±9.35	**2.07**±1.00 **3.23**±1.30	**82.1**±5.7 **0.00**±0.00	**15.2**±6.1 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.36±2.21 5.6±2.1	35.12±0.91
			V-L	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 34.55±9.34	2.35±1.21 3.79±1.67	**82.1**±5.7 **0.00**±0.00	**15.2**±6.1 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.36±2.21 6.2±2.3	**35.07**±0.91
			V-C	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 34.55±9.34	2.35±1.21 3.79±1.67	**82.1**±5.7 **0.00**±0.00	**15.2**±6.1 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.36±2.21 6.2±2.3	**35.07**±0.91
		50	V-U	0.82±0.05 **0.17**±0.01	**0.92**±0.09 10.21±2.41	**2.12**±1.03 **3.37**±1.35	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.38±2.25 5.8±2.3	35.11±0.91
			V-Δ	0.82±0.05 **0.17**±0.01	**0.92**±0.09 10.21±2.41	**2.12**±1.03 **3.37**±1.35	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.38±2.25 5.8±2.3	35.11±0.91
			V-L	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 10.18±2.40	2.37±1.24 3.82±1.71	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.38±2.25 6.3±2.3	**35.06**±0.91
			V-C	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 10.18±2.40	2.37±1.24 3.82±1.71	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.38±2.25 6.3±2.3	**35.06**±0.91
		100	V-U	0.83±0.05 **0.17**±0.01	0.93±0.09 6.21±1.38	**2.04**±1.00 **3.28**±1.33	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.8	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.8±2.2	35.12±0.90
			V-Δ	0.83±0.05 **0.17**±0.01	0.93±0.09 6.21±1.38	**2.04**±1.00 **3.28**±1.33	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.8	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.8±2.2	35.12±0.90
			V-L	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 6.20±1.38	2.36±1.22 3.80±1.70	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.06**±0.91
			V-C	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 6.20±1.38	2.36±1.22 3.80±1.70	**82.0**±5.8 **0.00**±0.00	**15.3**±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.06**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.10	200	V-U	0.84±0.05 **0.17**±0.01	0.93±0.08 4.29±0.99	**1.78**±0.91 **2.82**±1.19	77.4±8.7 **0.00**±0.00	**15.2**±6.3 7.4±6.4	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.6±2.2	35.19±0.90
			V-Δ	0.84±0.05 **0.17**±0.01	0.93±0.08 4.29±0.99	**1.78**±0.91 **2.82**±1.19	77.4±8.7 **0.00**±0.00	**15.2**±6.3 7.4±6.4	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.6±2.2	35.19±0.90
			V-L	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 4.27±0.97	2.34±1.20 3.78±1.66	**82.0**±5.8 **0.00**±0.00	15.3±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.07**±0.91
			V-C	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 4.27±0.97	2.34±1.20 3.78±1.66	**82.0**±5.8 **0.00**±0.00	15.3±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.07**±0.91
		∞	V-U	0.87±0.04 **0.17**±0.01	0.94±0.08 2.68±0.63	**1.49**±0.92 **2.30**±1.18	57.1±6.1 **0.00**±0.00	**15.2**±6.2 27.7±1.4	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.3±2.2	35.27±0.92
			V-Δ	0.87±0.04 **0.17**±0.01	0.94±0.08 2.68±0.63	**1.49**±0.92 **2.30**±1.18	57.1±6.1 **0.00**±0.00	**15.2**±6.2 27.7±1.4	**0.03**±0.01 **0.16**±0.01	5.39±2.26 5.3±2.2	35.27±0.92
			V-L	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 2.69±0.63	2.33±1.20 3.77±1.67	**82.0**±5.8 **0.00**±0.00	15.3±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.07**±0.91
			V-C	**0.81**±0.06 **0.17**±0.01	**0.92**±0.09 2.69±0.63	2.33±1.20 3.77±1.67	**82.0**±5.8 **0.00**±0.00	15.3±6.3 **2.7**±0.7	**0.03**±0.01 **0.16**±0.01	5.39±2.26 6.2±2.3	**35.07**±0.91
0.10	0.15	10	V-U	0.81±0.05 **0.17**±0.01	**0.92**±0.09 36.26±9.35	**2.26**±1.15 **3.53**±1.49	**83.5**±5.2 **0.00**±0.00	**13.7**±5.6 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.83±2.04 6.0±2.4	35.07±0.92
			V-Δ	0.81±0.05 **0.17**±0.01	**0.92**±0.09 36.26±9.35	**2.26**±1.15 **3.53**±1.49	**83.5**±5.2 **0.00**±0.00	**13.7**±5.6 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.83±2.04 6.0±2.4	35.07±0.92
			V-L	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 36.15±9.25	2.61±1.39 4.13±1.87	83.4±5.2 **0.00**±0.00	**13.7**±5.6 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.83±2.04 6.6±2.4	**35.01**±0.92
			V-C	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 36.15±9.25	2.61±1.39 4.13±1.87	83.4±5.2 **0.00**±0.00	**13.7**±5.6 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.83±2.04 6.6±2.4	**35.01**±0.92
		50	V-U	0.81±0.05 **0.17**±0.01	**0.92**±0.09 10.63±2.37	**2.33**±1.18 **3.66**±1.53	**83.3**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.1±2.4	35.06±0.91
			V-Δ	0.81±0.05 **0.17**±0.01	**0.92**±0.09 10.63±2.37	**2.33**±1.18 **3.66**±1.53	**83.3**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.1±2.4	35.06±0.91
			V-L	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 10.64±2.41	2.62±1.41 4.15±1.90	**83.3**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
			V-C	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 10.64±2.41	2.62±1.41 4.15±1.90	**83.3**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
		100	V-U	0.81±0.05 **0.17**±0.01	**0.92**±0.09 6.47±1.40	**2.28**±1.12 **3.61**±1.46	**83.4**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.2±2.4	35.07±0.90
			V-Δ	0.81±0.05 **0.17**±0.01	**0.92**±0.09 6.47±1.40	**2.28**±1.12 **3.61**±1.46	**83.4**±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.2±2.4	35.07±0.90
			V-L	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 6.48±1.39	2.62±1.39 4.16±1.88	83.3±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.4	**35.01**±0.91
			V-C	**0.79**±0.06 **0.17**±0.01	**0.92**±0.09 6.48±1.39	2.62±1.39 4.16±1.88	83.3±5.5 **0.00**±0.00	**13.9**±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.4	**35.01**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.10	0.15	200	V-U	0.83±0.05 **0.17**±0.01	0.93±0.09 4.48±1.00	**1.97**±1.09 **3.12**±1.42	79.9±8.2 **0.00**±0.00	**13.8**±5.9 6.3±5.5	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.0±2.4	35.13±0.91
			V-Δ	0.83±0.05 **0.17**±0.01	0.93±0.09 4.48±1.00	**1.97**±1.09 **3.12**±1.42	79.9±8.2 **0.00**±0.00	**13.8**±5.9 6.3±5.5	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.0±2.4	35.13±0.91
			V-L	**0.80**±0.06 **0.17**±0.01	**0.92**±0.09 4.49±1.00	2.61±1.40 4.13±1.89	**83.3**±5.5 **0.00**±0.00	13.9±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
			V-C	**0.80**±0.06 **0.17**±0.01	**0.92**±0.09 4.49±1.00	2.61±1.40 4.13±1.89	**83.3**±5.5 **0.00**±0.00	13.9±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
		∞	V-U	0.86±0.05 **0.17**±0.01	0.94±0.09 2.80±0.66	**1.67**±1.08 **2.55**±1.39	58.5±5.8 **0.00**±0.00	**13.8**±5.9 27.7±1.4	**0.03**±0.01 **0.16**±0.01	4.87±2.14 5.8±2.4	35.23±0.92
			V-Δ	0.86±0.05 **0.17**±0.01	0.94±0.09 2.80±0.66	**1.67**±1.08 **2.55**±1.39	58.5±5.8 **0.00**±0.00	**13.8**±5.9 27.7±1.4	**0.03**±0.01 **0.16**±0.01	4.87±2.14 5.8±2.4	35.23±0.92
			V-L	**0.80**±0.06 **0.17**±0.01	**0.92**±0.09 2.79±0.66	2.60±1.40 4.12±1.90	**83.3**±5.5 **0.00**±0.00	13.9±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
			V-C	**0.80**±0.06 **0.17**±0.01	**0.92**±0.09 2.79±0.66	2.60±1.40 4.12±1.90	**83.3**±5.5 **0.00**±0.00	13.9±5.9 **2.8**±0.7	**0.03**±0.01 **0.16**±0.01	4.87±2.14 6.6±2.5	**35.01**±0.91
	0.20	10	V-U	0.80±0.06 **0.17**±0.01	0.92±0.09 37.45±9.41	**2.49**±1.44 **3.92**±1.99	**84.7**±5.1 **0.00**±0.02	**12.4**±5.5 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.38±2.00 6.5±2.6	35.02±0.93
			V-Δ	0.80±0.06 **0.17**±0.01	0.92±0.09 37.45±9.41	**2.49**±1.44 **3.92**±1.99	**84.7**±5.1 **0.00**±0.02	**12.4**±5.5 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.38±2.00 6.5±2.6	35.02±0.93
			V-L	**0.78**±0.07 **0.17**±0.01	**0.91**±0.09 37.36±9.30	2.86±1.71 4.57±2.42	**84.7**±5.1 **0.00**±0.02	**12.4**±5.5 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.38±2.00 7.1±2.8	**34.96**±0.93
			V-C	**0.78**±0.07 **0.17**±0.01	**0.91**±0.09 37.36±9.30	2.86±1.71 4.57±2.42	**84.7**±5.1 **0.00**±0.02	**12.4**±5.5 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.38±2.00 7.1±2.8	**34.96**±0.93
		50	V-U	0.80±0.05 **0.17**±0.01	**0.91**±0.09 11.04±2.39	**2.55**±1.28 **4.03**±1.66	**84.7**±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.7±2.5	35.00±0.92
			V-Δ	0.80±0.05 **0.17**±0.01	**0.91**±0.09 11.04±2.39	**2.55**±1.28 **4.03**±1.66	**84.7**±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.7±2.5	35.00±0.92
			V-L	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 11.04±2.39	2.87±1.55 4.57±2.11	84.6±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.1±2.7	**34.96**±0.92
			V-C	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 11.04±2.39	2.87±1.55 4.57±2.11	84.6±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.1±2.7	**34.96**±0.92
		100	V-U	0.80±0.05 **0.17**±0.01	0.92±0.09 6.73±1.37	**2.50**±1.19 **3.96**±1.56	**84.7**±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.7±2.4	35.01±0.91
			V-Δ	0.80±0.05 **0.17**±0.01	0.92±0.09 6.73±1.37	**2.50**±1.19 **3.96**±1.56	**84.7**±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.7±2.4	35.01±0.91
			V-L	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 6.72±1.35	2.86±1.53 4.57±2.09	84.6±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.1±2.6	**34.95**±0.92
			V-C	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 6.72±1.35	2.86±1.53 4.57±2.09	84.6±5.2 **0.00**±0.00	**12.5**±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.1±2.6	**34.95**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.10$ $\delta = 0.20$	200	V-U	0.82±0.05 **0.17**±0.01	0.92±0.09 4.64±0.98	**2.19**±1.18 **3.47**±1.54	82.0±7.3 **0.00**±0.00	**12.4**±5.6 5.6±4.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.4±2.4	35.08±0.92
		V-Δ	0.82±0.05 **0.17**±0.01	0.92±0.09 4.64±0.98	**2.19**±1.18 **3.47**±1.54	82.0±7.3 **0.00**±0.00	**12.4**±5.6 5.6±4.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.4±2.4	35.08±0.92
		V-L	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 4.66±0.97	2.83±1.53 4.50±2.09	**84.7**±5.2 **0.00**±0.00	12.5±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.0±2.6	**34.96**±0.92
		V-C	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 4.66±0.97	2.83±1.53 4.50±2.09	**84.7**±5.2 **0.00**±0.00	12.5±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.0±2.6	**34.96**±0.92
	∞	V-U	0.85±0.05 **0.17**±0.01	0.93±0.09 2.90±0.68	**1.84**±1.19 **2.82**±1.54	59.8±5.5 **0.00**±0.00	**12.4**±5.6 27.7±1.4	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.3±2.5	35.19±0.93
		V-Δ	0.85±0.05 **0.17**±0.01	0.93±0.09 2.90±0.68	**1.84**±1.19 **2.82**±1.54	59.8±5.5 **0.00**±0.00	**12.4**±5.6 27.7±1.4	**0.03**±0.01 **0.16**±0.01	4.39±2.02 6.3±2.5	35.19±0.93
		V-L	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 2.90±0.67	2.82±1.54 4.49±2.10	**84.6**±5.2 **0.00**±0.00	12.5±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.0±2.6	**34.96**±0.92
		V-C	**0.78**±0.06 **0.17**±0.01	**0.91**±0.09 2.90±0.67	2.82±1.54 4.49±2.10	**84.6**±5.2 **0.00**±0.00	12.5±5.6 **2.9**±0.7	**0.03**±0.01 **0.16**±0.01	4.39±2.02 7.0±2.6	**34.96**±0.92
$\alpha = 0.15$ $\delta = 0.025$	10	V-U	0.79±0.05 **0.17**±0.01	0.91±0.09 58.72±11.05	**2.75**±1.11 **4.29**±1.35	**83.9**±3.3 **0.00**±0.02	**13.2**±3.6 2.9±0.6	**0.03**±0.01 **0.16**±0.01	4.63±1.28 7.9±2.8	34.95±0.90
		V-Δ	0.79±0.05 **0.17**±0.01	0.91±0.09 58.72±11.05	**2.75**±1.11 **4.29**±1.35	**83.9**±3.3 **0.00**±0.02	**13.2**±3.6 2.9±0.6	**0.03**±0.01 **0.16**±0.01	4.63±1.28 7.9±2.8	34.95±0.90
		V-L	0.75±0.06 **0.17**±0.02	0.90±0.09 58.55±11.42	3.62±1.60 5.61±2.15	**83.9**±3.3 **0.00**±0.02	**13.2**±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.28 9.2±2.8	34.81±0.90
		V-C	**0.74**±0.06 **0.17**±0.02	**0.89**±0.09 58.53±11.45	3.71±1.63 5.82±2.22	**83.9**±3.3 **0.00**±0.02	**13.2**±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.28 9.6±2.8	**34.80**±0.89
	50	V-U	0.78±0.04 **0.17**±0.01	0.91±0.09 15.23±2.63	**2.86**±1.05 **4.45**±1.21	**84.0**±3.3 **0.00**±0.00	**13.2**±3.6 2.9±0.6	**0.03**±0.01 **0.15**±0.01	4.63±1.29 8.1±2.6	34.94±0.90
		V-Δ	0.78±0.04 **0.17**±0.01	0.91±0.09 15.23±2.63	**2.86**±1.05 **4.45**±1.21	**84.0**±3.3 **0.00**±0.00	**13.2**±3.6 2.9±0.6	**0.03**±0.01 **0.15**±0.01	4.63±1.29 8.1±2.6	34.94±0.90
		V-L	0.75±0.05 **0.17**±0.02	0.90±0.09 15.24±2.66	3.60±1.43 5.60±1.76	83.9±3.3 **0.00**±0.00	**13.2**±3.6 **2.8**±0.6	0.02±0.01 **0.15**±0.01	4.63±1.29 9.2±2.6	34.81±0.90
		V-C	**0.74**±0.05 **0.17**±0.02	**0.89**±0.09 15.23±2.67	3.69±1.49 5.80±1.90	83.9±3.3 **0.00**±0.00	**13.2**±3.6 **2.8**±0.6	0.02±0.01 **0.15**±0.01	4.63±1.29 9.6±2.6	**34.80**±0.89
	100	V-U	0.79±0.04 **0.17**±0.01	0.91±0.09 8.81±1.42	**2.63**±1.03 **4.14**±1.24	**83.9**±3.4 **0.00**±0.00	**13.2**±3.6 3.0±0.8	**0.03**±0.01 **0.16**±0.01	4.63±1.29 8.1±2.6	35.00±0.90
		V-Δ	0.79±0.04 **0.17**±0.01	0.91±0.09 8.81±1.42	**2.63**±1.03 **4.14**±1.24	**83.9**±3.4 **0.00**±0.00	**13.2**±3.6 3.0±0.8	**0.03**±0.01 **0.16**±0.01	4.63±1.29 8.1±2.6	35.00±0.90
		V-L	0.75±0.05 **0.17**±0.02	**0.90**±0.09 8.76±1.51	3.51±1.40 5.45±1.76	**83.9**±3.3 **0.00**±0.00	**13.2**±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.29 9.2±2.6	34.82±0.90
		V-C	**0.74**±0.05 **0.17**±0.02	**0.90**±0.09 8.75±1.51	3.66±1.51 5.76±1.95	**83.9**±3.3 **0.00**±0.00	**13.2**±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.29 9.6±2.6	**34.81**±0.89

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.025	200	V-U	0.84±0.04 **0.17**±0.01	0.94±0.08 5.89±0.91	**1.87**±0.90 **2.99**±1.22	66.3±4.9 **0.00**±0.00	**13.1**±3.5 20.6±3.6	**0.03**±0.01 **0.16**±0.01	4.63±1.29 7.7±2.5	35.27±0.91
			V-Δ	0.84±0.04 **0.17**±0.01	0.94±0.08 5.89±0.91	**1.87**±0.90 **2.99**±1.22	66.3±4.9 **0.00**±0.00	**13.1**±3.5 20.6±3.6	**0.03**±0.01 **0.16**±0.01	4.63±1.29 7.7±2.5	35.27±0.91
			V-L	0.76±0.05 **0.17**±0.01	**0.90**±0.09 5.88±1.01	3.31±1.38 5.09±1.73	**83.9**±3.4 **0.00**±0.00	13.2±3.6 2.9±0.7	**0.03**±0.01 0.15±0.01	4.63±1.29 9.0±2.6	34.85±0.90
			V-C	**0.75**±0.06 **0.17**±0.02	**0.90**±0.09 5.88±1.01	3.64±1.51 5.73±1.96	**83.9**±3.3 **0.00**±0.00	13.2±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.29 9.5±2.6	**34.81**±0.89
		∞	V-U	0.89±0.03 **0.17**±0.01	0.96±0.08 3.53±0.60	**1.32**±0.82 **1.99**±1.03	40.0±3.5 **0.00**±0.00	**13.0**±3.5 47.0±1.4	**0.03**±0.01 **0.16**±0.01	4.63±1.29 7.2±2.8	35.44±0.94
			V-Δ	0.89±0.03 **0.17**±0.01	0.96±0.08 3.53±0.60	**1.32**±0.82 **1.99**±1.03	40.0±3.5 **0.00**±0.00	**13.0**±3.5 47.0±1.4	**0.03**±0.01 **0.16**±0.01	4.63±1.29 7.2±2.8	35.44±0.94
			V-L	0.79±0.05 **0.17**±0.01	0.91±0.09 3.55±0.60	3.00±1.37 4.46±1.70	70.4±4.2 **0.00**±0.00	13.2±3.6 16.4±2.2	**0.03**±0.01 0.15±0.01	4.63±1.29 8.9±2.6	34.92±0.91
			V-C	**0.75**±0.06 **0.17**±0.02	**0.90**±0.09 3.55±0.60	3.63±1.51 5.72±1.96	**83.9**±3.3 **0.00**±0.00	13.2±3.6 **2.8**±0.6	0.02±0.01 0.15±0.01	4.63±1.29 9.5±2.6	**34.81**±0.89
	0.05	10	V-U	0.77±0.05 **0.17**±0.01	0.90±0.09 61.93±11.41	**3.07**±1.14 **4.79**±1.35	**85.4**±3.1 **0.00**±0.00	**11.6**±3.4 **2.9**±0.7	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.5±2.8	34.88±0.91
			V-Δ	0.77±0.05 **0.17**±0.01	0.90±0.09 61.93±11.41	**3.07**±1.14 **4.79**±1.35	**85.4**±3.1 **0.00**±0.00	**11.6**±3.4 **2.9**±0.6	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.5±2.8	34.88±0.91
			V-L	**0.73**±0.06 **0.17**±0.01	**0.89**±0.09 61.66±11.75	3.98±1.63 6.17±2.19	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 9.9±2.9	34.74±0.91
			V-C	**0.73**±0.06 **0.17**±0.01	**0.89**±0.09 61.64±11.74	4.08±1.67 6.42±2.31	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 10.3±2.9	**34.73**±0.90
		50	V-U	0.76±0.05 **0.17**±0.01	0.90±0.09 16.11±2.75	**3.19**±1.18 **5.02**±1.47	**85.4**±3.1 **0.00**±0.00	**11.6**±3.4 3.0±0.7	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.9±2.7	34.86±0.90
			V-Δ	0.76±0.05 **0.17**±0.01	0.90±0.09 16.11±2.75	**3.19**±1.18 **5.02**±1.47	**85.4**±3.1 **0.00**±0.00	**11.6**±3.4 3.0±0.7	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.9±2.7	34.86±0.90
			V-L	0.73±0.06 **0.17**±0.01	**0.89**±0.09 16.10±2.81	3.97±1.60 6.20±2.17	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 10.0±2.8	34.74±0.90
			V-C	**0.72**±0.06 **0.17**±0.01	**0.89**±0.09 16.10±2.79	4.08±1.65 6.43±2.29	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 10.3±2.8	**34.73**±0.90
		100	V-U	0.78±0.04 **0.17**±0.02	0.91±0.09 9.32±1.51	**2.93**±1.10 **4.64**±1.37	**85.4**±3.2 **0.00**±0.00	**11.6**±3.4 3.0±0.7	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.8±2.7	34.92±0.91
			V-Δ	0.78±0.04 **0.17**±0.02	0.91±0.09 9.32±1.51	**2.93**±1.10 **4.64**±1.37	**85.4**±3.2 **0.00**±0.00	**11.6**±3.4 3.0±0.7	**0.03**±0.01 **0.15**±0.01	4.08±1.20 8.8±2.7	34.92±0.91
			V-L	**0.73**±0.05 **0.17**±0.02	**0.89**±0.09 9.27±1.57	3.87±1.51 6.04±2.05	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 9.8±2.8	34.75±0.90
			V-C	**0.73**±0.06 **0.17**±0.02	**0.89**±0.09 9.27±1.56	4.04±1.62 6.38±2.29	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 **0.15**±0.01	4.08±1.20 10.3±2.8	**34.74**±0.90

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.05	200	V-U	0.83±0.04 **0.17**±0.01	0.93±0.08 6.19±1.02	**2.14**±0.93 **3.43**±1.27	69.3±4.7 **0.00**±0.00	**11.5**±3.3 19.2±3.5	**0.03**±0.01 **0.16**±0.01	4.08±1.20 8.4±2.6	35.21±0.91
			V-Δ	0.83±0.04 **0.17**±0.01	0.93±0.08 6.19±1.02	**2.14**±0.93 **3.43**±1.27	69.3±4.7 **0.00**±0.00	**11.5**±3.3 19.2±3.5	**0.03**±0.01 **0.16**±0.01	4.08±1.20 8.4±2.6	35.21±0.91
			V-L	0.75±0.05 **0.17**±0.02	**0.89**±0.09 6.18±1.01	3.70±1.43 5.71±1.91	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 0.15±0.01	4.08±1.20 9.7±2.7	34.78±0.90
			V-C	**0.73**±0.06 **0.17**±0.02	**0.89**±0.09 6.17±1.01	4.02±1.60 6.34±2.25	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 0.15±0.01	4.08±1.20 10.2±2.8	**34.74**±0.90
		∞	V-U	0.88±0.03 **0.17**±0.01	0.95±0.08 3.69±0.63	**1.51**±0.89 **2.30**±1.15	41.5±3.3 **0.00**±0.00	**11.5**±3.3 47.1±1.3	**0.03**±0.01 **0.16**±0.01	4.08±1.20 7.8±2.8	35.39±0.94
			V-Δ	0.88±0.03 **0.17**±0.01	0.95±0.08 3.69±0.63	**1.51**±0.89 **2.30**±1.15	41.5±3.3 **0.00**±0.00	**11.5**±3.3 47.1±1.3	**0.03**±0.01 **0.16**±0.01	4.08±1.20 7.8±2.8	35.39±0.94
			V-L	0.78±0.05 **0.17**±0.02	0.90±0.09 3.70±0.63	3.38±1.49 5.01±1.95	72.0±4.0 **0.00**±0.02	11.6±3.4 16.4±2.2	0.02±0.01 0.15±0.01	4.08±1.20 9.5±2.8	34.85±0.92
			V-C	**0.73**±0.06 **0.17**±0.02	**0.89**±0.09 3.70±0.63	4.02±1.60 6.34±2.26	**85.4**±3.1 **0.00**±0.02	11.7±3.4 **2.9**±0.6	0.02±0.01 0.15±0.01	4.08±1.20 10.1±2.8	**34.74**±0.90
0.15	0.10	10	V-U	0.75±0.05 **0.17**±0.01	0.89±0.09 64.98±12.20	**3.45**±1.28 **5.38**±1.64	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.34±1.03 9.3±3.0	34.79±0.91
			V-Δ	0.75±0.05 **0.17**±0.01	0.89±0.09 64.99±12.22	**3.45**±1.28 **5.38**±1.64	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.34±1.03 9.3±3.0	34.79±0.91
			V-L	0.71±0.06 **0.17**±0.02	**0.88**±0.09 64.47±12.27	4.41±1.79 6.86±2.44	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.34±1.03 10.6±3.0	34.64±0.91
			V-C	**0.70**±0.07 **0.17**±0.02	**0.88**±0.09 64.44±12.29	4.52±1.83 7.13±2.60	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.34±1.03 11.0±3.0	**34.63**±0.91
		50	V-U	0.74±0.05 **0.17**±0.01	0.89±0.09 16.94±2.69	**3.60**±1.28 **5.64**±1.69	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 9.7±2.8	34.77±0.91
			V-Δ	0.74±0.05 **0.17**±0.01	0.89±0.09 16.94±2.69	**3.60**±1.28 **5.64**±1.69	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 9.7±2.8	34.77±0.91
			V-L	0.71±0.06 **0.17**±0.02	**0.88**±0.09 16.88±2.73	4.41±1.75 6.92±2.43	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 10.7±2.9	34.65±0.91
			V-C	**0.70**±0.06 **0.17**±0.02	**0.88**±0.09 16.88±2.72	4.52±1.80 7.15±2.57	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 11.0±2.9	**34.63**±0.91
		100	V-U	0.75±0.05 **0.17**±0.02	0.90±0.09 9.96±1.55	**3.36**±1.18 **5.29**±1.54	**87.3**±2.8 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 9.6±2.7	34.82±0.92
			V-Δ	0.75±0.05 **0.17**±0.02	0.90±0.09 9.96±1.55	**3.36**±1.18 **5.29**±1.54	**87.3**±2.8 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 9.6±2.7	34.82±0.92
			V-L	0.71±0.06 **0.17**±0.02	**0.88**±0.09 9.85±1.53	4.32±1.66 6.76±2.25	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 10.6±2.8	34.66±0.91
			V-C	**0.70**±0.06 **0.17**±0.02	**0.88**±0.09 9.84±1.52	4.49±1.77 7.11±2.52	**87.3**±2.7 **0.00**±0.02	**9.6**±2.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.35±1.04 11.0±2.8	**34.64**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.10	200	V-U	0.80±0.04 **0.17**±0.01	0.92±0.09 6.52±1.02	**2.49**±1.10 **4.00**±1.62	72.7±4.3 **0.00**±0.02	**9.5**±2.9 17.8±3.3	**0.03**±0.01 **0.16**±0.01	3.35±1.04 9.3±2.7	35.11±0.93
			V-Δ	0.80±0.04 **0.17**±0.01	0.92±0.09 6.52±1.02	**2.49**±1.10 **4.00**±1.62	72.7±4.3 **0.00**±0.02	**9.5**±2.9 17.8±3.3	**0.03**±0.01 **0.16**±0.01	3.35±1.04 9.3±2.7	35.11±0.93
			V-L	0.72±0.06 **0.17**±0.02	**0.88**±0.09 6.46±1.04	4.14±1.64 6.39±2.22	**87.3**±2.7 **0.00**±0.02	9.6±2.9 **3.1**±0.7	0.02±0.01 0.15±0.01	3.35±1.04 10.4±2.8	34.68±0.91
			V-C	**0.70**±0.06 **0.17**±0.02	**0.88**±0.09 6.47±1.04	4.47±1.76 7.06±2.50	**87.3**±2.7 **0.00**±0.02	9.6±2.9 **3.1**±0.7	0.02±0.01 0.15±0.01	3.35±1.04 10.9±2.8	**34.64**±0.90
		∞	V-U	0.87±0.04 **0.17**±0.01	0.95±0.09 3.85±0.63	**1.76**±1.02 **2.70**±1.41	43.4±3.0 **0.00**±0.02	**9.4**±2.9 47.1±1.3	**0.03**±0.01 **0.16**±0.01	3.35±1.04 8.6±3.0	35.32±0.95
			V-Δ	0.87±0.04 **0.17**±0.01	0.95±0.09 3.85±0.63	**1.76**±1.02 **2.70**±1.41	43.4±3.0 **0.00**±0.02	**9.4**±2.9 47.1±1.3	**0.03**±0.01 **0.16**±0.01	3.35±1.04 8.6±3.0	35.32±0.95
			V-L	0.75±0.06 **0.17**±0.02	0.89±0.09 3.86±0.61	3.79±1.66 5.64±2.24	74.1±3.8 **0.00**±0.02	9.6±2.9 16.3±2.2	0.02±0.01 0.15±0.01	3.35±1.04 10.3±2.8	34.76±0.93
			V-C	**0.70**±0.06 **0.17**±0.02	**0.88**±0.09 3.86±0.61	4.46±1.76 7.06±2.51	**87.3**±2.7 **0.00**±0.02	9.6±2.9 **3.1**±0.7	0.02±0.01 0.15±0.01	3.35±1.04 10.8±2.8	**34.64**±0.90
0.15	0.15	10	V-U	0.73±0.05 **0.17**±0.02	0.89±0.09 67.31±12.66	**3.74**±1.27 **5.83**±1.64	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.93 9.9±3.1	34.74±0.91
			V-Δ	0.73±0.05 **0.17**±0.02	0.89±0.09 67.31±12.66	**3.74**±1.27 **5.83**±1.64	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.93 9.9±3.1	34.74±0.91
			V-L	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 66.97±12.68	4.74±1.83 7.36±2.51	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.93 11.1±3.1	34.59±0.91
			V-C	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 66.94±12.68	4.84±1.88 7.66±2.70	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.93 11.6±3.1	**34.57**±0.91
		50	V-U	0.73±0.05 **0.17**±0.02	0.89±0.09 17.70±2.89	**3.93**±1.29 **6.17**±1.70	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 10.3±2.9	34.72±0.91
			V-Δ	0.73±0.05 **0.17**±0.02	0.89±0.09 17.70±2.89	**3.93**±1.29 **6.17**±1.70	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 10.3±2.9	34.72±0.91
			V-L	**0.69**±0.06 **0.17**±0.02	**0.87**±0.09 17.56±2.90	4.75±1.79 7.45±2.50	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 11.3±3.1	34.58±0.91
			V-C	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 17.57±2.89	4.85±1.86 7.69±2.67	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 11.6±3.1	**34.57**±0.92
		100	V-U	0.74±0.05 **0.17**±0.02	0.89±0.09 10.36±1.62	**3.67**±1.19 **5.77**±1.57	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 10.2±2.8	34.77±0.91
			V-Δ	0.74±0.05 **0.17**±0.02	0.89±0.09 10.36±1.62	**3.67**±1.19 **5.77**±1.57	**88.3**±2.5 **0.00**±0.02	**8.5**±2.6 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 10.2±2.8	34.77±0.91
			V-L	0.70±0.06 **0.17**±0.02	**0.87**±0.09 10.24±1.63	4.67±1.75 7.27±2.39	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 11.2±3.0	34.60±0.91
			V-C	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 10.24±1.63	4.82±1.87 7.64±2.69	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.98±0.94 11.6±3.0	**34.57**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.15	200	V-U	0.79±0.05 **0.17**±0.01	0.92±0.09 6.76±1.01	**2.82**±1.18 **4.52**±1.71	74.7±4.0 **0.00**±0.02	**8.5**±2.6 16.9±3.1	**0.03**±0.01 **0.15**±0.01	2.98±0.94 9.8±2.9	35.04±0.93
			V-Δ	0.79±0.05 **0.17**±0.01	0.92±0.09 6.76±1.01	**2.82**±1.18 **4.52**±1.71	74.7±4.0 **0.00**±0.02	**8.5**±2.6 16.8±3.1	**0.03**±0.01 **0.15**±0.01	2.98±0.94 9.8±2.9	35.04±0.93
			V-L	0.71±0.06 **0.17**±0.02	0.88±0.09 6.73±1.05	4.48±1.75 6.93±2.38	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	0.02±0.01 **0.15**±0.01	2.98±0.94 11.0±3.0	34.62±0.91
			V-C	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 6.74±1.05	4.79±1.86 7.59±2.66	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	0.02±0.01 **0.15**±0.01	2.98±0.94 11.5±3.0	**34.58**±0.91
		∞	V-U	0.86±0.04 **0.17**±0.01	0.94±0.09 3.96±0.66	**1.96**±1.08 **3.01**±1.49	44.5±2.8 **0.00**±0.02	**8.4**±2.6 47.1±1.3	**0.03**±0.01 **0.16**±0.01	2.98±0.94 9.2±3.1	35.28±0.94
			V-Δ	0.86±0.04 **0.17**±0.01	0.94±0.09 3.96±0.66	**1.96**±1.08 **3.01**±1.49	44.5±2.8 **0.00**±0.02	**8.4**±2.6 47.1±1.3	**0.03**±0.01 **0.16**±0.01	2.98±0.94 9.2±3.1	35.28±0.94
			V-L	0.74±0.06 **0.17**±0.02	0.89±0.09 3.98±0.66	4.10±1.77 6.11±2.42	75.3±3.6 **0.00**±0.03	8.5±2.6 16.2±2.3	0.02±0.01 0.15±0.01	2.98±0.94 10.8±3.1	34.70±0.94
			V-C	**0.69**±0.07 **0.17**±0.02	**0.87**±0.09 3.99±0.65	4.79±1.86 7.58±2.67	**88.3**±2.5 **0.00**±0.03	8.6±2.6 **3.1**±0.7	0.02±0.01 0.15±0.01	2.98±0.94 11.4±3.0	**34.58**±0.91
	0.20	10	V-U	0.72±0.06 **0.17**±0.02	0.88±0.09 68.41±12.55	**3.94**±1.34 6.20±1.88	89.0±2.4 **0.00**±0.02	**7.8**±2.5 3.2±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.88 10.4±3.4	34.69±0.92
			V-Δ	0.72±0.06 **0.17**±0.02	0.88±0.09 68.41±12.55	**3.94**±1.34 **6.19**±1.88	89.0±2.4 **0.00**±0.02	**7.8**±2.5 3.2±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.88 10.4±3.4	34.69±0.92
			V-L	0.68±0.07 **0.17**±0.02	**0.87**±0.09 68.12±12.40	5.03±2.05 7.83±2.90	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.88 11.6±3.5	34.53±0.94
			V-C	**0.67**±0.08 **0.17**±0.02	**0.87**±0.09 68.13±12.38	5.14±2.11 8.14±3.10	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.88 11.9±3.5	**34.52**±0.94
		50	V-U	0.71±0.06 **0.17**±0.02	0.88±0.09 18.12±2.95	**4.19**±1.38 **6.57**±1.96	89.0±2.4 **0.00**±0.03	**7.8**±2.5 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.9±3.3	34.66±0.92
			V-Δ	0.71±0.06 **0.17**±0.02	0.88±0.09 18.12±2.95	**4.19**±1.38 **6.57**±1.95	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.9±3.3	34.66±0.92
			V-L	0.68±0.07 **0.17**±0.02	**0.87**±0.09 18.02±2.88	5.08±2.00 7.99±2.87	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 11.9±3.5	34.52±0.93
			V-C	**0.67**±0.07 **0.17**±0.02	**0.87**±0.09 18.00±2.90	5.18±2.07 8.21±3.06	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 12.1±3.5	**34.51**±0.93
		100	V-U	0.72±0.05 **0.17**±0.02	0.89±0.09 10.60±1.72	**3.94**±1.31 6.20±1.88	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 3.2±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.7±3.2	34.71±0.92
			V-Δ	0.72±0.05 **0.17**±0.02	0.89±0.09 10.60±1.72	**3.94**±1.31 **6.19**±1.88	**89.1**±2.4 **0.00**±0.03	**7.8**±2.5 3.2±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.7±3.2	34.71±0.92
			V-L	0.68±0.07 **0.17**±0.02	**0.87**±0.09 10.54±1.69	4.99±1.98 7.80±2.82	89.0±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 11.7±3.4	34.54±0.93
			V-C	**0.67**±0.07 **0.17**±0.02	**0.87**±0.09 10.54±1.68	5.14±2.08 8.14±3.08	89.0±2.4 **0.00**±0.03	**7.8**±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 12.1±3.5	**34.52**±0.93

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.20	200 V-U	0.77±0.05 **0.17**±0.02	0.91±0.09 6.95±1.08	**3.09**±1.27 **4.94**±1.96	76.3±3.7 **0.00**±0.02	**7.7**±2.5 16.0±3.0	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.4±3.2	34.98±0.94
		200 V-Δ	0.77±0.05 **0.17**±0.01	0.91±0.09 6.95±1.08	**3.09**±1.27 **4.94**±1.96	76.3±3.7 **0.00**±0.02	**7.7**±2.5 16.0±3.0	**0.02**±0.01 **0.15**±0.01	2.72±0.89 10.4±3.2	34.98±0.94
		200 V-L	0.69±0.07 **0.17**±0.02	**0.87**±0.09 6.91±1.12	4.79±1.99 7.42±2.80	**89.1**±2.4 **0.00**±0.03	7.8±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 11.5±3.4	34.56±0.94
		200 V-C	**0.67**±0.07 **0.17**±0.02	**0.87**±0.09 6.91±1.12	5.10±2.08 8.08±3.07	89.0±2.4 **0.00**±0.03	7.8±2.5 **3.1**±0.8	**0.02**±0.01 **0.15**±0.01	2.72±0.89 12.0±3.4	**34.53**±0.94
		∞ V-U	0.85±0.05 **0.17**±0.01	0.94±0.09 4.06±0.73	**2.15**±1.21 **3.31**±1.79	45.2±2.7 **0.00**±0.02	**7.7**±2.4 47.1±1.4	**0.03**±0.01 **0.16**±0.01	2.72±0.89 9.8±3.5	35.24±0.96
		∞ V-Δ	0.85±0.05 **0.17**±0.01	0.94±0.09 4.06±0.73	**2.15**±1.21 **3.31**±1.79	45.2±2.7 **0.00**±0.02	**7.7**±2.4 47.1±1.4	**0.03**±0.01 **0.16**±0.01	2.72±0.89 9.8±3.5	35.24±0.96
		∞ V-L	0.73±0.07 **0.17**±0.02	0.88±0.09 4.07±0.73	4.40±2.02 6.58±2.86	76.1±3.6 **0.00**±0.03	7.8±2.5 16.1±2.4	0.02±0.01 0.15±0.01	2.72±0.89 11.3±3.5	34.64±0.96
		∞ V-C	**0.67**±0.08 **0.17**±0.02	**0.87**±0.09 4.07±0.73	5.10±2.08 8.07±3.08	**89.1**±2.4 **0.00**±0.03	7.8±2.5 **3.1**±0.8	0.02±0.01 0.15±0.01	2.72±0.89 11.9±3.4	**34.53**±0.94
0.20	0.025	10 V-U	0.75±0.04 **0.17**±0.02	0.90±0.09 87.07±12.66	**3.41**±0.93 **5.38**±1.10	**87.2**±1.9 **0.00**±0.02	**9.7**±1.9 **3.1**±0.6	**0.02**±0.01 **0.15**±0.01	3.39±0.70 10.6±3.0	34.81±0.89
		10 V-Δ	0.75±0.04 **0.17**±0.02	0.90±0.09 87.07±12.69	**3.41**±0.92 5.39±1.09	**87.2**±1.9 **0.00**±0.04	**9.7**±1.9 **3.1**±0.6	**0.02**±0.01 **0.15**±0.01	3.39±0.70 10.6±3.0	34.81±0.89
		10 V-L	0.69±0.05 **0.17**±0.02	0.87±0.09 86.31±12.50	4.81±1.33 7.49±1.72	87.1±1.9 0.01±0.04	9.8±2.0 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.39±0.70 12.5±2.9	34.56±0.88
		10 V-C	**0.65**±0.06 **0.17**±0.02	**0.86**±0.09 86.03±12.83	5.83±1.86 9.21±2.63	**87.2**±1.9 0.01±0.06	9.8±2.0 **3.1**±0.6	**0.02**±0.01 0.14±0.01	3.39±0.70 14.3±3.1	**34.42**±0.87
		50 V-U	0.75±0.04 **0.17**±0.02	0.90±0.09 21.63±2.80	**3.50**±1.01 **5.51**±1.19	**87.1**±2.0 **0.00**±0.02	**9.7**±2.0 3.2±0.7	**0.02**±0.01 **0.15**±0.01	3.39±0.70 11.2±2.9	34.81±0.88
		50 V-Δ	0.74±0.04 **0.17**±0.02	0.90±0.09 21.63±2.80	**3.50**±1.01 **5.51**±1.18	**87.1**±2.0 **0.00**±0.04	**9.7**±2.0 3.2±0.7	**0.02**±0.01 **0.15**±0.01	3.39±0.70 11.2±2.9	34.81±0.88
		50 V-L	0.68±0.05 **0.17**±0.02	0.87±0.09 21.51±2.91	4.95±1.33 7.74±1.76	**87.1**±1.9 0.01±0.04	9.8±2.0 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	3.39±0.70 12.9±2.9	34.53±0.87
		50 V-C	**0.65**±0.06 **0.17**±0.02	**0.86**±0.09 21.44±3.04	5.86±1.81 9.24±2.56	**87.1**±1.9 0.01±0.06	9.8±2.0 **3.1**±0.6	**0.02**±0.01 0.14±0.01	3.39±0.70 14.4±3.1	**34.41**±0.87
		100 V-U	0.79±0.04 **0.17**±0.01	0.92±0.09 12.37±1.68	**2.92**±0.96 **4.47**±1.16	76.0±3.8 **0.00**±0.02	**9.6**±1.9 14.3±3.3	**0.03**±0.01 **0.15**±0.01	3.39±0.70 11.1±2.9	34.99±0.91
		100 V-Δ	0.79±0.04 **0.17**±0.01	0.92±0.09 12.37±1.68	**2.92**±0.96 **4.47**±1.16	76.0±3.8 **0.00**±0.04	**9.6**±1.9 14.3±3.3	**0.03**±0.01 **0.15**±0.01	3.39±0.70 11.1±2.9	34.99±0.91
		100 V-L	0.70±0.04 **0.17**±0.02	0.87±0.09 12.17±1.56	4.65±1.23 7.26±1.57	**87.2**±1.9 0.01±0.04	9.8±2.0 **3.1**±0.7	0.02±0.01 **0.15**±0.01	3.39±0.70 12.6±2.8	34.57±0.88
		100 V-C	**0.65**±0.06 **0.17**±0.02	**0.86**±0.09 12.08±1.70	5.83±1.81 9.21±2.57	87.1±1.9 0.01±0.06	9.8±2.0 **3.1**±0.6	0.02±0.01 0.14±0.01	3.39±0.70 14.3±3.1	**34.42**±0.87

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.20	0.025	200	V-U	0.86±0.03 **0.17**±0.01	0.94±0.08 7.89±0.93	**2.04**±0.91 **3.12**±1.17	50.6±3.0 **0.00**±0.02	**9.5**±1.9 39.8±2.4	**0.03**±0.01 **0.16**±0.01	3.39±0.70 10.5±2.9	35.33±0.93
			V-Δ	0.86±0.03 **0.17**±0.01	0.94±0.08 7.89±0.93	**2.04**±0.91 **3.12**±1.17	50.6±3.0 **0.00**±0.04	**9.5**±1.9 39.8±2.4	**0.03**±0.01 **0.16**±0.01	3.39±0.70 10.5±2.9	35.33±0.93
			V-L	0.76±0.04 **0.17**±0.01	0.90±0.09 7.85±0.93	3.46±1.17 5.59±1.57	75.5±2.7 **0.00**±0.04	9.7±1.9 14.8±2.1	0.02±0.01 0.15±0.01	3.39±0.70 12.3±2.9	34.85±0.91
			V-C	**0.65**±0.06 **0.17**±0.02	**0.86**±0.09 7.84±0.98	5.80±1.85 9.16±2.63	**87.1**±1.9 0.01±0.06	9.8±2.0 **3.1**±0.6	0.02±0.01 0.14±0.01	3.39±0.70 14.2±3.0	**34.42**±0.87
		∞	V-U	0.92±0.02 **0.17**±0.01	0.97±0.08 4.59±0.60	**1.02**±0.62 **1.67**±0.82	25.0±2.0 **0.00**±0.02	**9.5**±1.9 65.5±1.0	**0.03**±0.01 **0.16**±0.01	3.39±0.70 9.3±3.4	35.60±0.94
			V-Δ	0.92±0.02 **0.17**±0.01	0.97±0.08 4.59±0.59	**1.02**±0.62 **1.67**±0.82	25.0±2.0 **0.00**±0.04	**9.5**±1.9 65.5±1.0	**0.03**±0.01 **0.16**±0.01	3.39±0.70 9.3±3.4	35.60±0.94
			V-L	0.84±0.04 **0.17**±0.01	0.93±0.09 4.57±0.59	2.46±1.06 3.75±1.43	45.6±2.4 **0.00**±0.04	9.6±1.9 44.8±1.6	0.02±0.01 0.15±0.01	3.39±0.70 11.7±3.2	35.11±0.94
			V-C	**0.65**±0.06 **0.17**±0.02	**0.86**±0.09 4.57±0.60	5.80±1.85 9.16±2.63	**87.1**±1.9 0.01±0.06	9.8±2.0 **3.1**±0.6	0.02±0.01 0.14±0.01	3.39±0.70 14.2±3.0	**34.42**±0.87
	0.05	10	V-U	0.73±0.04 **0.17**±0.02	0.89±0.09 90.83±13.10	**3.77**±0.99 **5.97**±1.20	**88.4**±1.8 **0.00**±0.02	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.66 11.3±3.0	34.73±0.89
			V-Δ	0.73±0.04 **0.17**±0.02	0.89±0.09 90.82±13.11	**3.77**±0.97 **5.97**±1.19	**88.4**±1.8 **0.00**±0.02	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.66 11.3±3.0	34.73±0.89
			V-L	0.67±0.05 **0.17**±0.02	0.86±0.09 89.87±13.08	5.14±1.40 8.10±1.81	88.3±1.8 0.01±0.09	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.66 13.1±3.1	34.49±0.89
			V-C	**0.63**±0.06 **0.17**±0.02	**0.85**±0.09 89.84±13.14	6.13±1.85 9.77±2.53	88.3±1.9 0.01±0.11	**8.5**±1.9 **3.1**±0.7	**0.02**±0.01 0.14±0.01	2.95±0.66 14.8±3.1	**34.36**±0.89
		50	V-U	0.72±0.04 **0.17**±0.02	0.89±0.09 22.57±3.05	3.97±1.02 6.22±1.21	**88.3**±1.8 **0.00**±0.02	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.67 12.1±3.0	34.72±0.89
			V-Δ	0.72±0.04 **0.17**±0.02	0.89±0.09 22.57±3.05	**3.96**±1.01 **6.21**±1.21	**88.3**±1.8 **0.00**±0.02	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.67 12.1±3.0	34.72±0.89
			V-L	0.67±0.05 **0.17**±0.02	0.86±0.09 22.33±3.14	5.31±1.34 8.29±1.74	**88.3**±1.9 0.01±0.11	**8.5**±1.9 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.67 13.4±3.1	34.47±0.88
			V-C	**0.63**±0.06 **0.17**±0.02	**0.85**±0.09 22.27±3.16	6.13±1.84 9.74±2.54	**88.3**±1.9 0.01±0.11	**8.5**±1.9 **3.1**±0.7	**0.02**±0.01 0.14±0.01	2.95±0.67 14.8±3.1	**34.36**±0.88
		100	V-U	0.77±0.04 **0.17**±0.02	0.91±0.09 12.90±1.69	**3.30**±1.00 **5.13**±1.25	80.4±3.6 **0.00**±0.02	**8.4**±1.8 11.2±3.1	**0.02**±0.01 **0.15**±0.01	2.95±0.67 11.7±3.0	34.89±0.91
			V-Δ	0.77±0.04 **0.17**±0.01	0.91±0.09 12.90±1.69	**3.30**±1.00 **5.13**±1.25	80.4±3.6 **0.00**±0.02	**8.4**±1.8 11.1±3.1	**0.02**±0.01 **0.15**±0.01	2.95±0.67 11.7±3.0	34.89±0.91
			V-L	0.68±0.05 **0.17**±0.02	0.87±0.09 12.78±1.78	5.01±1.30 7.87±1.66	**88.3**±1.9 0.01±0.11	8.5±1.9 **3.1**±0.7	**0.02**±0.01 **0.15**±0.01	2.95±0.67 13.4±3.0	34.51±0.89
			V-C	**0.63**±0.06 **0.17**±0.02	**0.85**±0.09 12.66±1.82	6.09±1.84 9.68±2.54	**88.3**±1.9 0.01±0.11	8.5±1.9 **3.1**±0.7	**0.02**±0.01 0.14±0.01	2.95±0.67 14.7±3.1	**34.37**±0.88

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.05	200	V-U	0.84±0.03 **0.17**±0.01	0.94±0.08 8.13±0.95	**2.32**±0.97 **3.57**±1.24	53.1±3.0 **0.00**±0.02	**8.3**±1.8 38.6±2.4	**0.03**±0.01 **0.15**±0.01	2.95±0.67 11.1±2.9	35.26±0.92
			V-Δ	0.84±0.03 **0.17**±0.01	0.94±0.08 8.13±0.95	**2.32**±0.96 **3.57**±1.23	53.1±3.0 **0.00**±0.02	**8.3**±1.8 38.6±2.4	**0.03**±0.01 **0.15**±0.01	2.95±0.67 11.1±2.9	35.26±0.92
			V-L	0.74±0.04 **0.17**±0.01	0.89±0.09 8.15±0.96	3.81±1.25 6.22±1.68	77.9±2.6 **0.00**±0.04	8.5±1.8 13.6±2.0	0.02±0.01 **0.15**±0.01	2.95±0.67 13.0±2.9	34.77±0.92
			V-C	**0.63**±0.06 **0.17**±0.02	**0.85**±0.09 8.09±1.03	6.07±1.85 9.67±2.56	**88.3**±1.9 0.01±0.11	8.5±1.9 **3.1**±0.7	0.02±0.01 0.14±0.01	2.95±0.67 14.7±3.1	**34.37**±0.88
		∞	V-U	0.91±0.02 **0.17**±0.01	0.97±0.08 4.72±0.59	**1.14**±0.66 **1.90**±0.91	26.2±1.9 **0.00**±0.02	**8.3**±1.8 65.5±1.0	**0.03**±0.01 **0.16**±0.01	2.95±0.67 10.0±3.4	35.56±0.93
			V-Δ	0.91±0.02 **0.17**±0.01	0.97±0.08 4.73±0.59	**1.14**±0.66 **1.90**±0.90	26.2±1.9 **0.00**±0.02	**8.3**±1.8 65.5±1.0	**0.03**±0.01 **0.16**±0.01	2.95±0.67 10.0±3.4	35.56±0.93
			V-L	0.82±0.04 **0.17**±0.01	0.92±0.09 4.67±0.60	2.66±1.09 4.11±1.47	46.9±2.4 **0.00**±0.04	8.4±1.8 44.7±1.6	0.02±0.01 0.15±0.01	2.95±0.67 12.3±3.3	35.05±0.95
			V-C	**0.63**±0.06 **0.17**±0.02	**0.85**±0.09 4.66±0.61	6.07±1.86 9.66±2.57	**88.3**±1.9 0.01±0.11	8.5±1.9 **3.1**±0.7	0.02±0.01 0.14±0.01	2.95±0.67 14.7±3.1	**34.37**±0.89
0.20	0.10	10	V-U	0.71±0.05 **0.17**±0.02	0.88±0.09 94.37±13.40	**4.19**±1.04 **6.65**±1.29	**89.7**±1.8 **0.00**±0.03	**7.1**±1.7 3.2±0.8	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.3±3.4	34.65±0.89
			V-Δ	0.70±0.05 **0.17**±0.02	0.88±0.09 94.38±13.39	**4.19**±1.04 **6.65**±1.30	**89.7**±1.8 **0.00**±0.05	**7.1**±1.7 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.3±3.4	34.65±0.89
			V-L	0.65±0.05 **0.17**±0.02	0.86±0.09 93.88±13.76	5.60±1.49 8.79±1.98	**89.7**±1.8 0.01±0.10	7.2±1.7 3.2±0.7	**0.02**±0.01 0.14±0.01	2.48±0.60 13.9±3.3	34.39±0.89
			V-C	**0.61**±0.07 **0.17**±0.02	**0.84**±0.09 93.25±13.62	6.70±2.06 10.61±3.02	**89.7**±1.8 0.03±0.14	7.2±1.7 **3.1**±0.7	0.01±0.01 0.14±0.01	2.48±0.60 15.6±3.4	**34.26**±0.91
		50	V-U	0.70±0.05 **0.17**±0.02	0.88±0.09 23.52±3.02	**4.41**±1.11 **6.92**±1.37	**89.7**±1.8 **0.00**±0.03	**7.1**±1.7 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.48±0.60 13.0±3.3	34.62±0.91
			V-Δ	0.70±0.05 **0.17**±0.02	0.88±0.09 23.52±3.01	**4.41**±1.11 **6.92**±1.38	**89.7**±1.8 **0.00**±0.05	**7.1**±1.7 3.2±0.7	**0.02**±0.01 **0.15**±0.01	2.48±0.60 13.0±3.3	34.62±0.91
			V-L	0.64±0.05 **0.17**±0.02	0.85±0.09 23.36±3.13	5.71±1.45 9.01±1.95	**89.7**±1.8 0.01±0.12	7.2±1.7 3.2±0.7	**0.02**±0.01 0.14±0.01	2.48±0.60 14.3±3.2	34.38±0.89
			V-C	**0.61**±0.07 **0.17**±0.02	**0.84**±0.09 23.21±3.16	6.71±2.03 10.62±2.95	**89.7**±1.8 0.03±0.15	7.2±1.7 **3.1**±0.7	0.01±0.01 0.14±0.01	2.48±0.60 15.7±3.3	**34.27**±0.91
		100	V-U	0.74±0.05 **0.17**±0.01	0.89±0.09 13.65±1.74	**3.82**±1.07 **5.92**±1.44	84.8±3.5 **0.00**±0.02	**7.1**±1.7 8.1±3.2	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.8±3.3	34.79±0.92
			V-Δ	0.74±0.05 **0.17**±0.01	0.89±0.09 13.65±1.74	**3.82**±1.07 **5.92**±1.44	84.8±3.5 **0.00**±0.04	**7.1**±1.7 8.1±3.2	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.8±3.3	34.79±0.92
			V-L	0.65±0.05 **0.17**±0.02	0.86±0.09 13.49±1.79	5.47±1.41 8.61±1.82	89.6±1.8 0.02±0.12	7.2±1.7 3.2±0.7	**0.02**±0.01 0.14±0.01	2.48±0.60 14.2±3.1	34.42±0.90
			V-C	**0.61**±0.07 **0.17**±0.02	**0.84**±0.09 13.30±1.85	6.65±2.04 10.53±2.96	**89.7**±1.8 0.02±0.14	7.2±1.7 **3.1**±0.7	0.01±0.01 0.14±0.01	2.48±0.60 15.6±3.4	**34.27**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.20	0.10	200	V-U	0.82±0.04 **0.17**±0.01	0.93±0.08 8.43±0.95	**2.70**±1.07 **4.17**±1.43	55.7±2.8 **0.00**±0.02	**7.0**±1.7 37.3±2.3	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.2±3.3	35.17±0.93
			V-Δ	0.82±0.04 **0.17**±0.01	0.93±0.08 8.44±0.95	2.71±1.07 4.18±1.43	55.7±2.8 **0.00**±0.04	**7.0**±1.7 37.2±2.3	**0.02**±0.01 **0.15**±0.01	2.48±0.60 12.2±3.3	35.17±0.93
			V-L	0.71±0.05 **0.17**±0.02	0.88±0.09 8.46±1.05	4.28±1.36 6.97±1.86	80.4±2.4 0.01±0.05	7.1±1.7 12.5±2.0	**0.02**±0.01 **0.15**±0.01	2.48±0.60 13.8±3.2	34.65±0.93
			V-C	**0.61**±0.07 **0.17**±0.02	**0.84**±0.09 8.43±1.07	6.65±2.04 10.53±2.97	**89.7**±1.8 0.02±0.14	7.2±1.7 **3.1**±0.7	0.01±0.01 0.14±0.01	2.48±0.60 15.6±3.3	**34.27**±0.91
		∞	V-U	0.90±0.03 **0.17**±0.01	0.96±0.08 4.87±0.62	**1.34**±0.76 **2.24**±1.10	27.5±1.8 **0.00**±0.02	**7.0**±1.6 65.5±1.0	**0.03**±0.01 **0.16**±0.01	2.48±0.60 11.2±3.7	35.50±0.94
			V-Δ	0.90±0.03 **0.17**±0.01	0.96±0.08 4.87±0.62	**1.34**±0.76 **2.24**±1.10	27.5±1.8 **0.00**±0.04	**7.0**±1.6 65.5±1.0	**0.03**±0.01 **0.16**±0.01	2.48±0.60 11.2±3.7	35.50±0.94
			V-L	0.81±0.04 **0.17**±0.02	0.91±0.09 4.83±0.65	2.96±1.22 4.61±1.73	48.3±2.4 0.01±0.05	7.1±1.7 44.6±1.6	0.02±0.01 0.15±0.01	2.48±0.60 13.0±3.5	34.98±0.97
			V-C	**0.61**±0.07 **0.17**±0.02	**0.84**±0.09 4.83±0.65	6.65±2.04 10.52±2.97	**89.7**±1.8 0.02±0.14	7.2±1.7 **3.1**±0.7	0.01±0.01 0.14±0.01	2.48±0.60 15.5±3.3	**34.27**±0.91
	0.15	10	V-U	0.69±0.05 **0.17**±0.02	0.88±0.09 96.56±13.56	**4.49**±1.14 **7.12**±1.43	**90.5**±1.8 **0.02**±0.18	**6.3**±1.7 **3.2**±0.8	**0.02**±0.01 **0.15**±0.01	2.18±0.61 12.9±3.6	34.61±0.90
			V-Δ	0.69±0.05 **0.17**±0.02	0.88±0.09 96.57±13.56	**4.49**±1.14 **7.12**±1.44	**90.5**±1.8 **0.02**±0.20	**6.3**±1.7 **3.2**±0.8	**0.02**±0.01 **0.15**±0.01	2.18±0.61 12.9±3.6	34.60±0.90
			V-L	0.63±0.06 **0.17**±0.02	0.85±0.09 95.66±13.12	5.88±1.59 9.25±2.16	**90.5**±1.8 **0.02**±0.18	**6.3**±1.7 **3.2**±0.8	**0.02**±0.01 0.14±0.01	2.18±0.61 14.4±3.4	34.35±0.91
			V-C	**0.59**±0.07 **0.17**±0.02	**0.84**±0.09 94.98±13.45	7.07±2.13 11.22±3.17	**90.5**±1.8 0.05±0.25	**6.3**±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.18±0.61 16.1±3.5	**34.22**±0.91
		50	V-U	0.68±0.05 **0.17**±0.02	0.87±0.09 24.21±3.20	**4.77**±1.18 **7.45**±1.52	**90.5**±1.8 **0.02**±0.18	**6.3**±1.7 3.3±0.8	**0.02**±0.01 **0.15**±0.01	2.19±0.61 13.6±3.5	34.57±0.90
			V-Δ	0.68±0.05 **0.17**±0.02	0.87±0.09 24.21±3.20	**4.77**±1.18 **7.45**±1.52	**90.5**±1.8 **0.02**±0.20	**6.3**±1.7 **3.2**±0.8	**0.02**±0.01 **0.15**±0.01	2.19±0.61 13.6±3.5	34.57±0.90
			V-L	0.63±0.06 **0.17**±0.02	0.85±0.09 23.89±3.08	6.06±1.56 9.54±2.09	**90.5**±1.8 0.03±0.21	**6.3**±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.19±0.61 14.9±3.4	34.33±0.91
			V-C	**0.59**±0.07 **0.17**±0.02	**0.84**±0.09 23.80±3.17	7.11±2.10 11.27±3.11	**90.5**±1.8 0.05±0.25	**6.3**±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.19±0.61 16.2±3.6	**34.22**±0.92
		100	V-U	0.72±0.05 **0.17**±0.02	0.89±0.09 14.02±1.86	4.16±1.10 6.44±1.52	87.0±3.4 **0.02**±0.18	**6.2**±1.7 6.7±3.0	**0.02**±0.01 **0.15**±0.01	2.19±0.61 13.5±3.4	34.73±0.91
			V-Δ	0.72±0.05 **0.17**±0.02	0.89±0.09 14.01±1.87	**4.15**±1.11 **6.43**±1.52	87.1±3.4 **0.02**±0.20	**6.2**±1.7 6.7±3.0	**0.02**±0.01 **0.15**±0.01	2.19±0.61 13.5±3.4	34.73±0.91
			V-L	0.64±0.05 **0.17**±0.02	0.85±0.09 13.86±1.80	5.85±1.52 9.19±1.99	**90.5**±1.8 0.03±0.21	6.3±1.7 **3.2**±0.8	**0.02**±0.01 0.14±0.01	2.19±0.61 14.7±3.3	34.37±0.91
			V-C	**0.59**±0.07 **0.17**±0.02	**0.84**±0.09 13.67±1.77	7.06±2.12 11.19±3.13	**90.5**±1.8 0.04±0.25	6.3±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.19±0.61 16.1±3.5	**34.22**±0.91

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.20	0.15	200	V-U	0.80±0.04 **0.17**±0.01	0.92±0.08 8.68±1.01	**2.99**±1.17 **4.63**±1.56	57.5±2.9 **0.02**±0.18	**6.2**±1.7 36.4±2.4	**0.02**±0.01 **0.15**±0.01	2.19±0.61 12.9±3.4	35.14±0.93
			V-Δ	0.80±0.04 **0.17**±0.01	0.92±0.08 8.68±1.01	**2.99**±1.17 **4.63**±1.56	57.5±2.9 **0.02**±0.20	**6.2**±1.7 36.4±2.4	**0.02**±0.01 **0.15**±0.01	2.19±0.61 12.9±3.4	35.14±0.93
			V-L	0.69±0.05 **0.17**±0.02	0.87±0.09 8.67±1.13	4.66±1.50 7.59±2.11	82.1±2.5 **0.02**±0.18	6.3±1.7 11.7±2.1	**0.02**±0.01 **0.15**±0.01	2.19±0.61 14.5±3.4	34.60±0.94
			V-C	**0.59**±0.07 **0.17**±0.02	**0.84**±0.09 8.63±1.06	7.05±2.12 11.17±3.15	**90.5**±1.8 0.05±0.26	6.3±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.19±0.61 16.2±3.5	**34.23**±0.91
		∞	V-U	0.89±0.03 **0.17**±0.01	0.96±0.08 4.95±0.68	**1.47**±0.85 **2.46**±1.23	28.3±1.9 **0.02**±0.17	**6.1**±1.7 65.6±1.0	**0.03**±0.01 **0.16**±0.01	2.19±0.61 11.8±3.9	35.49±0.94
			V-Δ	0.89±0.03 **0.17**±0.01	0.96±0.08 4.95±0.68	**1.47**±0.85 **2.46**±1.23	28.3±1.9 **0.02**±0.20	**6.1**±1.7 65.6±1.0	**0.03**±0.01 **0.16**±0.01	2.19±0.61 11.8±3.9	35.49±0.94
			V-L	0.79±0.05 **0.17**±0.02	0.91±0.09 4.94±0.68	3.21±1.32 4.99±1.94	49.2±2.4 **0.02**±0.18	6.2±1.7 44.6±1.6	0.02±0.01 0.15±0.01	2.19±0.61 13.6±3.7	34.94±0.96
			V-C	**0.59**±0.07 **0.17**±0.02	**0.84**±0.09 4.93±0.68	7.04±2.11 11.14±3.14	**90.5**±1.8 0.04±0.25	6.3±1.7 **3.2**±0.8	0.01±0.01 0.14±0.01	2.19±0.61 16.1±3.5	**34.23**±0.91
	0.20	10	V-U	0.68±0.05 **0.17**±0.02	0.87±0.09 98.28±13.82	**4.69**±1.19 **7.42**±1.52	**90.9**±1.6 **0.02**±0.20	**5.9**±1.5 3.3±0.7	**0.02**±0.01 **0.15**±0.01	2.04±0.54 13.4±3.7	34.58±0.89
			V-Δ	0.68±0.05 **0.17**±0.02	0.87±0.09 98.29±13.78	**4.69**±1.19 **7.42**±1.52	**90.9**±1.6 **0.02**±0.20	**5.9**±1.5 **3.2**±0.8	**0.02**±0.01 **0.15**±0.01	2.04±0.54 13.4±3.7	34.58±0.89
			V-L	0.62±0.06 **0.17**±0.02	0.85±0.09 97.10±13.54	6.14±1.63 9.66±2.22	**90.9**±1.7 0.03±0.21	**5.9**±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 15.0±3.7	34.32±0.91
			V-C	**0.58**±0.07 **0.17**±0.02	**0.83**±0.09 96.76±13.68	7.36±2.23 11.65±3.28	**90.9**±1.7 0.06±0.26	**5.9**±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 16.5±3.8	**34.18**±0.91
		50	V-U	0.67±0.05 **0.17**±0.02	0.87±0.09 24.57±3.19	**5.03**±1.15 **7.89**±1.54	**90.9**±1.6 **0.02**±0.20	**5.9**±1.6 3.3±0.8	**0.02**±0.01 **0.15**±0.01	2.04±0.54 14.0±3.6	34.53±0.90
			V-Δ	0.67±0.05 **0.17**±0.02	0.87±0.09 24.56±3.17	5.04±1.19 7.91±1.57	**90.9**±1.7 **0.02**±0.20	**5.9**±1.6 3.3±0.8	**0.02**±0.01 **0.15**±0.01	2.04±0.54 14.0±3.6	34.53±0.90
			V-L	0.62±0.06 **0.17**±0.02	0.84±0.09 24.33±3.09	6.38±1.57 10.03±2.14	**90.9**±1.7 0.04±0.21	**5.9**±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 15.3±3.6	34.28±0.90
			V-C	**0.58**±0.07 **0.17**±0.02	**0.83**±0.09 24.28±3.09	7.44±2.16 11.73±3.19	**90.9**±1.7 0.07±0.27	**5.9**±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 16.6±3.8	**34.18**±0.91
		100	V-U	0.70±0.05 **0.17**±0.02	0.88±0.09 14.40±1.91	**4.42**±1.12 **6.89**±1.51	88.3±3.1 **0.02**±0.20	**5.8**±1.5 5.9±2.7	**0.02**±0.01 **0.15**±0.01	2.04±0.54 14.0±3.5	34.68±0.92
			V-Δ	0.70±0.05 **0.17**±0.02	0.88±0.09 14.40±1.91	**4.42**±1.11 6.90±1.50	88.3±3.1 **0.02**±0.20	**5.8**±1.5 5.8±2.7	**0.02**±0.01 **0.15**±0.01	2.04±0.54 14.0±3.5	34.68±0.92
			V-L	0.63±0.06 **0.17**±0.02	0.85±0.09 14.21±1.91	6.18±1.60 9.72±2.17	**90.9**±1.7 0.03±0.21	5.9±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 15.3±3.6	34.33±0.91
			V-C	**0.58**±0.07 **0.17**±0.02	**0.83**±0.09 14.10±1.88	7.39±2.20 11.68±3.21	**90.9**±1.7 0.06±0.26	5.9±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 16.7±3.9	**34.18**±0.92

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.20$ $\delta = 0.20$	200 V-U	0.79±0.04 **0.17**±0.01	0.92±0.08 8.81±1.08	**3.17**±1.25 **4.93**±1.63	58.2±2.8 **0.02**±0.20	**5.8**±1.5 36.0±2.3	**0.02**±0.01 **0.15**±0.01	2.04±0.54 13.4±3.6	35.10±0.93
	200 V-Δ	0.79±0.04 **0.17**±0.01	0.92±0.08 8.81±1.10	**3.17**±1.25 **4.93**±1.66	58.2±2.8 **0.02**±0.20	**5.8**±1.5 36.0±2.3	**0.02**±0.01 **0.15**±0.01	2.04±0.54 13.4±3.6	35.10±0.93
	200 V-L	0.68±0.05 **0.17**±0.02	0.87±0.09 8.83±1.14	4.94±1.58 7.95±2.15	82.8±2.5 0.03±0.21	5.9±1.5 11.3±2.1	**0.02**±0.01 **0.15**±0.01	2.04±0.54 15.1±3.7	34.56±0.95
	200 V-C	**0.58**±0.07 **0.17**±0.02	**0.83**±0.09 8.83±1.09	7.35±2.21 11.61±3.22	**90.9**±1.7 0.06±0.26	5.9±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 16.6±3.8	**34.19**±0.92
	∞ V-U	0.89±0.03 **0.17**±0.01	0.96±0.08 5.02±0.72	**1.59**±0.86 **2.67**±1.27	28.7±1.8 **0.02**±0.19	**5.7**±1.5 65.6±1.0	**0.02**±0.01 **0.16**±0.01	2.04±0.54 12.3±4.1	35.48±0.93
	∞ V-Δ	0.89±0.03 **0.17**±0.01	0.96±0.08 5.02±0.72	**1.59**±0.87 **2.67**±1.27	28.7±1.8 **0.02**±0.20	**5.7**±1.5 65.6±1.1	**0.02**±0.01 **0.16**±0.01	2.04±0.54 12.3±4.1	35.48±0.93
	∞ V-L	0.79±0.05 **0.17**±0.02	0.91±0.09 5.02±0.73	3.40±1.39 5.29±1.96	49.6±2.3 0.03±0.20	5.8±1.5 44.6±1.7	**0.02**±0.01 0.15±0.01	2.04±0.54 14.2±4.0	34.92±0.98
	∞ V-C	**0.58**±0.07 **0.17**±0.02	**0.83**±0.09 5.02±0.74	7.33±2.21 11.58±3.24	**90.9**±1.7 0.06±0.26	5.9±1.6 **3.2**±0.8	0.01±0.01 0.14±0.01	2.04±0.54 16.5±3.8	**34.19**±0.92
$\alpha = 0.25$ $\delta = 0.025$	10 V-U	0.83±0.03 **0.17**±0.01	0.93±0.08 115.61±12.59	**1.90**±0.67 **3.23**±0.74	81.2±7.1 **0.00**±0.02	**7.8**±1.3 11.1±7.0	**0.03**±0.01 **0.16**±0.01	2.74±0.47 11.1±3.4	35.15±0.90
	10 V-Δ	0.83±0.03 **0.17**±0.01	0.93±0.08 115.62±12.63	**1.90**±0.67 **3.23**±0.74	81.2±7.1 **0.00**±0.02	**7.8**±1.3 11.0±7.0	**0.03**±0.01 **0.16**±0.01	2.74±0.47 11.1±3.4	35.15±0.90
	10 V-L	0.71±0.04 **0.17**±0.02	0.88±0.09 114.36±14.39	4.19±0.97 6.66±1.05	**88.9**±1.4 0.01±0.04	7.9±1.3 3.2±0.7	0.02±0.01 0.15±0.01	2.74±0.47 13.8±3.4	34.66±0.88
	10 V-C	**0.57**±0.05 **0.17**±0.02	**0.83**±0.09 112.51±13.86	7.83±1.77 12.15±2.36	**88.9**±1.4 0.05±0.19	8.0±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.74±0.47 18.2±2.9	**34.12**±0.89
	50 V-U	0.89±0.02 **0.17**±0.01	0.96±0.08 28.37±2.97	1.19±0.59 2.28±0.77	47.8±4.0 **0.00**±0.00	**7.7**±1.3 44.5±3.9	**0.03**±0.01 **0.16**±0.01	2.74±0.47 11.8±3.3	35.43±0.85
	50 V-Δ	0.89±0.02 **0.17**±0.01	0.96±0.08 28.37±2.96	**1.18**±0.59 **2.27**±0.77	47.8±4.0 **0.00**±0.00	**7.7**±1.3 44.5±3.9	**0.03**±0.01 **0.16**±0.01	2.74±0.47 11.7±3.3	35.43±0.85
	50 V-L	0.72±0.04 **0.17**±0.02	0.89±0.09 27.43±3.23	3.99±0.93 6.40±1.20	88.5±1.9 0.01±0.05	7.9±1.3 3.6±1.3	0.02±0.01 0.15±0.01	2.74±0.47 14.1±3.5	34.70±0.89
	50 V-C	**0.57**±0.05 **0.17**±0.02	**0.83**±0.09 26.99±3.13	7.83±1.74 12.14±2.30	**88.9**±1.4 0.05±0.19	8.0±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.74±0.47 18.3±2.9	**34.13**±0.89
	100 V-U	0.92±0.02 **0.17**±0.01	0.97±0.08 16.38±1.82	**0.75**±0.52 **1.44**±0.71	29.8±3.3 **0.00**±0.00	**7.7**±1.3 62.5±3.0	**0.03**±0.01 **0.17**±0.01	2.74±0.47 11.2±3.4	35.62±0.88
	100 V-Δ	0.92±0.02 **0.17**±0.01	0.97±0.08 16.38±1.82	**0.75**±0.52 **1.44**±0.71	29.8±3.3 **0.00**±0.00	**7.7**±1.3 62.5±3.0	**0.03**±0.01 **0.17**±0.01	2.74±0.47 11.2±3.4	35.62±0.88
	100 V-L	0.82±0.03 **0.17**±0.01	0.92±0.08 15.88±1.88	2.48±0.91 4.12±1.18	58.7±2.7 **0.00**±0.02	7.8±1.3 33.6±2.5	0.02±0.01 0.15±0.01	2.74±0.47 13.8±3.2	35.07±0.90
	100 V-C	**0.58**±0.05 **0.17**±0.02	**0.83**±0.09 15.15±1.83	7.74±1.71 11.99±2.17	**88.9**±1.4 0.05±0.19	8.0±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.74±0.47 18.3±2.9	**34.14**±0.88

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.25	0.025	200	V-U	0.95±0.01 **0.17**±0.01	0.98±0.08 9.75±0.93	**0.43**±0.40 **0.93**±0.62	17.7±2.1 **0.00**±0.00	**7.6**±1.3 74.6±1.8	**0.04**±0.01 **0.17**±0.01	2.74±0.47 10.4±3.5	35.72±0.88
			V-Δ	0.95±0.01 **0.17**±0.01	0.98±0.08 9.75±0.93	**0.43**±0.40 **0.93**±0.62	17.7±2.1 **0.00**±0.00	**7.6**±1.3 74.6±1.8	**0.04**±0.01 **0.17**±0.01	2.74±0.47 10.4±3.5	35.72±0.88
			V-L	0.87±0.02 **0.17**±0.01	0.95±0.08 9.72±1.04	1.69±0.79 2.87±1.06	37.9±2.2 **0.00**±0.02	7.7±1.3 54.4±1.8	0.03±0.01 0.16±0.01	2.74±0.47 13.3±3.5	35.38±0.89
			V-C	**0.59**±0.05 **0.17**±0.02	**0.83**±0.09 9.55±0.97	7.52±1.72 11.58±2.24	**88.9**±1.4 0.05±0.17	8.0±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.74±0.47 17.9±2.9	**34.16**±0.89
		∞	V-U	0.97±0.01 **0.17**±0.01	1.00±0.08 5.48±0.61	**0.13**±0.20 **0.47**±0.42	8.5±1.4 **0.00**±0.00	**7.6**±1.3 83.9±0.8	**0.05**±0.01 **0.17**±0.01	2.74±0.47 8.8±4.0	35.85±0.88
			V-Δ	0.97±0.01 **0.17**±0.01	1.00±0.08 5.48±0.61	**0.13**±0.20 **0.47**±0.42	8.5±1.4 **0.00**±0.00	**7.6**±1.3 83.9±0.8	**0.05**±0.01 **0.17**±0.01	2.74±0.47 8.8±4.0	35.85±0.88
			V-L	0.93±0.02 **0.17**±0.01	0.98±0.08 5.47±0.61	0.76±0.52 1.55±0.77	19.3±1.5 **0.00**±0.02	7.7±1.3 73.0±0.8	0.03±0.01 0.16±0.01	2.74±0.47 11.9±4.0	35.65±0.89
			V-C	**0.59**±0.05 **0.17**±0.02	**0.83**±0.09 5.46±0.60	7.47±1.73 11.40±2.32	**88.9**±1.4 0.05±0.17	8.0±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.74±0.47 17.7±3.0	**34.17**±0.89
	0.05	10	V-U	0.80±0.04 **0.17**±0.01	0.92±0.08 119.14±13.25	**2.38**±0.72 **3.98**±0.79	86.0±4.9 **0.00**±0.02	**6.6**±1.3 7.4±4.8	**0.03**±0.01 **0.16**±0.01	2.31±0.46 12.2±3.6	35.04±0.90
			V-Δ	0.80±0.04 **0.17**±0.01	0.92±0.08 119.13±13.28	2.39±0.71 **3.98**±0.79	86.0±4.9 **0.00**±0.02	**6.6**±1.3 7.4±4.8	**0.03**±0.01 **0.16**±0.01	2.31±0.46 12.2±3.6	35.04±0.90
			V-L	0.68±0.04 **0.17**±0.02	0.87±0.09 117.86±14.60	4.66±1.02 7.40±1.16	**90.2**±1.5 0.01±0.04	6.7±1.3 3.2±0.8	0.02±0.01 0.15±0.01	2.31±0.46 14.6±3.5	34.59±0.88
			V-C	**0.55**±0.06 **0.17**±0.02	**0.82**±0.09 115.70±14.07	8.33±1.84 12.89±2.48	90.1±1.5 0.08±0.26	6.8±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.31±0.46 18.8±3.0	**34.04**±0.90
		50	V-U	0.86±0.02 **0.17**±0.01	0.95±0.08 29.33±3.19	**1.56**±0.65 **2.93**±0.94	54.3±3.9 **0.00**±0.00	**6.5**±1.3 39.2±3.7	**0.03**±0.01 **0.16**±0.01	2.31±0.46 13.0±3.5	35.33±0.83
			V-Δ	0.86±0.02 **0.17**±0.01	0.95±0.08 29.34±3.18	**1.56**±0.65 **2.93**±0.94	54.3±3.9 **0.00**±0.00	**6.5**±1.3 39.2±3.7	**0.03**±0.01 **0.16**±0.01	2.31±0.46 13.0±3.5	35.33±0.83
			V-L	0.69±0.04 **0.17**±0.02	0.87±0.09 28.56±3.30	4.55±1.01 7.33±1.27	90.0±1.7 0.01±0.05	6.7±1.3 3.4±1.0	0.02±0.01 0.15±0.01	2.31±0.46 15.1±3.5	34.60±0.89
			V-C	**0.55**±0.05 **0.17**±0.02	**0.82**±0.09 27.95±3.16	8.41±1.80 13.02±2.40	**90.1**±1.5 0.08±0.27	6.8±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.31±0.46 19.1±3.0	**34.03**±0.90
		100	V-U	0.91±0.02 **0.17**±0.01	0.97±0.08 17.02±1.89	**0.98**±0.61 **1.94**±0.86	33.7±3.1 **0.00**±0.00	**6.5**±1.3 59.8±2.8	**0.03**±0.01 **0.16**±0.01	2.31±0.46 12.3±3.5	35.57±0.88
			V-Δ	0.91±0.02 **0.17**±0.01	0.97±0.08 17.01±1.90	**0.98**±0.61 **1.94**±0.86	33.7±3.1 **0.00**±0.00	**6.5**±1.3 59.8±2.8	**0.03**±0.01 **0.16**±0.01	2.31±0.46 12.3±3.5	35.57±0.88
			V-L	0.79±0.03 **0.17**±0.01	0.91±0.08 16.53±1.93	2.90±0.98 4.82±1.32	62.5±2.5 **0.00**±0.02	6.6±1.3 30.9±2.2	0.02±0.01 0.15±0.01	2.31±0.46 14.9±3.4	34.98±0.90
			V-C	**0.56**±0.05 **0.17**±0.02	**0.82**±0.09 15.89±1.92	8.30±1.79 12.85±2.35	**90.1**±1.5 0.08±0.26	6.8±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.31±0.46 19.1±3.0	**34.05**±0.90

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\text{m},\ T_s = 0.1\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
$\alpha = 0.25$	$\delta = 0.05$	200	V-U	0.94±0.02 **0.17**±0.01	0.98±0.08 10.00±0.97	**0.60**±0.49 **1.22**±0.71	19.8±2.2 **0.00**±0.02	**6.5**±1.3 73.7±1.8	**0.04**±0.01 **0.17**±0.01	2.31±0.46 11.3±3.8	35.69±0.88
			V-Δ	0.94±0.02 **0.17**±0.01	0.98±0.08 10.00±0.97	**0.60**±0.49 **1.22**±0.71	19.8±2.2 **0.00**±0.02	**6.5**±1.3 73.7±1.8	**0.04**±0.01 **0.17**±0.01	2.31±0.46 11.3±3.8	35.69±0.88
			V-L	0.86±0.03 **0.17**±0.01	0.94±0.08 10.05±1.03	1.99±0.84 3.38±1.17	39.9±2.2 **0.00**±0.02	**6.5**±1.3 53.5±1.8	0.02±0.01 0.15±0.01	2.31±0.46 14.4±3.6	35.32±0.88
			V-C	**0.57**±0.05 **0.17**±0.02	**0.82**±0.09 9.88±0.97	8.03±1.80 12.39±2.37	**90.1**±1.5 0.07±0.25	6.8±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.31±0.46 18.6±3.0	**34.08**±0.90
		∞	V-U	0.97±0.01 **0.17**±0.01	1.00±0.08 5.61±0.59	**0.20**±0.28 **0.65**±0.54	9.7±1.4 **0.00**±0.00	**6.4**±1.3 83.9±0.9	**0.04**±0.01 **0.17**±0.01	2.31±0.46 9.7±4.3	35.83±0.89
			V-Δ	0.97±0.01 **0.17**±0.01	1.00±0.08 5.61±0.59	**0.20**±0.28 0.66±0.54	9.7±1.5 **0.00**±0.00	**6.4**±1.3 83.9±0.9	**0.04**±0.01 **0.17**±0.01	2.31±0.46 9.7±4.3	35.83±0.89
			V-L	0.92±0.02 **0.17**±0.01	0.98±0.08 5.64±0.61	0.91±0.60 1.86±0.93	20.5±1.5 **0.00**±0.02	6.5±1.3 73.0±0.8	0.03±0.01 0.16±0.01	2.31±0.46 12.9±4.1	35.62±0.89
			V-C	**0.57**±0.05 **0.17**±0.02	**0.82**±0.09 5.59±0.62	7.95±1.79 12.15±2.45	**90.1**±1.5 0.06±0.23	6.8±1.3 **3.1**±0.7	0.01±0.01 0.14±0.01	2.31±0.46 18.4±3.0	**34.08**±0.90
	$\delta = 0.10$	10	V-U	0.77±0.04 **0.17**±0.01	0.91±0.08 122.99±13.66	**2.87**±0.82 **4.77**±0.91	89.1±3.4 **0.01**±0.15	**5.7**±1.1 5.3±3.2	**0.03**±0.01 **0.15**±0.01	1.99±0.40 13.6±4.0	34.92±0.89
			V-Δ	0.77±0.04 **0.17**±0.01	0.91±0.08 122.99±13.67	**2.87**±0.82 **4.77**±0.91	89.1±3.4 **0.01**±0.15	**5.7**±1.1 5.3±3.2	**0.03**±0.01 **0.15**±0.01	1.99±0.40 13.6±4.0	34.92±0.89
			V-L	0.66±0.05 **0.17**±0.02	0.87±0.09 121.62±14.51	5.14±1.09 8.21±1.32	**91.0**±1.4 0.02±0.20	**5.7**±1.1 3.3±0.8	0.02±0.01 **0.15**±0.01	1.99±0.40 15.7±3.6	34.51±0.88
			V-C	**0.53**±0.06 **0.17**±0.02	**0.81**±0.09 119.71±13.69	8.94±1.88 13.84±2.67	**91.0**±1.4 0.12±0.36	5.8±1.2 **3.2**±0.8	0.01±0.01 0.13±0.01	1.99±0.40 19.6±3.2	**33.97**±0.88
		50	V-U	0.83±0.03 **0.17**±0.01	0.94±0.08 30.38±3.34	**2.04**±0.77 **3.76**±1.02	61.0±3.5 **0.01**±0.15	**5.6**±1.1 33.4±3.4	**0.03**±0.01 **0.16**±0.01	1.99±0.40 14.3±3.8	35.20±0.83
			V-Δ	0.83±0.03 **0.17**±0.01	0.94±0.08 30.38±3.34	**2.04**±0.77 **3.76**±1.02	61.0±3.5 **0.01**±0.15	**5.6**±1.1 33.4±3.4	**0.03**±0.01 **0.16**±0.01	1.99±0.40 14.3±3.8	35.20±0.83
			V-L	0.66±0.05 **0.17**±0.02	0.86±0.09 29.67±3.41	5.19±1.11 8.32±1.44	90.9±1.5 0.03±0.21	5.7±1.1 3.4±1.0	0.02±0.01 0.14±0.01	1.99±0.40 16.4±3.6	34.48±0.88
			V-C	**0.53**±0.06 **0.17**±0.02	**0.81**±0.09 29.05±3.19	9.00±1.93 13.98±2.76	**91.0**±1.4 0.13±0.37	5.8±1.2 **3.2**±0.8	0.01±0.01 0.13±0.01	1.99±0.40 19.9±3.3	**33.96**±0.88
		100	V-U	0.89±0.02 **0.17**±0.01	0.96±0.08 17.82±1.97	**1.24**±0.71 **2.40**±0.94	37.6±3.2 **0.01**±0.15	**5.6**±1.1 56.8±3.0	**0.03**±0.01 **0.16**±0.01	1.99±0.40 13.6±3.9	35.49±0.86
			V-Δ	0.89±0.02 **0.17**±0.01	0.96±0.08 17.82±1.97	**1.24**±0.71 **2.40**±0.94	37.6±3.2 **0.01**±0.15	**5.6**±1.1 56.8±3.0	**0.03**±0.01 **0.16**±0.01	1.99±0.40 13.6±3.9	35.49±0.86
			V-L	0.77±0.03 **0.17**±0.01	0.90±0.08 17.40±1.98	3.33±1.06 5.60±1.49	66.3±2.6 0.02±0.19	5.7±1.1 28.0±2.4	0.02±0.01 0.15±0.01	1.99±0.40 16.2±3.6	34.88±0.89
			V-C	**0.53**±0.06 **0.17**±0.02	**0.81**±0.09 16.77±1.98	8.90±1.88 13.83±2.58	**91.0**±1.4 0.12±0.36	5.8±1.2 **3.2**±0.8	0.01±0.01 0.13±0.01	1.99±0.40 19.8±3.3	**33.98**±0.87

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.25	0.10	200	V-U	0.93±0.02 **0.17**±0.01	0.98±0.08 10.42±1.04	**0.82**±0.60 **1.60**±0.87	21.8±2.1 **0.01**±0.14	**5.5**±1.1 72.7±1.9	**0.03**±0.01 **0.16**±0.01	1.99±0.40 13.1±4.3	35.65±0.87
			V-Δ	0.93±0.02 **0.17**±0.01	0.98±0.08 10.42±1.04	**0.82**±0.60 **1.60**±0.87	21.8±2.1 **0.01**±0.14	**5.5**±1.1 72.7±1.9	**0.03**±0.01 **0.16**±0.01	1.99±0.40 13.1±4.3	35.65±0.87
			V-L	0.84±0.03 **0.17**±0.01	0.94±0.08 10.33±1.07	2.34±1.02 3.96±1.34	42.1±2.3 0.02±0.19	5.6±1.1 52.3±2.0	0.02±0.01 0.15±0.01	1.99±0.40 15.5±3.8	35.26±0.88
			V-C	**0.54**±0.06 **0.17**±0.02	**0.81**±0.09 10.25±1.01	8.60±1.87 13.31±2.59	**91.0**±1.4 0.11±0.35	5.8±1.2 **3.2**±0.8	0.01±0.01 0.13±0.01	1.99±0.40 19.5±3.2	**34.00**±0.88
		∞	V-U	0.96±0.01 **0.17**±0.01	0.99±0.08 5.77±0.64	**0.30**±0.37 **0.85**±0.64	10.6±1.4 **0.01**±0.14	**5.5**±1.1 83.9±0.9	**0.04**±0.01 **0.17**±0.01	1.99±0.40 11.1±4.7	35.82±0.89
			V-Δ	0.96±0.01 **0.17**±0.01	0.99±0.08 5.77±0.64	**0.30**±0.37 **0.85**±0.64	10.6±1.4 **0.01**±0.14	**5.5**±1.1 83.9±0.9	**0.04**±0.01 **0.17**±0.01	1.99±0.40 11.1±4.7	35.82±0.89
			V-L	0.91±0.02 **0.17**±0.01	0.97±0.08 5.80±0.65	1.06±0.68 2.17±1.05	21.4±1.4 0.02±0.19	**5.5**±1.1 73.1±0.9	0.03±0.01 0.16±0.01	1.99±0.40 14.0±4.2	35.60±0.89
			V-C	**0.55**±0.06 **0.17**±0.02	**0.82**±0.09 5.78±0.65	8.49±1.87 12.98±2.65	**91.0**±1.4 0.10±0.34	5.8±1.2 **3.2**±0.8	0.01±0.01 0.14±0.01	1.99±0.40 19.2±3.2	**34.01**±0.89
	0.15	10	V-U	0.75±0.04 **0.17**±0.01	0.90±0.08 125.31±14.20	**3.21**±0.89 **5.37**±1.01	90.5±2.7 **0.01**±0.15	**4.9**±0.9 4.6±2.5	**0.02**±0.01 **0.15**±0.01	1.74±0.32 14.4±4.0	34.85±0.90
			V-Δ	0.75±0.04 **0.17**±0.01	0.90±0.08 125.31±14.20	**3.21**±0.89 **5.37**±1.01	90.5±2.7 **0.01**±0.15	**4.9**±0.9 4.6±2.5	**0.02**±0.01 **0.15**±0.01	1.74±0.32 14.4±4.0	34.85±0.90
			V-L	0.65±0.05 **0.17**±0.02	0.86±0.08 123.80±14.81	5.52±1.10 8.79±1.34	**91.7**±1.2 0.03±0.20	5.0±0.9 3.3±0.8	**0.02**±0.01 0.14±0.01	1.74±0.32 16.3±3.5	34.46±0.89
			V-C	**0.52**±0.06 **0.17**±0.02	**0.81**±0.09 122.03±13.45	9.31±1.97 14.40±2.71	**91.7**±1.2 0.14±0.36	5.1±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.74±0.32 20.1±3.2	**33.93**±0.88
		50	V-U	0.81±0.03 **0.17**±0.01	0.94±0.08 30.95±3.27	**2.32**±0.81 **4.29**±1.08	65.0±3.4 **0.01**±0.15	**4.9**±0.9 30.1±3.3	**0.03**±0.01 **0.16**±0.01	1.74±0.32 15.3±3.8	35.12±0.84
			V-Δ	0.81±0.03 **0.17**±0.01	0.94±0.08 30.95±3.26	**2.32**±0.81 **4.29**±1.08	65.0±3.4 **0.01**±0.15	**4.9**±0.9 30.1±3.3	**0.03**±0.01 **0.16**±0.01	1.74±0.32 15.3±3.8	35.12±0.84
			V-L	0.64±0.05 **0.17**±0.02	0.86±0.09 30.41±3.49	5.54±1.19 8.94±1.47	**91.7**±1.2 0.03±0.21	5.0±0.9 3.3±0.9	0.02±0.01 0.14±0.01	1.74±0.32 17.0±3.6	34.43±0.88
			V-C	**0.52**±0.06 **0.17**±0.02	**0.81**±0.09 29.80±3.27	9.35±1.96 14.52±2.74	**91.7**±1.2 0.14±0.36	5.1±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.74±0.32 20.3±3.3	**33.92**±0.88
		100	V-U	0.88±0.02 **0.17**±0.01	0.96±0.08 18.27±2.06	**1.44**±0.78 **2.82**±1.12	40.0±3.0 **0.01**±0.15	**4.9**±0.9 55.2±2.8	**0.03**±0.01 **0.16**±0.01	1.74±0.32 14.8±4.1	35.44±0.85
			V-Δ	0.88±0.02 **0.17**±0.01	0.96±0.08 18.26±2.06	**1.44**±0.78 **2.82**±1.11	40.0±3.0 **0.01**±0.15	**4.9**±0.9 55.1±2.8	**0.03**±0.01 **0.16**±0.01	1.74±0.32 14.8±4.1	35.44±0.86
			V-L	0.75±0.04 **0.17**±0.01	0.90±0.08 17.75±1.95	3.66±1.28 6.17±1.72	68.8±2.4 0.02±0.20	5.0±0.9 26.3±2.3	0.02±0.01 0.15±0.01	1.74±0.32 16.8±3.5	34.81±0.89
			V-C	**0.52**±0.06 **0.17**±0.02	**0.81**±0.09 17.24±1.96	9.23±1.94 14.33±2.65	**91.7**±1.2 0.14±0.37	5.1±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.74±0.32 20.3±3.3	**33.94**±0.87

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.15	200	V-U	0.92±0.02 **0.17**±0.01	0.97±0.08 10.56±1.05	**0.99**±0.67 **1.88**±0.96	23.0±2.0 **0.01**±0.14	**4.8**±0.9 72.1±1.8	**0.03**±0.01 **0.16**±0.01	1.74±0.32 14.3±4.2	35.62±0.88
			V-Δ	0.92±0.02 **0.17**±0.01	0.97±0.08 10.55±1.06	**0.99**±0.67 **1.88**±0.96	23.0±2.0 **0.01**±0.14	**4.8**±0.9 72.1±1.8	**0.03**±0.01 **0.16**±0.01	1.74±0.32 14.3±4.2	35.62±0.88
			V-L	0.83±0.03 **0.17**±0.01	0.93±0.08 10.59±1.04	2.63±1.10 4.45±1.48	43.4±2.0 0.02±0.20	4.9±0.9 51.7±1.9	0.02±0.01 0.15±0.01	1.74±0.32 16.4±3.7	35.22±0.88
			V-C	**0.53**±0.06 **0.17**±0.02	**0.81**±0.09 10.41±1.09	8.94±1.93 13.83±2.57	**91.7**±1.2 0.12±0.35	5.1±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.74±0.32 20.0±3.2	**33.96**±0.88
		∞	V-U	0.96±0.02 **0.17**±0.01	0.99±0.08 5.86±0.65	**0.36**±0.42 **1.01**±0.72	11.2±1.3 **0.01**±0.14	**4.8**±0.9 84.0±0.9	**0.04**±0.01 **0.17**±0.01	1.74±0.32 12.4±4.8	35.81±0.89
			V-Δ	0.96±0.02 **0.17**±0.01	0.99±0.08 5.85±0.66	**0.36**±0.42 **1.01**±0.72	11.2±1.3 **0.01**±0.14	**4.8**±0.9 84.0±0.9	**0.04**±0.01 **0.17**±0.01	1.74±0.32 12.4±4.8	35.81±0.89
			V-L	0.90±0.02 **0.17**±0.01	0.97±0.08 5.90±0.62	1.19±0.72 2.43±1.13	22.1±1.2 0.02±0.19	**4.8**±0.9 73.1±0.9	0.02±0.01 0.16±0.01	1.74±0.32 14.9±4.1	35.58±0.89
			V-C	**0.54**±0.06 **0.17**±0.02	**0.81**±0.09 5.86±0.65	8.80±1.93 13.45±2.67	**91.7**±1.2 0.11±0.33	5.1±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.74±0.32 19.6±3.2	**33.98**±0.89
0.25	0.20	10	V-U	0.73±0.04 **0.17**±0.01	0.90±0.08 126.95±14.65	**3.46**±0.95 **5.79**±1.02	91.3±2.4 **0.01**±0.15	**4.3**±0.9 4.4±2.3	**0.02**±0.01 **0.15**±0.01	1.53±0.31 15.1±4.3	34.82±0.89
			V-Δ	0.73±0.04 **0.17**±0.01	0.90±0.08 126.94±14.73	**3.46**±0.95 **5.79**±1.02	91.3±2.4 **0.01**±0.15	**4.3**±0.9 4.4±2.3	**0.02**±0.01 **0.15**±0.01	1.53±0.31 15.1±4.3	34.82±0.89
			V-L	0.63±0.05 **0.17**±0.02	0.86±0.08 125.42±14.26	5.74±1.13 9.23±1.47	**92.3**±1.1 0.03±0.21	4.4±0.9 3.3±0.8	0.01±0.01 0.14±0.01	1.53±0.31 17.1±3.7	34.42±0.88
			V-C	**0.50**±0.06 **0.17**±0.01	**0.80**±0.09 123.83±13.91	9.67±2.09 14.97±2.94	**92.3**±1.1 0.18±0.41	4.5±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.53±0.31 20.8±3.2	**33.88**±0.89
		50	V-U	0.79±0.03 **0.17**±0.01	0.93±0.08 31.50±3.27	**2.57**±0.88 4.71±1.12	67.7±3.3 **0.01**±0.15	**4.3**±0.9 28.0±3.2	**0.02**±0.01 **0.16**±0.01	1.53±0.31 16.2±4.0	35.06±0.83
			V-Δ	0.79±0.03 **0.17**±0.01	0.93±0.08 31.48±3.21	**2.57**±0.88 **4.70**±1.12	67.7±3.3 **0.01**±0.15	**4.3**±0.9 28.0±3.2	**0.02**±0.01 **0.16**±0.01	1.53±0.31 16.2±4.0	35.06±0.83
			V-L	0.63±0.05 **0.17**±0.02	0.85±0.08 30.97±3.33	5.86±1.20 9.48±1.53	92.2±1.5 0.04±0.21	4.4±0.9 3.4±1.2	0.01±0.01 0.14±0.01	1.53±0.31 17.8±3.7	34.38±0.87
			V-C	**0.50**±0.06 **0.17**±0.02	**0.80**±0.09 30.46±3.36	9.74±2.12 15.10±2.94	**92.3**±1.1 0.18±0.41	4.5±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.53±0.31 21.1±3.3	**33.86**±0.89
		100	V-U	0.86±0.03 **0.17**±0.01	0.95±0.08 18.69±1.98	**1.62**±0.82 3.16±1.21	41.7±2.9 **0.01**±0.15	**4.3**±0.9 54.1±2.7	**0.03**±0.01 **0.16**±0.01	1.53±0.31 15.6±4.4	35.40±0.85
			V-Δ	0.86±0.02 **0.17**±0.01	0.95±0.08 18.69±1.99	**1.62**±0.82 **3.15**±1.21	41.7±2.9 **0.01**±0.15	**4.3**±0.9 54.0±2.7	**0.03**±0.01 **0.16**±0.01	1.53±0.31 15.6±4.4	35.40±0.85
			V-L	0.73±0.04 **0.17**±0.01	0.89±0.08 18.17±2.01	3.90±1.28 6.60±1.79	70.3±2.2 0.02±0.20	4.4±0.9 25.3±2.1	0.02±0.01 0.15±0.01	1.53±0.31 17.6±3.6	34.77±0.88
			V-C	**0.50**±0.06 **0.17**±0.01	**0.80**±0.09 17.72±1.96	9.61±2.08 14.92±2.83	**92.3**±1.1 0.19±0.41	4.5±0.9 **3.2**±0.8	0.00±0.01 0.13±0.01	1.53±0.31 21.0±3.3	**33.88**±0.87

Table III (continued).

Corridor **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.04	0.00±0.00	30.69±13.51 N-A	N-A 22.32±10.04	−0.27±0.05 −0.02±0.01	N-A N-A	70.00±0.00
$c = 0$	0.00±0.00 0.17±0.02	0.66±0.07	28.00±2.99 49.65±2.89	100.0±0.0 15.27±4.17	−0.03±0.01 0.10±0.01	0.00±0.00 78.8±8.6	32.45±0.77
$c_{\max}$	1.00±0.00 0.17±0.01	1.00±0.08	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.05±0.01 0.17±0.01	N-A N-A	35.91±0.92
ACI	0.22±0.04 0.17±0.02	0.73±0.08	17.68±1.59 30.21±0.72	99.1±0.0 2.74±1.43	−0.01±0.01 0.12±0.01	0.30±0.00 57.4±10.2	33.13±0.81
QT	0.30±0.03 0.17±0.02	0.78±0.09	15.19±1.31 26.54±1.03	99.7±0.1 0.61±0.52	−0.00±0.00 0.13±0.01	0.10±0.00 60.1±11.5	33.44±0.83

Corridor $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.02\,\mathrm{m},\ T_s = 0.1\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.25$ $\delta = 0.20$	200 V-U	0.91±0.02 **0.17**±0.01	0.97±0.08 10.70±1.05	**1.10**±0.72 2.08±1.04	23.9±1.9 **0.01**±0.15	**4.2**±0.8 71.8±1.7	**0.03**±0.01 **0.16**±0.01	1.53±0.31 15.1±4.5	35.60±0.87
	200 V-Δ	0.91±0.02 **0.17**±0.01	0.97±0.08 10.70±1.04	**1.10**±0.72 **2.07**±1.04	24.0±1.9 **0.01**±0.15	**4.2**±0.8 71.8±1.7	**0.03**±0.01 **0.16**±0.01	1.53±0.31 15.1±4.5	35.60±0.87
	200 V-L	0.82±0.03 **0.17**±0.01	0.93±0.08 10.71±1.08	2.82±1.12 4.74±1.51	44.3±2.0 0.03±0.21	4.3±0.9 51.4±1.8	0.02±0.01 0.15±0.01	1.53±0.31 17.1±4.0	35.19±0.89
	200 V-C	**0.52**±0.06 **0.17**±0.01	**0.81**±0.09 10.63±1.06	9.32±2.05 14.39±2.80	**92.3**±1.1 0.17±0.40	4.5±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.53±0.31 20.6±3.3	**33.91**±0.89
	∞ V-U	0.95±0.02 **0.17**±0.01	0.99±0.08 5.95±0.66	**0.42**±0.44 **1.14**±0.77	11.7±1.3 **0.01**±0.15	**4.2**±0.8 84.0±1.0	**0.04**±0.01 **0.17**±0.01	1.53±0.31 13.2±5.2	35.80±0.89
	∞ V-Δ	0.95±0.02 **0.17**±0.01	0.99±0.08 5.95±0.66	**0.42**±0.44 **1.14**±0.77	11.8±1.3 **0.01**±0.15	**4.2**±0.8 84.0±1.0	**0.04**±0.01 **0.17**±0.01	1.53±0.31 13.2±5.2	35.80±0.89
	∞ V-L	0.90±0.03 **0.17**±0.01	0.97±0.08 5.97±0.68	1.29±0.76 2.64±1.22	22.6±1.2 0.02±0.19	4.3±0.8 73.1±1.0	0.02±0.01 0.16±0.01	1.53±0.31 15.6±4.5	35.56±0.89
	∞ V-C	**0.52**±0.06 **0.17**±0.01	**0.81**±0.09 5.98±0.66	9.15±2.05 13.94±2.92	**92.3**±1.1 0.15±0.39	4.5±0.9 **3.2**±0.8	0.01±0.01 0.13±0.01	1.53±0.31 20.3±3.3	**33.93**±0.90

B. *Maze*, $\alpha_0 = 0.30$

TABLE IV

FULL SWEEP, MAZE, $\alpha_0 = 0.30$. *File:* APPENDICES/GENERATED_N200/CATALOG/PAPER/SWEEP_MAZE_A030.TEX. *Data:* [F]; FAMILIES MAIN_GRID_200, MATCHED_BASELINES_200; 505 VIEWS (IDS IN [C]). *Restore:* \NTABLE{../CATALOG/PAPER/SWEEP_MAZE_A030}; NOT USED IN THE CURRENT EDITIONS.

$\pm$: one sample SD over the trajectories that enter the mean; $^+$: decision fraction; $\downarrow/\uparrow$: lower/higher is better; bold: lowest ($\downarrow$) or highest ($\uparrow$) mean of the four certified variants at one setting; N-A: not applicable.

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$

Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$

α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.05	0.025	10	V-U	**0.97**±0.03 **0.17**±0.02	**0.78**±0.03 2.30±2.64	0.09±0.22 0.40±0.60	25.8±22.5 **0.00**±0.00	**72.7**±23.3 1.5±1.2	**0.03**±0.00 **0.22**±0.01	34.20±10.65 1.5±0.7	**54.21**±0.39
			V-Δ	**0.97**±0.03 **0.17**±0.02	**0.78**±0.03 2.30±2.64	**0.08**±0.21 **0.39**±0.59	25.9±22.5 **0.00**±0.00	**72.7**±23.3 1.5±1.1	**0.03**±0.00 **0.22**±0.01	34.20±10.65 1.5±0.7	**54.21**±0.38
			V-L	**0.97**±0.03 **0.17**±0.02	**0.78**±0.03 2.30±2.64	0.09±0.22 0.41±0.60	**26.3**±22.4 **0.00**±0.00	**72.7**±23.3 **1.1**±1.0	**0.03**±0.00 **0.22**±0.01	34.20±10.65 1.5±0.7	**54.21**±0.39
			V-C	**0.97**±0.03 **0.17**±0.02	**0.78**±0.03 2.30±2.64	**0.08**±0.21 0.41±0.60	**26.3**±22.5 **0.00**±0.00	**72.7**±23.3 **1.1**±0.9	**0.03**±0.00 **0.22**±0.01	34.20±10.65 1.5±0.7	**54.21**±0.38
		50	V-U	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.32±1.16	0.09±0.22 **0.38**±0.58	25.3±22.2 **0.00**±0.00	**73.0**±23.2 1.7±1.3	**0.03**±0.00 **0.22**±0.01	34.46±10.69 1.5±0.7	**54.21**±0.39
			V-Δ	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.32±1.16	**0.08**±0.21 **0.38**±0.57	25.3±22.2 **0.00**±0.00	**73.0**±23.2 1.7±1.3	**0.03**±0.00 **0.22**±0.01	34.46±10.69 1.5±0.7	**54.21**±0.39
			V-L	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.32±1.16	0.09±0.22 0.40±0.60	**25.9**±22.3 **0.00**±0.00	**73.0**±23.2 1.1±1.0	**0.03**±0.00 **0.22**±0.01	34.46±10.69 1.5±0.8	**54.21**±0.39
			V-C	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.32±1.16	0.09±0.21 0.40±0.59	**25.9**±22.3 **0.00**±0.00	**73.0**±23.2 **1.0**±0.9	**0.03**±0.00 **0.22**±0.01	34.46±10.69 1.5±0.8	**54.21**±0.39
		100	V-U	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.04±0.84	0.09±0.22 **0.38**±0.58	24.9±22.0 **0.00**±0.00	**73.2**±23.2 1.9±1.4	**0.03**±0.00 **0.22**±0.01	34.59±10.73 1.5±0.7	**54.21**±0.39
			V-Δ	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.04±0.84	**0.08**±0.21 **0.38**±0.57	24.9±22.0 **0.00**±0.00	**73.2**±23.2 1.9±1.4	**0.03**±0.00 **0.22**±0.01	34.59±10.73 1.5±0.7	**54.21**±0.39
			V-L	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.04±0.84	0.09±0.22 0.40±0.60	**25.7**±22.3 **0.00**±0.00	**73.2**±23.2 **1.1**±1.0	**0.03**±0.00 **0.22**±0.01	34.59±10.73 1.5±0.8	**54.21**±0.39
			V-C	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 1.04±0.84	0.09±0.21 0.40±0.59	**25.7**±22.3 **0.00**±0.00	**73.2**±23.2 **1.1**±0.9	**0.03**±0.00 **0.22**±0.01	34.59±10.73 1.5±0.8	**54.21**±0.39

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.05	0.025	200	V-U	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.22 0.38±0.58	24.4±22.0 **0.00**±0.00	**73.4**±23.3 2.2±1.5	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.39
			V-Δ	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	**0.08**±0.21 **0.37**±0.57	24.4±22.0 **0.00**±0.00	**73.4**±23.3 2.2±1.5	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.38
			V-L	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.22 0.40±0.60	**25.6**±22.5 **0.00**±0.00	**73.4**±23.3 1.1±1.0	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.39
			V-C	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.21 0.40±0.60	**25.6**±22.5 **0.00**±0.00	**73.4**±23.3 **1.0**±1.0	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.38
		∞	V-U	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.22 0.38±0.58	24.4±22.0 **0.00**±0.00	**73.4**±23.3 2.2±1.5	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.39
			V-Δ	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	**0.08**±0.21 **0.37**±0.57	24.4±22.0 **0.00**±0.00	**73.4**±23.3 2.2±1.5	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.38
			V-L	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.22 0.40±0.60	**25.6**±22.5 **0.00**±0.00	**73.4**±23.3 1.1±1.0	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.39
			V-C	**0.97**±0.03 **0.17**±0.02	**0.79**±0.03 0.81±0.56	0.09±0.21 0.40±0.60	**25.6**±22.5 **0.00**±0.00	**73.4**±23.3 **1.0**±1.0	**0.03**±0.00 **0.22**±0.01	34.44±10.74 1.5±0.7	**54.22**±0.38
	0.05	10	V-U	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 3.01±2.97	**0.10**±0.25 **0.57**±0.72	34.9±23.5 **0.00**±0.00	**63.2**±24.2 1.9±1.2	**0.03**±0.00 **0.21**±0.01	30.60±10.97 1.6±0.9	**54.16**±0.40
			V-Δ	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 3.01±2.97	**0.10**±0.26 **0.57**±0.72	34.9±23.5 **0.00**±0.00	**63.2**±24.2 1.9±1.2	**0.03**±0.00 **0.21**±0.01	30.60±10.97 1.6±0.9	**54.16**±0.40
			V-L	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 3.01±2.97	**0.10**±0.25 0.58±0.73	**35.4**±23.3 **0.00**±0.00	**63.2**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	30.86±11.18 1.6±0.9	**54.16**±0.40
			V-C	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 3.01±2.97	**0.10**±0.26 0.58±0.73	**35.4**±23.4 **0.00**±0.00	**63.2**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	30.86±11.18 1.6±0.9	**54.16**±0.40
		50	V-U	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.61±1.16	**0.10**±0.26 **0.57**±0.74	34.4±23.3 **0.00**±0.00	**63.5**±24.2 2.1±1.3	**0.03**±0.00 **0.21**±0.01	30.66±10.88 1.6±1.0	**54.15**±0.40
			V-Δ	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.61±1.16	**0.10**±0.26 **0.57**±0.74	34.4±23.3 **0.00**±0.00	**63.5**±24.2 2.1±1.3	**0.03**±0.00 **0.21**±0.01	30.66±10.88 1.6±1.0	**54.15**±0.40
			V-L	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.61±1.16	0.11±0.26 0.59±0.74	35.0±23.3 **0.00**±0.00	**63.5**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	30.92±11.09 1.6±1.0	**54.15**±0.40
			V-C	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.61±1.16	0.11±0.27 0.59±0.74	**35.1**±23.3 **0.00**±0.00	**63.5**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	30.92±11.09 1.6±1.0	**54.15**±0.40
		100	V-U	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.25±0.84	**0.10**±0.25 **0.55**±0.71	34.0±23.1 **0.00**±0.00	**63.7**±24.2 2.4±1.4	**0.03**±0.00 **0.21**±0.01	30.75±10.92 1.6±0.9	**54.16**±0.40
			V-Δ	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.25±0.84	**0.10**±0.25 **0.55**±0.72	34.0±23.2 **0.00**±0.00	**63.7**±24.2 2.4±1.4	**0.03**±0.00 **0.21**±0.01	30.75±10.92 1.6±0.9	**54.16**±0.39
			V-L	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.25±0.84	**0.10**±0.25 0.58±0.72	**34.9**±23.3 **0.00**±0.00	**63.7**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	31.02±11.12 1.6±0.9	**54.16**±0.40
			V-C	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 1.25±0.84	**0.10**±0.26 0.58±0.73	**34.9**±23.3 **0.00**±0.00	**63.7**±24.2 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	31.02±11.12 1.6±0.9	**54.16**±0.39

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$								
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.05	0.05	200	V-U	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 **0.53**±0.71	33.4±23.2 **0.00**±0.00	**63.9**±24.4 2.8±1.4	**0.03**±0.00 **0.21**±0.01	30.74±11.02 1.5±0.9	**54.16**±0.40
			V-Δ	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 0.54±0.71	33.4±23.2 **0.00**±0.00	**63.9**±24.4 2.7±1.4	**0.03**±0.00 **0.21**±0.01	30.74±11.02 1.5±0.9	**54.16**±0.40
			V-L	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 0.57±0.73	**34.7**±23.5 **0.00**±0.00	**63.9**±24.4 **1.4**±1.1	**0.03**±0.00 **0.21**±0.01	31.01±11.22 1.6±0.9	**54.16**±0.40
			V-C	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.26 0.57±0.73	**34.7**±23.5 **0.00**±0.00	**63.9**±24.4 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	31.01±11.22 1.6±0.9	**54.16**±0.40
		∞	V-U	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 **0.53**±0.71	33.4±23.2 **0.00**±0.00	**63.9**±24.4 2.8±1.4	**0.03**±0.00 **0.21**±0.01	30.74±11.02 1.5±0.9	**54.16**±0.40
			V-Δ	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 0.54±0.71	33.4±23.2 **0.00**±0.00	**63.9**±24.4 2.7±1.4	**0.03**±0.00 **0.21**±0.01	30.74±11.02 1.5±0.9	**54.16**±0.40
			V-L	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.25 0.57±0.73	**34.7**±23.5 **0.00**±0.00	**63.9**±24.4 **1.4**±1.1	**0.03**±0.00 **0.21**±0.01	31.01±11.22 1.6±0.9	**54.16**±0.40
			V-C	**0.96**±0.03 **0.17**±0.02	**0.78**±0.03 0.98±0.54	**0.10**±0.26 0.57±0.73	**34.7**±23.5 **0.00**±0.00	**63.9**±24.4 **1.4**±1.0	**0.03**±0.00 **0.21**±0.01	31.01±11.22 1.6±0.9	**54.16**±0.40
	0.10	10	V-U	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 4.20±3.61	**0.18**±0.35 **0.77**±0.84	44.3±23.7 **0.00**±0.00	**53.7**±24.4 2.0±1.1	**0.03**±0.00 **0.21**±0.01	26.14±10.73 1.8±1.0	54.05±0.42
			V-Δ	**0.95**±0.04 **0.17**±0.02	**0.77**±0.04 4.20±3.61	**0.18**±0.35 **0.77**±0.84	44.3±23.7 **0.00**±0.00	**53.7**±24.4 2.0±1.1	**0.03**±0.00 **0.21**±0.01	26.14±10.73 1.8±1.0	54.05±0.42
			V-L	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 4.20±3.61	0.19±0.36 0.78±0.84	**44.5**±23.6 **0.00**±0.00	**53.7**±24.4 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.14±10.73 1.8±1.0	54.05±0.43
			V-C	**0.95**±0.04 **0.17**±0.02	**0.77**±0.04 4.20±3.61	**0.18**±0.35 0.78±0.84	**44.5**±23.6 **0.00**±0.00	**53.7**±24.4 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.14±10.73 1.8±1.0	**54.04**±0.43
		50	V-U	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 2.05±1.33	**0.19**±0.37 **0.75**±0.85	43.5±23.8 **0.00**±0.00	**54.3**±24.7 2.2±1.2	**0.03**±0.00 **0.21**±0.01	26.36±10.84 1.8±1.1	**54.05**±0.42
			V-Δ	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 2.05±1.33	**0.19**±0.36 **0.75**±0.85	43.5±23.9 **0.00**±0.00	**54.3**±24.7 2.1±1.2	**0.03**±0.00 **0.21**±0.01	26.36±10.84 1.8±1.1	**54.05**±0.42
			V-L	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 2.05±1.33	**0.19**±0.37 0.77±0.85	43.9±23.9 **0.00**±0.00	**54.3**±24.7 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.36±10.84 1.8±1.1	**54.05**±0.42
			V-C	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 2.05±1.33	**0.19**±0.37 0.77±0.85	**44.0**±23.9 **0.00**±0.00	**54.3**±24.7 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.36±10.84 1.8±1.1	**54.05**±0.42
		100	V-U	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.45±0.93	**0.17**±0.35 **0.72**±0.83	42.8±24.0 **0.00**±0.00	**54.6**±24.9 2.6±1.3	**0.03**±0.00 **0.21**±0.01	26.36±10.87 1.8±1.0	54.06±0.42
			V-Δ	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.45±0.93	**0.17**±0.34 **0.72**±0.83	42.9±24.0 **0.00**±0.00	**54.6**±24.8 2.5±1.3	**0.03**±0.00 **0.21**±0.01	26.36±10.87 1.8±1.0	54.06±0.43
			V-L	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.45±0.93	0.18±0.35 0.75±0.84	**43.7**±24.0 **0.00**±0.00	**54.6**±24.8 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.36±10.87 1.8±1.0	54.06±0.42
			V-C	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.45±0.93	**0.17**±0.34 0.75±0.85	**43.7**±24.1 **0.00**±0.00	**54.6**±24.8 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.36±10.87 1.8±1.0	**54.05**±0.43

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
$\alpha = 0.05$	$\delta = 0.10$	200	V-U	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.35 **0.71**±0.83	42.3±23.9 **0.00**±0.00	**54.7**±24.9 3.1±1.4	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.7±1.0	**54.06**±0.42
			V-Δ	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.34 **0.71**±0.83	42.3±23.9 **0.00**±0.00	**54.7**±24.9 3.0±1.3	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.7±1.0	**54.06**±0.43
			V-L	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	0.18±0.35 0.75±0.84	43.6±24.1 **0.00**±0.00	**54.7**±24.9 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.8±1.0	**54.06**±0.42
			V-C	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.34 0.75±0.85	**43.7**±24.1 **0.00**±0.00	**54.7**±24.9 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.8±1.0	**54.06**±0.43
		∞	V-U	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.35 **0.71**±0.83	42.3±23.9 **0.00**±0.00	**54.7**±24.9 3.1±1.4	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.7±1.0	**54.06**±0.42
			V-Δ	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.34 **0.71**±0.83	42.3±23.9 **0.00**±0.00	**54.7**±24.9 3.0±1.3	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.7±1.0	**54.06**±0.43
			V-L	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	0.18±0.35 0.75±0.84	43.6±24.1 **0.00**±0.00	**54.7**±24.9 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.8±1.0	**54.06**±0.42
			V-C	**0.95**±0.04 **0.17**±0.02	**0.77**±0.03 1.08±0.56	**0.17**±0.34 0.75±0.85	**43.7**±24.1 **0.00**±0.00	**54.7**±24.9 **1.7**±1.0	**0.03**±0.00 **0.21**±0.01	26.40±10.94 1.8±1.0	**54.06**±0.43
	$\delta = 0.15$	10	V-U	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 4.88±3.85	**0.21**±0.41 **0.90**±1.01	50.9±23.0 **0.00**±0.00	**46.8**±23.7 2.3±1.1	**0.03**±0.00 **0.21**±0.01	23.65±11.15 1.9±1.2	**53.96**±0.44
			V-Δ	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 4.88±3.85	0.22±0.42 **0.90**±1.01	50.9±23.0 **0.00**±0.00	**46.8**±23.7 2.2±1.1	**0.03**±0.00 **0.21**±0.01	23.65±11.15 1.9±1.2	**53.96**±0.44
			V-L	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 4.88±3.85	0.22±0.42 0.91±1.01	51.1±22.8 **0.00**±0.00	**46.8**±23.7 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.65±11.15 1.9±1.2	**53.96**±0.44
			V-C	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 4.88±3.85	0.22±0.43 0.91±1.02	**51.2**±22.8 **0.00**±0.00	**46.8**±23.7 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	23.65±11.15 1.9±1.2	**53.96**±0.44
		50	V-U	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 2.25±1.28	**0.22**±0.41 **0.89**±1.00	50.2±23.2 **0.00**±0.00	**47.4**±24.0 2.4±1.2	**0.03**±0.00 **0.21**±0.01	23.81±11.24 1.9±1.2	**53.96**±0.44
			V-Δ	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 2.24±1.27	**0.22**±0.42 **0.89**±1.01	50.2±23.2 **0.00**±0.00	**47.4**±24.0 2.4±1.2	**0.03**±0.00 **0.21**±0.01	23.81±11.24 1.9±1.2	**53.96**±0.44
			V-L	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 2.25±1.28	**0.22**±0.42 0.91±1.01	**50.6**±23.1 **0.00**±0.00	**47.4**±24.0 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.81±11.24 1.9±1.2	**53.96**±0.44
			V-C	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 2.24±1.27	**0.22**±0.43 0.91±1.01	**50.6**±23.2 **0.00**±0.00	**47.4**±24.0 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	23.81±11.24 1.9±1.2	**53.96**±0.44
		100	V-U	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.74±0.93	**0.21**±0.41 **0.88**±0.99	49.7±23.1 **0.00**±0.00	**47.5**±24.1 2.8±1.4	**0.03**±0.00 **0.21**±0.01	23.89±11.30 1.9±1.2	**53.97**±0.44
			V-Δ	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.74±0.93	0.22±0.42 **0.88**±1.00	49.7±23.1 **0.00**±0.00	**47.5**±24.1 2.8±1.4	**0.03**±0.00 **0.21**±0.01	23.89±11.30 1.9±1.2	**53.97**±0.44
			V-L	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.74±0.93	0.22±0.41 0.91±1.01	50.4±23.2 **0.00**±0.00	**47.5**±24.1 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.89±11.30 2.0±1.2	**53.97**±0.44
			V-C	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.74±0.93	0.22±0.42 0.91±1.01	**50.5**±23.2 **0.00**±0.00	**47.5**±24.1 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	23.89±11.30 2.0±1.2	**53.97**±0.44

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.15	200	V-U	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.41 **0.86**±1.00	48.9±23.3 **0.00**±0.00	**47.7**±24.3 3.4±1.3	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-Δ	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.42 **0.86**±1.00	48.9±23.4 **0.00**±0.00	**47.7**±24.3 3.4±1.3	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-L	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.41 0.89±1.01	**50.3**±23.4 **0.00**±0.00	**47.7**±24.3 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-C	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.42 0.88±1.01	**50.3**±23.5 **0.00**±0.00	**47.7**±24.3 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
		∞	V-U	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.41 **0.86**±1.00	48.9±23.3 **0.00**±0.00	**47.7**±24.3 3.4±1.3	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-Δ	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.42 **0.86**±1.00	48.9±23.4 **0.00**±0.00	**47.7**±24.3 3.4±1.3	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-L	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.41 0.89±1.01	**50.3**±23.4 **0.00**±0.00	**47.7**±24.3 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
			V-C	**0.94**±0.04 **0.17**±0.02	**0.76**±0.04 1.19±0.56	**0.21**±0.42 0.88±1.01	**50.3**±23.5 **0.00**±0.00	**47.7**±24.3 **2.0**±1.1	**0.03**±0.00 **0.21**±0.01	23.83±11.32 1.9±1.2	**53.97**±0.44
	0.20	10	V-U	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 5.23±4.11	**0.25**±0.47 **1.03**±1.01	54.9±22.6 **0.00**±0.00	**42.8**±23.2 2.3±1.0	**0.03**±0.00 **0.21**±0.01	21.49±10.69 2.0±1.2	**53.91**±0.45
			V-Δ	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 5.20±4.12	0.26±0.48 **1.03**±1.00	54.9±22.6 **0.00**±0.00	**42.8**±23.2 2.3±1.0	**0.03**±0.00 **0.21**±0.01	21.49±10.69 2.0±1.2	**53.91**±0.45
			V-L	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 5.23±4.11	**0.25**±0.47 1.04±1.01	**55.1**±22.4 **0.00**±0.00	**42.8**±23.2 2.1±1.0	**0.03**±0.00 **0.21**±0.01	21.49±10.69 2.0±1.2	**53.91**±0.45
			V-C	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 5.20±4.12	0.26±0.48 1.04±1.00	**55.1**±22.4 **0.00**±0.00	**42.8**±23.2 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.49±10.69 2.0±1.2	**53.91**±0.45
		50	V-U	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 2.37±1.36	**0.26**±0.48 **1.04**±1.00	54.4±22.7 **0.00**±0.00	**43.1**±23.4 2.5±1.2	**0.03**±0.00 **0.21**±0.01	21.65±10.87 2.1±1.3	**53.91**±0.45
			V-Δ	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 2.37±1.36	0.27±0.48 1.05±0.99	54.5±22.7 **0.00**±0.00	**43.1**±23.4 2.4±1.2	**0.03**±0.00 **0.21**±0.01	21.65±10.87 2.1±1.3	**53.91**±0.45
			V-L	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 2.37±1.36	**0.26**±0.48 1.05±1.01	**54.9**±22.7 **0.00**±0.00	**43.1**±23.4 2.1±1.0	**0.03**±0.00 **0.21**±0.01	21.65±10.87 2.1±1.3	**53.91**±0.45
			V-C	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 2.37±1.36	0.27±0.48 1.06±1.00	**54.9**±22.7 **0.00**±0.00	**43.1**±23.4 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.65±10.87 2.1±1.3	**53.91**±0.45
		100	V-U	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.85±0.96	**0.25**±0.47 **1.01**±1.00	54.1±22.6 **0.00**±0.00	**43.2**±23.4 2.7±1.3	**0.03**±0.00 **0.21**±0.01	21.72±10.83 2.1±1.3	**53.91**±0.45
			V-Δ	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.84±0.96	0.26±0.48 1.02±0.99	54.1±22.6 **0.00**±0.00	**43.2**±23.4 2.7±1.3	**0.03**±0.00 **0.21**±0.01	21.72±10.83 2.1±1.3	**53.91**±0.45
			V-L	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.85±0.96	**0.25**±0.47 1.04±1.01	**54.7**±22.6 **0.00**±0.00	**43.2**±23.4 2.1±1.0	**0.03**±0.00 **0.21**±0.01	21.72±10.83 2.1±1.3	**53.91**±0.45
			V-C	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.84±0.96	0.26±0.48 1.04±1.00	**54.7**±22.6 **0.00**±0.00	**43.2**±23.4 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.72±10.83 2.1±1.3	**53.91**±0.45

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.05$ $\delta = 0.20$	200 V-U	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	**0.25**±0.47 **0.99**±1.00	53.0±23.0 **0.00**±0.00	**43.5**±23.9 3.4±1.2	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
	200 V-Δ	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	0.26±0.48 1.00±0.99	53.1±23.0 **0.00**±0.00	**43.5**±23.9 3.4±1.2	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	53.93±0.45
	200 V-L	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	**0.25**±0.47 1.02±1.01	54.4±23.1 **0.00**±0.00	**43.5**±23.9 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
	200 V-C	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	0.26±0.48 1.03±1.00	**54.5**±23.1 **0.00**±0.00	**43.5**±23.9 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
	∞ V-U	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	**0.25**±0.47 **0.99**±1.00	53.0±23.0 **0.00**±0.00	**43.5**±23.9 3.4±1.2	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
	∞ V-Δ	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	0.26±0.48 1.00±0.99	53.1±23.0 **0.00**±0.00	**43.5**±23.9 3.4±1.2	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	53.93±0.45
	∞ V-L	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	**0.25**±0.47 1.02±1.01	54.4±23.1 **0.00**±0.00	**43.5**±23.9 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
	∞ V-C	**0.93**±0.04 **0.17**±0.02	**0.75**±0.04 1.26±0.58	0.26±0.48 1.03±1.00	**54.5**±23.1 **0.00**±0.00	**43.5**±23.9 **2.0**±1.0	**0.03**±0.00 **0.21**±0.01	21.57±10.79 2.0±1.2	**53.92**±0.45
$\alpha = 0.10$ $\delta = 0.025$	10 V-U	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 18.48±7.04	0.56±0.56 **1.84**±1.10	68.2±10.6 **0.00**±0.00	**29.2**±10.7 2.6±1.1	**0.02**±0.00 **0.21**±0.01	15.69±5.82 3.9±1.8	53.56±0.46
	10 V-Δ	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 18.45±7.02	**0.55**±0.55 **1.84**±1.09	68.2±10.6 **0.00**±0.00	**29.2**±10.7 2.6±1.1	**0.02**±0.00 **0.21**±0.01	15.69±5.82 3.9±1.8	53.56±0.46
	10 V-L	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 18.47±7.09	0.63±0.62 2.03±1.17	68.2±10.4 **0.00**±0.00	29.3±10.7 2.5±0.7	**0.02**±0.00 **0.21**±0.01	15.69±5.82 4.3±1.7	**53.52**±0.47
	10 V-C	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 18.44±7.08	0.63±0.62 2.03±1.17	**68.3**±10.4 **0.00**±0.00	29.3±10.7 **2.4**±0.8	**0.02**±0.00 **0.21**±0.01	15.69±5.82 4.3±1.7	53.53±0.47
	50 V-U	**0.88**±0.04 **0.17**±0.02	0.72±0.04 5.95±1.91	0.56±0.55 **1.86**±1.10	68.1±10.4 **0.00**±0.00	**29.3**±10.6 2.6±0.9	**0.02**±0.00 **0.21**±0.01	15.71±5.77 4.0±1.8	53.57±0.46
	50 V-Δ	**0.88**±0.04 **0.17**±0.02	0.72±0.04 5.93±1.91	**0.55**±0.55 **1.86**±1.10	68.2±10.4 **0.00**±0.00	**29.3**±10.6 2.5±0.9	**0.02**±0.00 **0.21**±0.01	15.71±5.77 4.0±1.8	53.57±0.46
	50 V-L	**0.88**±0.04 **0.17**±0.02	**0.71**±0.04 5.94±1.91	0.64±0.62 2.06±1.19	68.2±10.3 **0.00**±0.00	**29.3**±10.6 2.5±0.7	**0.02**±0.00 **0.21**±0.01	15.71±5.77 4.4±1.7	**53.52**±0.47
	50 V-C	**0.88**±0.04 **0.17**±0.02	**0.71**±0.04 5.93±1.92	0.65±0.63 2.05±1.19	**68.3**±10.3 **0.00**±0.00	**29.3**±10.6 **2.4**±0.8	**0.02**±0.00 **0.21**±0.01	15.71±5.77 4.4±1.7	**53.52**±0.47
	100 V-U	0.89±0.04 **0.17**±0.02	**0.72**±0.04 3.68±1.05	**0.51**±0.53 **1.75**±1.05	66.7±11.5 **0.00**±0.00	**29.4**±10.9 3.9±3.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.9±1.8	53.60±0.45
	100 V-Δ	0.89±0.04 **0.17**±0.02	**0.72**±0.04 3.68±1.05	**0.51**±0.54 **1.75**±1.05	66.7±11.5 **0.00**±0.00	**29.4**±10.9 3.9±3.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.9±1.8	53.59±0.44
	100 V-L	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 3.69±1.04	0.62±0.61 2.02±1.19	68.1±10.6 **0.00**±0.00	**29.4**±10.9 2.5±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	**53.52**±0.47
	100 V-C	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 3.69±1.04	0.62±0.61 2.02±1.18	**68.2**±10.6 **0.00**±0.00	**29.4**±10.9 **2.4**±0.8	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	53.53±0.46

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.025	200	V-U	0.91±0.04 **0.17**±0.02	0.73±0.03 2.26±0.62	**0.43**±0.50 **1.43**±1.00	53.7±10.5 **0.00**±0.00	**29.3**±10.9 16.9±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.8±1.8	53.71±0.43
			V-Δ	0.91±0.04 **0.17**±0.02	0.73±0.03 2.26±0.62	**0.43**±0.50 **1.43**±1.00	53.8±10.6 **0.00**±0.00	**29.3**±10.9 16.9±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.8±1.8	53.71±0.43
			V-L	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 2.22±0.59	0.61±0.61 2.01±1.20	68.1±10.6 **0.00**±0.00	29.4±10.9 2.5±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	**53.53**±0.47
			V-C	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 2.22±0.59	0.62±0.62 2.01±1.19	**68.2**±10.6 **0.00**±0.00	29.4±10.9 **2.4**±0.8	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	**53.53**±0.47
		∞	V-U	0.91±0.04 **0.17**±0.02	0.73±0.03 2.19±0.58	**0.43**±0.50 **1.42**±1.00	53.7±10.5 **0.00**±0.00	**29.3**±10.9 16.9±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.8±1.8	53.71±0.43
			V-Δ	0.91±0.04 **0.17**±0.02	0.73±0.03 2.19±0.58	**0.43**±0.50 1.43±1.01	53.8±10.6 **0.00**±0.00	**29.3**±10.9 16.9±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 3.8±1.8	53.71±0.43
			V-L	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 2.20±0.57	0.61±0.61 2.01±1.20	68.1±10.6 **0.00**±0.00	29.4±10.9 2.5±0.7	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	**53.53**±0.47
			V-C	**0.88**±0.04 **0.17**±0.02	**0.72**±0.04 2.20±0.57	0.62±0.62 2.01±1.19	**68.2**±10.6 **0.00**±0.00	29.4±10.9 **2.4**±0.8	**0.02**±0.00 **0.21**±0.01	15.78±5.93 4.3±1.7	**53.53**±0.47
	0.05	10	V-U	0.87±0.05 **0.17**±0.02	0.71±0.04 20.44±7.36	**0.70**±0.67 **2.16**±1.18	72.5±8.9 **0.00**±0.00	**24.8**±9.0 2.7±0.9	**0.02**±0.00 **0.21**±0.01	13.27±4.88 4.4±1.9	53.46±0.44
			V-Δ	0.87±0.05 **0.17**±0.02	0.71±0.04 20.40±7.44	**0.70**±0.69 **2.16**±1.16	**72.6**±8.9 **0.00**±0.00	**24.8**±9.0 **2.6**±0.9	**0.02**±0.00 **0.21**±0.01	13.27±4.88 4.4±1.9	53.46±0.44
			V-L	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 20.41±7.35	0.76±0.72 2.34±1.25	72.5±8.9 **0.00**±0.00	**24.8**±9.0 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	13.27±4.88 4.7±1.9	**53.41**±0.45
			V-C	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 20.37±7.43	0.75±0.73 2.34±1.24	**72.6**±8.9 **0.00**±0.00	**24.8**±9.0 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	13.27±4.88 4.7±1.9	**53.41**±0.45
		50	V-U	0.87±0.05 **0.17**±0.02	**0.70**±0.04 6.50±1.87	**0.72**±0.68 **2.21**±1.18	72.4±8.8 **0.00**±0.00	**24.8**±9.0 2.7±0.8	**0.02**±0.00 **0.21**±0.01	13.28±4.84 4.5±1.9	53.44±0.43
			V-Δ	0.87±0.05 **0.17**±0.02	**0.70**±0.04 6.49±1.89	**0.72**±0.69 **2.21**±1.16	72.5±8.8 **0.00**±0.00	**24.8**±9.0 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	13.28±4.84 4.4±1.9	53.44±0.43
			V-L	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 6.50±1.86	0.78±0.73 2.36±1.25	72.5±8.8 **0.00**±0.00	**24.8**±9.0 2.7±0.8	**0.02**±0.00 **0.21**±0.01	13.28±4.84 4.7±1.9	**53.40**±0.44
			V-C	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 6.49±1.88	0.77±0.73 2.36±1.25	**72.6**±8.8 **0.00**±0.00	**24.8**±9.0 **2.6**±0.7	**0.02**±0.00 **0.21**±0.01	13.28±4.84 4.7±1.9	**53.40**±0.44
		100	V-U	0.87±0.04 **0.17**±0.02	0.71±0.04 3.96±1.02	0.68±0.62 2.10±1.08	71.7±9.4 **0.00**±0.00	**24.9**±9.2 3.4±2.6	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.4±1.9	53.48±0.42
			V-Δ	0.87±0.04 **0.17**±0.02	0.71±0.04 3.95±1.02	**0.67**±0.63 **2.09**±1.07	71.7±9.4 **0.00**±0.00	**24.9**±9.2 3.4±2.6	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.4±1.9	53.48±0.41
			V-L	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 3.96±1.02	0.77±0.71 2.35±1.23	72.4±9.0 **0.00**±0.00	**24.9**±9.2 2.7±0.8	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.7±1.9	**53.41**±0.44
			V-C	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 3.95±1.02	0.76±0.71 2.35±1.22	**72.5**±9.0 **0.00**±0.00	**24.9**±9.2 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.7±1.9	**53.41**±0.44

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.05	200	V-U	0.89±0.04 **0.17**±0.02	0.72±0.04 2.92±1.03	0.57±0.58 **1.74**±1.07	58.8±8.8 **0.00**±0.00	**24.8**±9.2 16.4±1.1	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.3±1.8	53.61±0.42
			V-Δ	0.89±0.04 **0.17**±0.02	0.72±0.04 2.92±1.03	**0.56**±0.59 1.75±1.07	58.9±8.8 **0.00**±0.00	**24.8**±9.2 16.3±1.1	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.3±1.8	53.61±0.41
			V-L	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 2.90±0.98	0.76±0.72 2.33±1.25	72.4±9.0 **0.00**±0.00	24.9±9.2 2.7±0.8	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.7±1.9	**53.41**±0.44
			V-C	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 2.90±0.98	0.75±0.72 2.33±1.24	**72.5**±9.0 **0.00**±0.00	24.9±9.2 **2.6**±0.7	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.6±1.9	**53.41**±0.44
		∞	V-U	0.89±0.04 **0.17**±0.02	0.72±0.04 2.33±0.59	0.57±0.58 **1.73**±1.07	58.1±8.9 **0.00**±0.00	**24.8**±9.2 17.1±0.7	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.2±1.9	53.61±0.42
			V-Δ	0.89±0.04 **0.17**±0.02	0.72±0.04 2.32±0.59	**0.56**±0.59 1.74±1.07	58.2±9.0 **0.00**±0.00	**24.8**±9.2 17.0±0.7	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.2±1.9	53.61±0.42
			V-L	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 2.32±0.59	0.76±0.72 2.33±1.25	72.4±9.0 **0.00**±0.00	24.9±9.2 2.7±0.8	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.6±1.9	**53.41**±0.44
			V-C	**0.86**±0.05 **0.17**±0.02	**0.70**±0.04 2.32±0.59	0.75±0.72 2.33±1.24	**72.5**±9.0 **0.00**±0.00	24.9±9.2 **2.6**±0.7	**0.02**±0.00 **0.21**±0.01	13.33±4.97 4.6±1.8	**53.41**±0.44
0.10	0.10	10	V-U	0.85±0.05 **0.17**±0.02	0.69±0.04 22.95±8.04	**0.97**±0.81 **2.69**±1.48	**76.7**±7.7 **0.00**±0.00	**20.5**±7.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	10.96±4.26 5.0±2.1	53.28±0.49
			V-Δ	**0.84**±0.05 **0.17**±0.02	0.69±0.04 22.91±8.10	0.98±0.81 **2.69**±1.46	**76.7**±7.7 **0.00**±0.00	**20.5**±7.9 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	10.96±4.26 5.0±2.1	53.29±0.49
			V-L	**0.84**±0.05 **0.17**±0.02	0.69±0.04 22.87±8.09	1.08±0.90 2.93±1.67	**76.7**±7.7 **0.00**±0.00	20.6±7.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	10.96±4.26 5.3±2.1	**53.25**±0.49
			V-C	**0.84**±0.05 **0.17**±0.02	**0.68**±0.04 22.85±8.14	1.08±0.92 2.93±1.64	**76.7**±7.7 **0.00**±0.00	20.6±7.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	10.96±4.26 5.3±2.1	**53.25**±0.49
		50	V-U	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 7.04±1.85	**0.95**±0.80 **2.72**±1.47	76.6±7.7 **0.00**±0.00	**20.6**±7.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	11.01±4.29 5.1±2.0	53.30±0.49
			V-Δ	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 7.03±1.87	0.96±0.81 **2.72**±1.45	**76.7**±7.8 **0.00**±0.00	**20.6**±7.9 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	11.01±4.29 5.1±2.0	53.29±0.49
			V-L	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 7.04±1.86	1.06±0.90 2.93±1.67	76.6±7.8 **0.00**±0.00	**20.6**±7.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	11.01±4.29 5.3±2.1	**53.26**±0.50
			V-C	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 7.03±1.88	1.07±0.92 2.92±1.63	**76.7**±7.8 **0.00**±0.00	**20.6**±7.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	11.01±4.29 5.3±2.1	**53.26**±0.50
		100	V-U	0.85±0.05 **0.17**±0.02	**0.69**±0.04 4.80±1.38	**0.89**±0.75 **2.60**±1.42	76.2±8.1 **0.00**±0.00	**20.7**±8.1 3.1±1.8	**0.02**±0.00 **0.21**±0.01	11.06±4.42 5.1±2.1	53.33±0.49
			V-Δ	0.85±0.05 **0.17**±0.02	**0.69**±0.04 4.79±1.39	0.91±0.76 **2.60**±1.40	76.3±8.1 **0.00**±0.00	**20.7**±8.2 3.0±1.9	**0.02**±0.00 **0.21**±0.01	11.05±4.42 5.1±2.1	53.32±0.49
			V-L	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 4.77±1.40	1.06±0.89 2.93±1.66	76.5±8.0 **0.00**±0.00	**20.7**±8.2 2.8±0.8	**0.02**±0.00 **0.21**±0.01	11.06±4.42 5.3±2.1	53.26±0.50
			V-C	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 4.76±1.40	1.07±0.91 2.92±1.63	**76.6**±8.0 **0.00**±0.00	**20.7**±8.2 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	11.05±4.42 5.3±2.1	**53.25**±0.50

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.10	200	V-U	0.87±0.05 **0.17**±0.02	0.70±0.04 3.16±1.07	**0.76**±0.74 **2.17**±1.39	64.0±7.8 **0.00**±0.00	**20.6**±8.1 15.3±2.1	**0.02**±0.00 **0.21**±0.01	11.06±4.42 4.9±2.0	53.47±0.47
			V-Δ	0.87±0.05 **0.17**±0.02	0.70±0.04 3.16±1.08	0.78±0.75 2.18±1.36	64.1±7.9 **0.00**±0.00	**20.6**±8.1 15.2±2.1	**0.02**±0.00 **0.21**±0.01	11.05±4.42 4.9±2.0	53.47±0.47
			V-L	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 3.15±1.05	1.04±0.90 2.87±1.68	76.5±8.0 **0.00**±0.00	20.7±8.1 2.8±0.8	**0.02**±0.00 **0.21**±0.01	11.06±4.42 5.2±2.1	**53.27**±0.50
			V-C	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 3.14±1.06	1.05±0.92 2.86±1.65	**76.6**±8.0 **0.00**±0.00	20.7±8.1 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	11.05±4.42 5.2±2.1	**53.27**±0.50
		∞	V-U	0.87±0.05 **0.17**±0.02	0.71±0.04 2.50±0.63	**0.76**±0.74 **2.14**±1.40	62.1±8.0 **0.00**±0.00	**20.6**±8.1 17.2±0.7	**0.02**±0.00 **0.21**±0.01	11.06±4.42 4.8±2.0	53.49±0.48
			V-Δ	0.87±0.05 **0.17**±0.02	0.71±0.04 2.50±0.63	0.77±0.76 2.15±1.38	62.3±8.0 **0.00**±0.00	**20.6**±8.1 17.1±0.7	**0.02**±0.00 **0.21**±0.01	11.05±4.42 4.8±2.0	53.48±0.48
			V-L	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 2.50±0.63	1.04±0.90 2.87±1.68	76.5±8.0 **0.00**±0.00	20.7±8.1 2.8±0.8	**0.02**±0.00 **0.21**±0.01	11.06±4.42 5.2±2.1	**53.27**±0.50
			V-C	**0.84**±0.05 **0.17**±0.02	**0.69**±0.04 2.50±0.63	1.05±0.92 2.86±1.65	**76.6**±8.0 **0.00**±0.00	20.7±8.1 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	11.05±4.42 5.2±2.1	**53.27**±0.50
	0.15	10	V-U	0.83±0.06 **0.17**±0.02	**0.68**±0.04 24.29±8.13	**1.07**±0.88 **3.06**±1.63	**79.1**±7.0 **0.00**±0.00	**18.3**±7.0 2.7±0.8	**0.02**±0.00 **0.21**±0.01	9.72±3.79 5.4±2.2	53.19±0.51
			V-Δ	0.83±0.06 **0.17**±0.02	**0.68**±0.04 24.23±8.08	1.09±0.90 **3.06**±1.63	**79.1**±7.0 **0.00**±0.00	**18.3**±7.0 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	9.72±3.79 5.4±2.2	53.19±0.52
			V-L	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 24.29±8.16	1.21±1.02 3.27±1.79	79.0±7.0 **0.00**±0.00	**18.3**±7.0 2.7±0.8	**0.02**±0.00 **0.21**±0.01	9.72±3.79 5.7±2.3	**53.14**±0.52
			V-C	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 24.23±8.10	1.21±1.03 3.27±1.79	**79.1**±6.9 **0.00**±0.00	**18.3**±7.0 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	9.72±3.79 5.6±2.2	**53.14**±0.53
		50	V-U	0.83±0.06 **0.17**±0.02	**0.68**±0.04 7.73±1.97	**1.09**±0.89 **3.11**±1.62	78.9±7.2 **0.00**±0.00	**18.4**±7.2 2.7±0.8	**0.02**±0.00 **0.21**±0.01	9.78±3.89 5.5±2.2	53.17±0.51
			V-Δ	0.83±0.06 **0.17**±0.02	**0.68**±0.04 7.69±1.97	1.11±0.91 **3.11**±1.62	**79.0**±7.2 **0.00**±0.00	**18.4**±7.2 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	9.78±3.89 5.5±2.2	53.16±0.52
			V-L	**0.82**±0.06 **0.17**±0.02	**0.68**±0.05 7.72±1.97	1.20±1.02 3.30±1.78	78.9±7.2 **0.00**±0.00	**18.4**±7.2 2.7±0.8	**0.02**±0.00 **0.21**±0.01	9.78±3.89 5.8±2.3	53.14±0.52
			V-C	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 7.70±1.95	1.21±1.03 3.30±1.77	**79.0**±7.2 **0.00**±0.00	**18.4**±7.2 **2.6**±0.8	**0.02**±0.00 **0.21**±0.01	9.78±3.89 5.7±2.3	**53.13**±0.53
		100	V-U	0.83±0.05 **0.17**±0.02	**0.68**±0.04 5.08±1.41	**1.03**±0.84 **2.96**±1.54	78.7±7.4 **0.00**±0.00	**18.4**±7.3 2.9±1.2	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.5±2.2	53.21±0.48
			V-Δ	0.83±0.05 **0.17**±0.02	**0.68**±0.04 5.05±1.39	1.05±0.87 **2.96**±1.54	78.8±7.4 **0.00**±0.00	**18.4**±7.3 2.8±1.3	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.5±2.2	53.20±0.49
			V-L	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 5.10±1.41	1.19±1.01 3.27±1.77	**78.9**±7.2 **0.00**±0.00	**18.4**±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.7±2.3	**53.15**±0.51
			V-C	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 5.07±1.39	1.19±1.02 3.27±1.77	**78.9**±7.2 **0.00**±0.00	**18.4**±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.7±2.3	**53.15**±0.52

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.10	0.15	200	V-U	0.86±0.05 **0.17**±0.02	0.70±0.04 3.34±1.10	**0.88**±0.82 **2.52**±1.55	67.3±7.5 **0.00**±0.00	**18.4**±7.3 14.3±2.9	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.4±2.3	53.35±0.49
			V-Δ	0.86±0.05 **0.17**±0.02	0.70±0.04 3.31±1.05	0.90±0.83 2.53±1.54	67.4±7.5 **0.00**±0.00	**18.4**±7.3 14.2±2.9	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.3±2.2	53.35±0.50
			V-L	**0.82**±0.06 **0.17**±0.02	**0.68**±0.05 3.32±1.10	1.18±1.01 3.24±1.78	**78.9**±7.2 **0.00**±0.00	**18.4**±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.7±2.3	53.16±0.52
			V-C	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 3.29±1.05	1.18±1.01 3.24±1.77	**78.9**±7.2 **0.00**±0.00	**18.4**±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.7±2.3	**53.15**±0.52
		∞	V-U	0.86±0.05 **0.17**±0.02	0.70±0.04 2.63±0.66	**0.87**±0.82 **2.45**±1.54	64.5±7.2 **0.00**±0.00	**18.3**±7.3 17.1±0.7	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.2±2.2	53.37±0.50
			V-Δ	0.86±0.05 **0.17**±0.02	0.70±0.04 2.62±0.65	0.88±0.83 2.46±1.54	64.7±7.3 **0.00**±0.00	**18.3**±7.3 17.0±0.7	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.2±2.2	53.37±0.50
			V-L	**0.82**±0.06 **0.17**±0.02	**0.68**±0.05 2.63±0.66	1.18±1.01 3.23±1.78	**78.9**±7.3 **0.00**±0.00	18.4±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.6±2.2	53.16±0.52
			V-C	**0.82**±0.06 **0.17**±0.02	**0.68**±0.04 2.62±0.65	1.18±1.01 3.23±1.78	**78.9**±7.2 **0.00**±0.00	18.4±7.3 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	9.81±3.95 5.6±2.2	**53.15**±0.52
	0.20	10	V-U	0.82±0.06 **0.17**±0.02	0.67±0.05 24.78±8.01	**1.34**±1.14 **3.52**±1.92	81.1±6.4 **0.00**±0.00	**16.1**±6.5 2.8±0.7	**0.02**±0.00 **0.21**±0.01	8.57±3.49 5.8±2.5	53.08±0.54
			V-Δ	0.82±0.06 **0.17**±0.02	0.67±0.05 24.87±8.19	1.35±1.14 3.53±1.93	**81.2**±6.4 **0.00**±0.00	**16.1**±6.5 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.57±3.49 5.8±2.6	53.07±0.55
			V-L	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 24.79±8.08	1.48±1.30 3.79±2.20	81.1±6.4 **0.00**±0.00	**16.1**±6.5 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.57±3.49 6.1±2.6	53.03±0.57
			V-C	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 24.89±8.27	1.49±1.31 3.81±2.21	**81.2**±6.4 **0.00**±0.00	**16.1**±6.5 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.57±3.49 6.1±2.6	**53.02**±0.58
		50	V-U	**0.81**±0.06 **0.17**±0.02	0.67±0.05 8.09±2.05	**1.37**±1.15 **3.57**±1.96	81.0±6.6 **0.00**±0.00	**16.2**±6.7 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.63±3.59 5.9±2.5	53.08±0.54
			V-Δ	**0.81**±0.06 **0.17**±0.02	0.67±0.05 8.11±2.11	1.39±1.17 **3.57**±1.96	**81.1**±6.6 **0.00**±0.00	**16.2**±6.7 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.63±3.59 5.9±2.5	53.08±0.55
			V-L	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 8.07±2.05	1.51±1.29 3.79±2.19	81.0±6.6 **0.00**±0.00	16.3±6.7 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.63±3.59 6.2±2.5	**53.03**±0.57
			V-C	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 8.09±2.12	1.51±1.31 3.80±2.20	**81.1**±6.6 **0.00**±0.00	16.3±6.7 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.63±3.59 6.2±2.5	**53.03**±0.57
		100	V-U	0.82±0.06 **0.17**±0.02	0.67±0.05 5.28±1.37	**1.29**±1.09 **3.41**±1.86	80.6±6.9 **0.00**±0.00	**16.3**±6.8 3.1±2.0	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.9±2.5	53.09±0.53
			V-Δ	0.82±0.06 **0.17**±0.02	0.67±0.05 5.29±1.42	1.30±1.10 3.43±1.87	80.7±6.9 **0.00**±0.00	**16.3**±6.8 3.0±2.0	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.9±2.5	53.10±0.53
			V-L	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 5.27±1.39	1.49±1.30 3.76±2.20	**81.0**±6.7 **0.00**±0.00	**16.3**±6.8 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.03**±0.57
			V-C	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 5.29±1.43	1.50±1.31 3.77±2.21	**81.0**±6.7 **0.00**±0.00	**16.3**±6.8 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.03**±0.57

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.10$ $\delta = 0.20$	200	V-U	0.84±0.06 **0.17**±0.02	0.68±0.04 3.42±1.09	**1.17**±1.09 **2.97**±1.86	69.7±6.9 **0.00**±0.00	**16.2**±6.8 14.1±3.4	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.8±2.5	53.23±0.53
		V-Δ	0.84±0.06 **0.17**±0.02	0.68±0.04 3.44±1.12	1.18±1.10 2.98±1.87	69.9±7.0 **0.00**±0.00	**16.2**±6.8 13.8±3.3	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.8±2.6	53.22±0.53
		V-L	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 3.42±1.10	1.47±1.30 3.73±2.21	**81.0**±6.7 **0.00**±0.00	16.3±6.8 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.04**±0.57
		V-C	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 3.44±1.13	1.48±1.31 3.75±2.22	**81.0**±6.7 **0.00**±0.00	16.3±6.8 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.04**±0.57
	∞	V-U	0.85±0.06 **0.17**±0.02	0.69±0.05 2.75±0.71	**1.14**±1.09 **2.88**±1.88	66.6±6.6 **0.00**±0.00	**16.2**±6.8 17.2±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.6±2.5	53.26±0.54
		V-Δ	0.85±0.06 **0.17**±0.02	0.69±0.05 2.75±0.71	1.15±1.11 **2.88**±1.89	66.8±6.7 **0.00**±0.00	**16.2**±6.8 17.0±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 5.7±2.5	53.25±0.54
		V-L	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 2.74±0.72	1.47±1.30 3.72±2.22	**81.0**±6.7 **0.00**±0.00	16.3±6.8 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.04**±0.58
		V-C	**0.81**±0.07 **0.17**±0.02	**0.66**±0.05 2.74±0.72	1.47±1.32 3.73±2.23	**81.0**±6.7 **0.00**±0.00	16.3±6.8 **2.7**±0.7	**0.02**±0.00 **0.21**±0.01	8.65±3.65 6.1±2.6	**53.04**±0.58
$\alpha = 0.15$ $\delta = 0.025$	10	V-U	0.82±0.04 **0.17**±0.02	0.67±0.04 38.73±8.76	1.24±0.79 **3.39**±1.28	79.4±4.6 **0.00**±0.00	**17.3**±4.2 3.3±1.8	**0.02**±0.00 **0.21**±0.01	9.21±2.24 6.9±2.3	53.12±0.42
		V-Δ	0.82±0.04 **0.17**±0.02	0.67±0.04 38.79±8.75	**1.23**±0.75 3.42±1.25	79.5±4.5 **0.00**±0.00	**17.3**±4.2 3.1±1.7	**0.02**±0.00 **0.21**±0.01	9.21±2.24 6.9±2.3	53.12±0.42
		V-L	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 38.34±8.59	1.87±1.10 4.58±1.76	79.7±4.2 **0.00**±0.00	17.4±4.2 2.9±0.7	**0.02**±0.00 **0.21**±0.01	9.21±2.24 8.2±2.3	**52.83**±0.46
		V-C	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 38.48±8.65	1.85±1.10 4.59±1.76	**79.8**±4.2 **0.00**±0.00	17.4±4.2 **2.8**±0.7	**0.02**±0.00 **0.21**±0.01	9.21±2.24 8.2±2.3	**52.83**±0.46
	50	V-U	0.82±0.04 **0.17**±0.02	0.67±0.04 11.13±2.14	**1.24**±0.81 **3.45**±1.24	79.0±5.0 **0.00**±0.00	**17.4**±4.3 3.6±2.3	**0.02**±0.00 **0.21**±0.01	9.24±2.29 7.1±2.2	53.11±0.43
		V-Δ	0.82±0.04 **0.17**±0.02	0.67±0.04 11.17±2.17	**1.24**±0.80 3.48±1.26	79.1±5.0 **0.00**±0.00	**17.4**±4.3 3.5±2.3	**0.02**±0.00 **0.21**±0.01	9.24±2.29 7.1±2.2	53.12±0.44
		V-L	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 10.98±2.25	1.84±1.12 4.54±1.77	**79.7**±4.3 **0.00**±0.00	17.5±4.3 2.9±0.8	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	**52.83**±0.46
		V-C	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 11.01±2.27	1.83±1.11 4.56±1.77	**79.7**±4.3 **0.00**±0.00	17.5±4.3 **2.8**±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	52.84±0.46
	100	V-U	0.81±0.04 **0.17**±0.02	0.67±0.04 6.65±1.36	**1.33**±0.80 **3.56**±1.28	79.0±5.1 **0.00**±0.00	**17.4**±4.3 3.6±2.3	**0.02**±0.00 **0.21**±0.01	9.24±2.29 7.2±2.3	53.08±0.42
		V-Δ	0.81±0.04 **0.17**±0.02	0.67±0.04 6.68±1.37	**1.33**±0.78 3.58±1.27	79.2±5.0 **0.00**±0.00	**17.4**±4.3 3.4±2.3	**0.02**±0.00 **0.21**±0.01	9.24±2.29 7.2±2.3	53.08±0.41
		V-L	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 6.69±1.45	1.84±1.11 4.55±1.74	**79.7**±4.3 **0.00**±0.00	17.5±4.3 2.9±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.3±2.3	**52.83**±0.46
		V-C	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 6.72±1.45	1.82±1.10 4.56±1.74	**79.7**±4.3 **0.00**±0.00	17.5±4.3 **2.8**±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.3±2.3	52.84±0.47

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.025	200	V-U	0.88±0.04 **0.17**±0.02	0.71±0.03 4.58±1.16	0.88±0.72 **2.28**±1.22	50.6±5.3 **0.00**±0.00	**17.2**±4.2 32.2±3.3	**0.02**±0.00 **0.21**±0.01	9.24±2.29 6.8±2.3	53.53±0.41
			V-Δ	0.88±0.04 **0.17**±0.02	0.71±0.03 4.60±1.18	**0.86**±0.70 2.31±1.21	50.9±5.3 **0.00**±0.00	**17.2**±4.2 31.9±3.2	**0.02**±0.00 **0.21**±0.01	9.24±2.29 6.9±2.3	53.53±0.42
			V-L	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 4.41±1.02	1.83±1.11 4.53±1.75	**79.7**±4.4 **0.00**±0.00	17.5±4.3 2.9±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	**52.84**±0.46
			V-C	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 4.42±1.04	1.81±1.10 4.53±1.75	**79.7**±4.3 **0.00**±0.00	17.5±4.3 **2.8**±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	52.85±0.47
		∞	V-U	0.89±0.04 **0.17**±0.02	0.72±0.03 3.34±0.61	0.85±0.73 **2.18**±1.23	46.6±4.3 **0.00**±0.00	**17.2**±4.2 36.2±0.6	**0.02**±0.00 **0.21**±0.01	9.24±2.29 6.6±2.3	53.56±0.41
			V-Δ	0.88±0.04 **0.17**±0.02	0.72±0.03 3.35±0.61	**0.84**±0.71 2.21±1.22	46.9±4.3 **0.00**±0.00	**17.2**±4.2 35.9±0.6	**0.02**±0.00 **0.21**±0.01	9.24±2.29 6.7±2.3	53.57±0.42
			V-L	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 3.34±0.60	1.82±1.12 4.52±1.76	**79.7**±4.4 **0.00**±0.00	17.5±4.3 2.9±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	**52.84**±0.46
			V-C	**0.78**±0.05 **0.17**±0.02	**0.65**±0.04 3.34±0.60	1.81±1.11 4.53±1.76	**79.7**±4.3 **0.00**±0.00	17.5±4.3 **2.8**±0.7	**0.02**±0.00 **0.21**±0.01	9.24±2.29 8.2±2.3	52.85±0.47
	0.05	10	V-U	0.80±0.05 **0.17**±0.02	0.66±0.04 41.20±8.94	**1.49**±0.91 **3.83**±1.39	81.8±3.9 **0.00**±0.00	**15.1**±3.8 3.1±1.3	**0.02**±0.00 **0.21**±0.01	8.00±2.05 7.4±2.4	52.97±0.47
			V-Δ	0.80±0.05 **0.17**±0.02	0.66±0.04 41.09±8.89	**1.49**±0.92 **3.83**±1.39	81.9±3.9 **0.00**±0.00	**15.1**±3.8 3.0±1.4	**0.02**±0.00 **0.21**±0.01	8.00±2.05 7.4±2.4	52.98±0.46
			V-L	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 40.92±9.04	2.16±1.27 5.13±1.99	**82.1**±3.8 **0.00**±0.00	15.2±3.8 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.00±2.05 8.6±2.5	**52.72**±0.52
			V-C	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 40.85±9.02	2.16±1.27 5.11±1.98	**82.1**±3.8 **0.00**±0.00	15.2±3.8 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.00±2.05 8.6±2.5	**52.72**±0.52
		50	V-U	0.80±0.05 **0.17**±0.02	0.66±0.04 11.64±2.31	**1.57**±0.86 **3.96**±1.34	81.6±4.3 **0.00**±0.00	**15.1**±3.9 3.3±1.7	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.7±2.4	52.97±0.44
			V-Δ	0.80±0.05 **0.17**±0.02	0.66±0.04 11.61±2.31	1.58±0.87 3.97±1.34	81.7±4.4 **0.00**±0.00	**15.1**±3.9 3.2±1.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.7±2.4	52.97±0.43
			V-L	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 11.66±2.33	2.16±1.28 5.11±2.01	**82.1**±4.0 **0.00**±0.00	15.2±3.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.7±2.4	**52.72**±0.52
			V-C	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 11.63±2.31	2.16±1.29 5.09±2.00	**82.1**±4.0 **0.00**±0.00	15.2±3.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.7±2.4	**52.72**±0.52
		100	V-U	0.79±0.05 **0.17**±0.02	0.65±0.04 7.34±1.38	**1.63**±0.89 **4.10**±1.41	81.8±4.1 **0.00**±0.00	**15.1**±3.9 3.1±1.3	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.9±2.5	52.94±0.44
			V-Δ	0.79±0.05 **0.17**±0.02	0.65±0.04 7.33±1.37	1.64±0.89 4.11±1.41	81.9±4.2 **0.00**±0.00	**15.1**±3.9 2.9±1.4	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.9±2.5	52.95±0.44
			V-L	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 7.23±1.39	2.14±1.28 5.10±2.00	82.0±4.0 **0.00**±0.00	15.2±3.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.7±2.5	**52.72**±0.52
			V-C	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 7.22±1.41	2.14±1.28 5.08±1.99	**82.1**±4.0 **0.00**±0.00	15.2±3.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.7±2.5	52.73±0.52

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.05	200	V-U	0.86±0.04 **0.17**±0.02	0.70±0.03 4.85±1.17	1.13±0.87 **2.69**±1.34	54.8±4.9 **0.00**±0.00	**15.0**±3.9 30.2±3.4	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.3±2.4	53.39±0.43
			V-Δ	0.86±0.04 **0.17**±0.02	0.70±0.03 4.84±1.18	**1.12**±0.87 2.70±1.34	55.0±4.9 **0.00**±0.00	**15.0**±3.9 30.0±3.4	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.3±2.4	53.39±0.43
			V-L	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 4.76±1.09	2.13±1.28 5.09±2.01	82.0±4.0 **0.00**±0.00	15.2±3.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.7±2.5	**52.73**±0.52
			V-C	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 4.75±1.10	2.13±1.28 5.07±2.00	**82.1**±4.0 **0.00**±0.00	15.2±3.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.6±2.5	**52.73**±0.51
		∞	V-U	0.87±0.04 **0.17**±0.02	0.71±0.03 3.47±0.63	**1.08**±0.86 **2.52**±1.33	48.9±3.9 **0.00**±0.00	**15.0**±3.9 36.1±0.7	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.2±2.5	53.44±0.43
			V-Δ	0.87±0.04 **0.17**±0.02	0.71±0.03 3.47±0.63	**1.08**±0.87 2.53±1.33	49.2±3.9 **0.00**±0.00	**15.0**±3.9 35.8±0.7	**0.02**±0.00 **0.21**±0.01	8.02±2.10 7.2±2.5	53.44±0.43
			V-L	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 3.46±0.62	2.13±1.28 5.07±2.02	82.0±4.0 **0.00**±0.00	15.2±3.9 2.8±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.6±2.5	**52.73**±0.52
			V-C	**0.76**±0.06 **0.17**±0.02	**0.63**±0.04 3.45±0.62	2.13±1.28 5.05±2.01	**82.1**±4.0 **0.00**±0.00	15.2±3.9 **2.7**±0.8	**0.02**±0.00 **0.21**±0.01	8.02±2.10 8.5±2.5	**52.73**±0.52
	0.10	10	V-U	0.77±0.05 **0.17**±0.02	0.64±0.04 44.49±8.57	1.92±1.07 **4.64**±1.68	84.4±3.8 **0.00**±0.00	**12.4**±3.4 3.2±1.6	**0.02**±0.00 **0.21**±0.01	6.58±1.83 8.4±2.8	52.84±0.43
			V-Δ	0.77±0.05 **0.17**±0.02	0.64±0.04 44.47±8.43	**1.90**±1.08 4.65±1.66	84.5±3.5 **0.00**±0.00	**12.4**±3.4 3.0±1.3	**0.02**±0.00 **0.21**±0.01	6.58±1.83 8.4±2.9	52.83±0.43
			V-L	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 43.80±8.34	2.70±1.54 6.07±2.47	**84.6**±3.5 **0.00**±0.00	12.5±3.4 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.58±1.83 9.6±2.8	52.54±0.52
			V-C	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 43.84±8.49	2.67±1.53 6.07±2.47	**84.6**±3.4 **0.00**±0.00	12.5±3.4 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.58±1.83 9.6±2.8	**52.53**±0.52
		50	V-U	0.77±0.05 **0.17**±0.02	0.64±0.04 12.63±2.21	2.06±1.07 **4.78**±1.72	84.4±3.8 **0.00**±0.00	**12.5**±3.6 3.1±1.1	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.7±2.8	52.81±0.45
			V-Δ	0.77±0.05 **0.17**±0.02	0.64±0.04 12.62±2.17	**2.05**±1.07 4.79±1.71	84.5±3.7 **0.00**±0.00	**12.5**±3.6 2.9±1.1	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.7±2.8	52.80±0.45
			V-L	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 12.41±2.19	2.71±1.52 6.04±2.45	84.5±3.6 **0.00**±0.00	12.6±3.6 2.9±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.7±2.8	**52.55**±0.53
			V-C	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 12.42±2.22	2.68±1.51 6.03±2.45	**84.6**±3.6 **0.00**±0.00	12.6±3.6 **2.8**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.7±2.8	**52.55**±0.53
		100	V-U	0.76±0.05 **0.17**±0.02	0.64±0.04 7.96±1.43	2.10±1.09 4.95±1.72	84.5±3.6 **0.00**±0.00	**12.5**±3.6 3.0±1.1	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.8±2.7	52.76±0.45
			V-Δ	0.76±0.05 **0.17**±0.02	0.64±0.04 7.96±1.42	**2.08**±1.09 **4.93**±1.72	**84.6**±3.6 **0.00**±0.00	**12.5**±3.6 **2.9**±1.1	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.8±2.7	52.75±0.45
			V-L	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 7.88±1.42	2.68±1.53 6.00±2.46	84.5±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.7±2.9	**52.55**±0.53
			V-C	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 7.90±1.44	2.65±1.53 6.00±2.46	**84.6**±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.7±2.9	**52.55**±0.53

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.10	200	V-U	0.84±0.04 **0.17**±0.02	0.68±0.03 5.36±1.17	1.44±1.02 **3.31**±1.66	58.6±4.9 **0.00**±0.00	**12.4**±3.5 29.0±3.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.3±2.9	53.28±0.43
			V-Δ	0.84±0.04 **0.17**±0.02	0.68±0.03 5.37±1.17	**1.43**±1.03 **3.31**±1.66	58.9±4.9 **0.00**±0.00	**12.4**±3.5 28.7±3.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.3±2.9	53.28±0.43
			V-L	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 5.13±1.16	2.67±1.54 5.97±2.47	84.5±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.6±2.9	**52.56**±0.53
			V-C	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 5.15±1.17	2.64±1.53 5.97±2.47	**84.6**±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.6±2.9	**52.56**±0.53
		∞	V-U	0.85±0.04 **0.17**±0.02	0.69±0.03 3.72±0.68	1.36±1.02 **3.08**±1.64	51.4±3.5 **0.00**±0.00	**12.4**±3.5 36.3±0.6	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.1±3.0	53.34±0.45
			V-Δ	0.85±0.04 **0.17**±0.02	0.69±0.03 3.72±0.68	**1.35**±1.03 **3.08**±1.64	51.6±3.5 **0.00**±0.00	**12.4**±3.5 36.0±0.6	**0.02**±0.00 **0.21**±0.01	6.61±1.91 8.1±3.0	53.34±0.45
			V-L	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 3.68±0.68	2.67±1.54 5.96±2.48	84.5±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.5±2.8	**52.56**±0.53
			V-C	**0.73**±0.06 **0.17**±0.02	**0.62**±0.04 3.68±0.68	2.64±1.53 5.96±2.48	**84.6**±3.6 **0.00**±0.00	12.6±3.6 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	6.61±1.91 9.6±2.8	**52.56**±0.53
	0.15	10	V-U	0.76±0.06 **0.17**±0.02	0.63±0.04 45.71±9.17	**2.18**±1.21 **5.13**±1.86	85.6±3.6 **0.00**±0.00	**11.2**±3.3 3.1±1.7	**0.02**±0.00 **0.21**±0.01	5.93±1.77 9.0±2.9	52.72±0.47
			V-Δ	0.76±0.05 **0.17**±0.02	0.63±0.04 45.68±9.02	2.20±1.22 5.14±1.86	85.7±3.6 **0.00**±0.00	**11.2**±3.3 3.1±1.7	**0.02**±0.00 **0.21**±0.01	5.93±1.77 9.0±2.9	52.72±0.45
			V-L	**0.71**±0.07 **0.17**±0.02	**0.61**±0.04 45.32±9.08	3.02±1.68 6.60±2.70	**85.8**±3.3 **0.00**±0.00	11.3±3.3 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.93±1.77 10.1±2.9	52.44±0.52
			V-C	**0.71**±0.07 **0.17**±0.02	**0.61**±0.04 45.27±9.01	3.04±1.67 6.62±2.72	**85.8**±3.3 **0.00**±0.00	11.3±3.3 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.93±1.77 10.1±2.9	**52.43**±0.52
		50	V-U	0.76±0.06 **0.17**±0.02	0.63±0.04 13.15±2.25	**2.22**±1.19 **5.21**±1.86	85.6±3.6 **0.00**±0.00	**11.3**±3.4 3.2±1.6	**0.02**±0.00 **0.21**±0.01	5.94±1.81 9.1±2.8	52.71±0.48
			V-Δ	0.75±0.06 **0.17**±0.02	0.63±0.04 13.13±2.20	2.24±1.19 5.23±1.88	85.6±3.6 **0.00**±0.00	**11.3**±3.4 3.1±1.5	**0.02**±0.00 **0.21**±0.01	5.94±1.81 9.1±2.8	52.70±0.48
			V-L	0.72±0.07 **0.17**±0.02	**0.61**±0.04 13.05±2.23	3.01±1.62 6.57±2.62	85.7±3.4 **0.00**±0.00	**11.3**±3.4 3.0±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.2±2.9	**52.43**±0.52
			V-C	**0.71**±0.07 **0.17**±0.02	**0.61**±0.04 13.01±2.21	3.01±1.62 6.60±2.64	**85.8**±3.4 **0.00**±0.00	**11.3**±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.2±2.9	**52.43**±0.52
		100	V-U	0.75±0.05 **0.17**±0.02	0.63±0.04 8.22±1.59	**2.34**±1.20 **5.40**±1.87	85.7±3.4 **0.00**±0.00	**11.3**±3.4 3.0±1.0	**0.02**±0.00 **0.21**±0.01	5.94±1.81 9.3±2.9	52.66±0.44
			V-Δ	0.75±0.05 **0.17**±0.02	0.63±0.04 8.21±1.59	2.35±1.20 5.42±1.89	**85.8**±3.4 **0.00**±0.00	**11.3**±3.4 3.0±0.9	**0.02**±0.00 **0.21**±0.01	5.94±1.81 9.3±2.9	52.66±0.44
			V-L	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 8.27±1.56	2.98±1.62 6.52±2.64	85.7±3.4 **0.00**±0.00	**11.3**±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.2±2.9	**52.44**±0.52
			V-C	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 8.26±1.54	2.98±1.61 6.54±2.66	**85.8**±3.4 **0.00**±0.00	**11.3**±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.2±2.9	**52.44**±0.53

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.15	200	V-U	0.82±0.05 **0.17**±0.02	0.68±0.04 5.72±1.12	**1.63**±1.09 **3.72**±1.78	60.8±4.8 **0.00**±0.00	**11.2**±3.4 28.1±3.8	**0.02**±0.00 **0.21**±0.01	5.94±1.81 8.9±2.8	53.19±0.44
			V-Δ	0.82±0.05 **0.17**±0.02	0.68±0.03 5.73±1.10	1.65±1.12 3.74±1.79	61.1±4.7 **0.00**±0.00	**11.2**±3.4 27.7±3.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 8.9±2.8	53.18±0.43
			V-L	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 5.54±1.09	2.97±1.62 6.50±2.65	85.7±3.4 **0.00**±0.00	11.3±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.1±2.9	52.45±0.53
			V-C	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 5.55±1.09	2.97±1.62 6.52±2.67	**85.8**±3.4 **0.00**±0.00	11.3±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.1±2.9	**52.44**±0.53
		∞	V-U	0.84±0.05 **0.17**±0.02	0.69±0.03 3.84±0.68	**1.56**±1.08 **3.44**±1.77	52.5±3.3 **0.00**±0.00	**11.1**±3.3 36.3±0.6	**0.02**±0.00 **0.21**±0.01	5.94±1.81 8.6±3.0	53.27±0.44
			V-Δ	0.84±0.05 **0.17**±0.02	0.69±0.03 3.84±0.68	**1.56**±1.11 3.46±1.78	52.8±3.3 **0.00**±0.00	**11.1**±3.3 36.0±0.6	**0.02**±0.00 **0.21**±0.01	5.94±1.81 8.6±3.0	53.26±0.43
			V-L	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 3.81±0.69	2.97±1.62 6.49±2.65	85.7±3.4 **0.00**±0.00	11.3±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.1±2.9	**52.45**±0.53
			V-C	**0.72**±0.07 **0.17**±0.02	**0.61**±0.04 3.81±0.69	2.97±1.62 6.51±2.67	**85.8**±3.4 **0.00**±0.00	11.3±3.4 **2.9**±0.7	**0.02**±0.00 **0.21**±0.01	5.94±1.81 10.1±2.9	**52.45**±0.53
	0.20	10	V-U	0.75±0.06 **0.17**±0.02	0.63±0.04 47.23±9.26	2.45±1.35 **5.55**±2.17	86.7±3.4 **0.00**±0.06	**10.3**±3.1 3.1±1.5	**0.02**±0.00 **0.21**±0.01	5.41±1.64 9.6±3.3	52.65±0.49
			V-Δ	0.74±0.06 **0.17**±0.02	0.63±0.04 47.22±9.31	**2.44**±1.34 5.58±2.18	86.7±3.4 **0.00**±0.06	**10.3**±3.1 3.0±1.5	**0.02**±0.00 **0.21**±0.01	5.41±1.64 9.6±3.3	52.64±0.49
			V-L	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 46.59±8.93	3.39±1.89 7.18±3.04	86.7±3.2 **0.00**±0.06	**10.3**±3.1 3.0±0.8	**0.02**±0.00 **0.21**±0.01	5.41±1.64 10.8±3.2	**52.33**±0.58
			V-C	**0.70**±0.08 **0.17**±0.02	**0.60**±0.04 46.55±8.91	3.38±1.87 7.21±3.05	**86.8**±3.2 **0.00**±0.06	**10.3**±3.1 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	5.41±1.64 10.8±3.3	**52.33**±0.59
		50	V-U	0.74±0.06 **0.17**±0.02	0.62±0.04 13.66±2.25	2.54±1.31 **5.79**±2.12	86.7±3.3 **0.00**±0.06	**10.3**±3.2 3.0±1.1	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.9±3.2	52.58±0.48
			V-Δ	0.74±0.06 **0.17**±0.02	0.62±0.04 13.63±2.28	**2.51**±1.31 5.81±2.13	**86.8**±3.3 **0.00**±0.06	**10.3**±3.2 2.9±1.1	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.9±3.2	52.58±0.49
			V-L	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 13.51±2.24	3.35±1.87 7.14±3.00	86.7±3.2 **0.00**±0.06	10.4±3.2 3.0±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.8±3.2	52.34±0.57
			V-C	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 13.47±2.22	3.35±1.85 7.15±3.01	**86.8**±3.2 **0.00**±0.06	10.4±3.2 **2.8**±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.8±3.3	**52.33**±0.58
		100	V-U	0.73±0.06 **0.17**±0.02	0.62±0.04 8.45±1.42	2.67±1.38 **5.95**±2.20	86.7±3.3 **0.00**±0.06	**10.3**±3.2 3.0±0.9	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.9±3.4	52.56±0.49
			V-Δ	0.73±0.06 **0.17**±0.02	0.62±0.04 8.44±1.45	**2.64**±1.37 5.97±2.21	**86.8**±3.3 **0.00**±0.06	**10.3**±3.2 **2.9**±0.9	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.9±3.4	52.56±0.50
			V-L	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 8.50±1.48	3.37±1.89 7.16±3.03	86.7±3.2 **0.00**±0.06	10.4±3.2 3.0±0.7	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.9±3.3	52.33±0.57
			V-C	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 8.49±1.49	3.36±1.88 7.17±3.04	**86.8**±3.2 **0.00**±0.06	10.4±3.2 **2.9**±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.9±3.3	**52.32**±0.58

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$								
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.20	200	V-U	0.81±0.05 **0.17**±0.02	0.67±0.04 5.91±1.20	**1.87**±1.24 **4.15**±2.13	62.6±5.1 **0.00**±0.05	**10.2**±3.1 27.2±4.2	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.5±3.3	53.10±0.44
			V-Δ	0.81±0.05 **0.17**±0.02	0.67±0.04 5.92±1.18	**1.87**±1.26 4.17±2.14	62.9±5.0 **0.00**±0.05	**10.2**±3.1 26.9±4.1	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.5±3.3	53.10±0.46
			V-L	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 5.78±1.14	3.31±1.89 7.06±3.03	86.7±3.2 **0.00**±0.06	10.4±3.2 2.9±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.7±3.3	**52.34**±0.58
			V-C	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 5.77±1.15	3.31±1.87 7.07±3.04	**86.8**±3.2 **0.00**±0.06	10.4±3.2 **2.8**±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.7±3.3	**52.34**±0.58
		∞	V-U	0.83±0.05 **0.17**±0.02	0.68±0.04 3.96±0.73	**1.75**±1.23 **3.82**±2.09	53.5±3.1 **0.00**±0.05	**10.2**±3.1 36.3±0.6	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.2±3.5	53.22±0.47
			V-Δ	0.83±0.05 **0.17**±0.02	0.68±0.03 3.96±0.73	**1.75**±1.24 3.83±2.10	53.8±3.1 **0.00**±0.05	**10.2**±3.1 36.0±0.6	**0.02**±0.00 **0.21**±0.01	5.42±1.68 9.2±3.4	53.21±0.48
			V-L	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 3.93±0.73	3.31±1.89 7.05±3.04	86.7±3.2 **0.00**±0.06	10.4±3.2 2.9±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.6±3.3	**52.34**±0.57
			V-C	**0.70**±0.08 **0.17**±0.02	**0.60**±0.05 3.93±0.74	3.30±1.87 7.07±3.04	**86.8**±3.2 **0.00**±0.06	10.4±3.2 **2.8**±0.8	**0.02**±0.00 **0.21**±0.01	5.42±1.68 10.6±3.3	**52.34**±0.58
0.20	0.025	10	V-U	0.87±0.04 **0.17**±0.02	0.71±0.03 61.38±9.44	**0.74**±0.56 **2.17**±0.82	68.1±11.9 **0.00**±0.00	**12.6**±2.5 19.3±11.2	**0.02**±0.00 **0.21**±0.01	6.78±1.55 7.5±2.9	53.50±0.40
			V-Δ	0.87±0.04 **0.17**±0.02	0.71±0.03 61.12±9.47	0.75±0.58 2.21±0.78	68.5±11.7 **0.00**±0.00	**12.6**±2.5 18.9±11.0	**0.02**±0.00 **0.21**±0.01	6.78±1.55 7.6±2.8	53.49±0.40
			V-L	0.70±0.05 **0.17**±0.02	**0.59**±0.04 58.79±9.36	3.37±1.38 7.23±2.07	84.1±2.6 **0.00**±0.00	12.9±2.6 3.0±0.8	**0.02**±0.00 **0.21**±0.01	6.75±1.35 11.7±2.8	52.28±0.46
			V-C	**0.69**±0.06 **0.17**±0.02	**0.59**±0.04 58.63±9.32	3.67±1.58 7.74±2.38	**84.2**±2.5 **0.00**±0.00	12.9±2.6 **2.9**±0.7	0.01±0.00 **0.21**±0.01	6.75±1.35 12.4±2.8	**52.20**±0.47
		50	V-U	0.91±0.02 **0.17**±0.02	0.74±0.03 16.40±2.36	**0.57**±0.47 **1.49**±0.68	43.8±6.2 **0.00**±0.00	**12.5**±2.5 43.6±5.8	**0.02**±0.00 **0.21**±0.01	6.80±1.61 7.7±2.8	53.80±0.34
			V-Δ	0.91±0.02 **0.17**±0.02	0.73±0.03 16.32±2.38	**0.57**±0.47 1.52±0.67	45.3±6.6 **0.00**±0.00	12.6±2.5 42.2±6.1	**0.02**±0.00 **0.21**±0.01	6.80±1.61 7.7±2.7	53.75±0.34
			V-L	0.70±0.05 **0.17**±0.02	**0.59**±0.04 15.76±2.14	3.32±1.33 7.18±2.03	84.1±2.6 **0.00**±0.00	12.9±2.6 3.0±0.8	**0.02**±0.00 **0.21**±0.01	6.75±1.36 11.8±2.8	52.28±0.46
			V-C	**0.69**±0.06 **0.17**±0.02	**0.59**±0.04 15.74±2.27	3.60±1.57 7.65±2.33	**84.2**±2.6 **0.00**±0.00	12.9±2.6 **2.9**±0.7	0.01±0.00 **0.21**±0.01	6.75±1.36 12.3±2.7	**52.21**±0.47
		100	V-U	0.95±0.02 **0.17**±0.02	0.75±0.03 9.94±1.50	**0.39**±0.40 **0.93**±0.64	26.0±4.5 **0.00**±0.00	**12.5**±2.5 61.5±4.0	**0.02**±0.00 **0.21**±0.01	6.91±2.49 7.3±3.0	54.00±0.34
			V-Δ	0.95±0.02 **0.17**±0.02	0.75±0.03 9.93±1.49	**0.39**±0.40 0.97±0.64	26.5±4.4 **0.00**±0.00	**12.5**±2.5 61.0±3.8	**0.02**±0.00 **0.21**±0.01	6.91±2.49 7.3±2.9	53.97±0.33
			V-L	0.71±0.05 **0.17**±0.02	0.60±0.04 9.57±1.41	3.23±1.28 6.87±1.92	84.1±2.6 **0.00**±0.00	12.9±2.6 3.0±0.7	**0.02**±0.00 **0.21**±0.01	6.75±1.36 11.7±2.8	52.32±0.45
			V-C	**0.69**±0.06 **0.17**±0.02	**0.59**±0.04 9.53±1.42	3.60±1.57 7.63±2.35	**84.2**±2.5 **0.00**±0.00	12.9±2.6 **2.9**±0.7	0.01±0.00 **0.21**±0.01	6.75±1.36 12.3±2.7	**52.21**±0.48

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^{+}_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\text{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.20	0.025	200	V-U	0.96±0.01 **0.17**±0.02	0.77±0.02 6.64±1.20	**0.28**±0.33 **0.70**±0.59	18.2±3.3 **0.00**±0.00	**12.5**±2.5 69.4±2.5	**0.03**±0.00 **0.21**±0.01	7.11±4.36 6.5±3.1	54.15±0.36
			V-Δ	0.96±0.01 **0.17**±0.02	0.77±0.02 6.64±1.20	**0.28**±0.33 **0.70**±0.59	18.2±3.3 **0.00**±0.00	**12.5**±2.5 69.4±2.5	**0.03**±0.00 **0.21**±0.01	7.11±4.36 6.5±3.1	54.15±0.36
			V-L	0.77±0.05 **0.17**±0.02	0.62±0.04 6.37±1.11	2.49±1.21 5.32±1.87	68.9±3.6 **0.00**±0.00	12.8±2.6 18.3±2.8	0.02±0.00 **0.21**±0.01	6.75±1.36 11.4±2.9	52.67±0.45
			V-C	**0.69**±0.06 **0.17**±0.02	**0.59**±0.04 6.32±1.10	3.59±1.57 7.64±2.35	**84.2**±2.5 **0.00**±0.00	12.9±2.6 **2.9**±0.7	0.01±0.00 **0.21**±0.01	6.75±1.36 12.4±2.8	**52.22**±0.48
		∞	V-U	0.98±0.01 **0.17**±0.02	0.79±0.02 4.32±0.63	**0.18**±0.28 **0.42**±0.45	10.0±2.3 **0.00**±0.00	**12.4**±2.5 77.5±0.4	**0.03**±0.00 **0.21**±0.01	6.69±1.19 5.5±3.3	54.27±0.37
			V-Δ	0.98±0.01 **0.17**±0.02	0.79±0.02 4.32±0.63	**0.18**±0.28 **0.42**±0.45	10.0±2.3 **0.00**±0.00	**12.4**±2.5 77.5±0.4	**0.03**±0.00 **0.21**±0.01	6.69±1.19 5.5±3.3	54.27±0.37
			V-L	0.78±0.05 **0.17**±0.02	0.63±0.04 4.36±0.59	2.43±1.21 5.06±1.89	61.7±2.7 **0.00**±0.00	12.8±2.6 25.5±0.9	0.02±0.00 **0.21**±0.01	6.75±1.36 11.2±3.0	52.73±0.44
			V-C	**0.69**±0.06 **0.17**±0.02	**0.59**±0.04 4.33±0.58	3.59±1.57 7.64±2.35	**84.2**±2.5 **0.00**±0.00	12.9±2.6 **2.9**±0.7	0.01±0.00 **0.21**±0.01	6.75±1.36 12.3±2.7	**52.22**±0.48
	0.05	10	V-U	0.85±0.04 **0.17**±0.02	0.70±0.04 64.45±9.67	1.07±0.61 **2.80**±0.92	74.5±10.3 **0.00**±0.00	**11.0**±2.3 14.5±9.7	**0.02**±0.00 **0.21**±0.01	5.90±1.49 8.5±3.0	53.36±0.39
			V-Δ	0.85±0.04 **0.17**±0.02	0.70±0.03 64.27±9.61	**1.05**±0.56 2.82±0.91	75.5±9.7 **0.00**±0.00	**11.0**±2.3 13.5±9.2	**0.02**±0.00 **0.21**±0.01	5.90±1.49 8.4±3.0	53.36±0.38
			V-L	0.68±0.06 **0.17**±0.02	**0.58**±0.04 61.33±9.07	3.68±1.50 7.88±2.16	85.8±2.4 **0.00**±0.00	11.2±2.4 **2.9**±0.9	0.01±0.00 **0.21**±0.01	5.87±1.26 12.4±2.9	52.19±0.46
			V-C	**0.66**±0.06 **0.17**±0.02	**0.58**±0.04 61.05±8.83	4.06±1.74 8.51±2.54	**85.9**±2.4 **0.00**±0.00	11.3±2.4 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.87±1.26 13.2±3.0	**52.09**±0.49
		50	V-U	0.89±0.03 **0.17**±0.02	0.72±0.03 17.02±2.48	**0.81**±0.58 **2.06**±0.87	51.9±6.3 **0.00**±0.00	**10.9**±2.4 37.2±6.0	**0.02**±0.00 **0.21**±0.01	5.92±1.54 8.8±3.1	53.65±0.37
			V-Δ	0.89±0.03 **0.17**±0.02	0.72±0.03 17.06±2.46	**0.81**±0.57 2.07±0.89	53.2±6.2 **0.00**±0.00	11.0±2.4 35.8±5.8	**0.02**±0.00 **0.21**±0.01	5.92±1.54 8.9±3.0	53.58±0.35
			V-L	0.68±0.06 **0.17**±0.02	**0.58**±0.04 16.57±2.27	3.75±1.51 7.90±2.23	85.7±2.4 **0.00**±0.00	11.3±2.4 3.0±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 12.6±3.0	52.16±0.48
			V-C	**0.67**±0.06 **0.17**±0.02	**0.58**±0.04 16.49±2.18	4.04±1.74 8.48±2.57	**85.8**±2.4 **0.00**±0.00	11.3±2.4 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 13.3±3.0	**52.08**±0.49
		100	V-U	0.93±0.02 **0.17**±0.02	0.74±0.02 10.54±1.61	**0.55**±0.50 **1.29**±0.79	30.9±4.3 **0.00**±0.00	**10.9**±2.3 58.2±3.7	**0.02**±0.00 **0.21**±0.01	6.03±2.45 8.1±3.3	53.92±0.37
			V-Δ	0.93±0.02 **0.17**±0.02	0.74±0.02 10.61±1.60	**0.55**±0.51 1.31±0.81	31.3±4.5 **0.00**±0.00	**10.9**±2.3 57.8±3.8	**0.02**±0.00 **0.21**±0.01	6.03±2.45 8.2±3.3	53.90±0.38
			V-L	0.69±0.06 **0.17**±0.02	**0.58**±0.04 10.24±1.47	3.60±1.41 7.62±2.11	**85.8**±2.4 **0.00**±0.00	11.2±2.4 3.0±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 12.5±3.1	52.20±0.47
			V-C	**0.67**±0.06 **0.17**±0.02	**0.58**±0.04 10.14±1.43	4.02±1.72 8.47±2.54	**85.8**±2.4 **0.00**±0.00	11.3±2.4 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 13.3±3.1	**52.08**±0.49

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.05	0.05	200	V-U	0.95±0.02 **0.17**±0.02	0.77±0.02 7.05±1.22	**0.42**±0.47 **0.95**±0.69	21.2±3.2 **0.00**±0.00	**10.8**±2.3 68.0±2.4	**0.03**±0.00 **0.21**±0.01	6.23±4.34 7.8±3.4	54.11±0.36
			V-Δ	0.95±0.02 **0.17**±0.02	0.77±0.02 7.05±1.22	**0.42**±0.47 **0.95**±0.69	21.2±3.2 **0.00**±0.00	**10.8**±2.3 67.9±2.4	**0.03**±0.00 **0.21**±0.01	6.23±4.33 7.8±3.4	54.11±0.37
			V-L	0.75±0.05 **0.17**±0.02	0.61±0.04 6.76±1.17	2.91±1.36 6.05±2.07	72.2±3.5 **0.00**±0.00	11.2±2.4 16.6±2.8	0.01±0.00 **0.21**±0.01	5.87±1.27 12.1±3.1	52.53±0.46
			V-C	**0.67**±0.06 **0.17**±0.02	**0.58**±0.04 6.65±1.14	4.01±1.73 8.42±2.56	**85.8**±2.4 **0.00**±0.00	11.3±2.4 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 13.2±3.0	**52.09**±0.49
		∞	V-U	0.97±0.01 **0.17**±0.02	0.79±0.02 4.45±0.66	**0.28**±0.39 **0.59**±0.58	11.6±2.3 **0.00**±0.00	**10.8**±2.3 77.6±0.3	**0.03**±0.00 **0.21**±0.01	5.81±1.09 6.2±3.7	54.24±0.37
			V-Δ	0.97±0.01 **0.17**±0.02	0.79±0.02 4.45±0.66	**0.28**±0.39 **0.59**±0.58	11.6±2.3 **0.00**±0.00	**10.8**±2.3 77.6±0.3	**0.03**±0.00 **0.21**±0.01	5.81±1.09 6.2±3.7	54.24±0.37
			V-L	0.76±0.05 **0.17**±0.02	0.62±0.04 4.51±0.64	2.79±1.34 5.70±2.06	63.5±2.6 **0.00**±0.00	11.2±2.4 25.4±0.9	0.01±0.00 **0.21**±0.01	5.87±1.27 11.9±3.1	52.63±0.48
			V-C	**0.67**±0.06 **0.17**±0.02	**0.58**±0.04 4.51±0.63	4.01±1.73 8.42±2.56	**85.8**±2.4 **0.00**±0.00	11.3±2.4 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.87±1.27 13.2±3.0	**52.09**±0.49
0.20	0.10	10	V-U	0.82±0.04 **0.17**±0.02	0.68±0.03 67.10±10.19	**1.34**±0.75 **3.43**±1.01	80.4±7.3 **0.00**±0.00	**9.2**±2.0 10.4±7.1	**0.02**±0.00 **0.21**±0.01	4.93±1.09 9.6±3.5	53.17±0.38
			V-Δ	0.82±0.04 **0.17**±0.02	0.68±0.03 67.34±10.23	1.38±0.73 3.47±1.00	81.2±6.8 **0.00**±0.00	**9.2**±2.0 9.6±6.6	**0.02**±0.00 **0.21**±0.01	4.93±1.09 9.7±3.5	53.19±0.40
			V-L	0.65±0.06 **0.17**±0.02	0.57±0.04 64.27±8.95	4.26±1.75 8.86±2.55	**87.6**±2.1 **0.00**±0.06	9.4±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 13.3±3.4	52.02±0.51
			V-C	**0.64**±0.07 **0.17**±0.02	**0.56**±0.04 63.88±9.10	4.60±1.93 9.55±2.94	**87.6**±2.2 **0.00**±0.06	9.5±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 14.1±3.5	**51.94**±0.54
		50	V-U	0.86±0.03 **0.17**±0.02	0.70±0.03 17.87±2.49	**1.10**±0.64 **2.77**±0.99	60.0±6.0 **0.00**±0.00	**9.2**±2.0 30.8±5.6	**0.02**±0.00 **0.21**±0.01	4.93±1.09 10.2±3.5	53.41±0.32
			V-Δ	0.86±0.03 **0.17**±0.02	0.70±0.03 17.86±2.48	1.16±0.65 2.82±1.02	61.6±5.7 **0.00**±0.00	**9.2**±2.0 29.2±5.3	**0.02**±0.00 **0.21**±0.01	4.93±1.09 10.3±3.4	53.41±0.35
			V-L	0.65±0.06 **0.17**±0.02	**0.57**±0.04 17.47±2.15	4.26±1.75 8.91±2.56	**87.6**±2.1 **0.00**±0.06	9.4±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 13.4±3.4	52.01±0.49
			V-C	**0.64**±0.07 **0.17**±0.02	**0.57**±0.04 17.32±2.18	4.57±1.91 9.53±2.93	87.5±2.2 **0.00**±0.06	9.5±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 14.1±3.5	**51.94**±0.53
		100	V-U	0.91±0.03 **0.17**±0.02	0.73±0.03 11.02±1.60	**0.79**±0.60 **1.80**±0.93	35.6±3.7 **0.00**±0.00	**9.1**±2.0 55.2±3.3	**0.02**±0.00 **0.21**±0.01	4.93±1.09 9.5±3.7	53.80±0.33
			V-Δ	0.91±0.03 **0.17**±0.02	0.73±0.03 11.03±1.59	0.80±0.61 1.82±0.95	36.1±3.7 **0.00**±0.00	**9.1**±2.0 54.8±3.2	**0.02**±0.00 **0.21**±0.01	4.93±1.09 9.5±3.6	53.79±0.33
			V-L	0.66±0.06 **0.17**±0.02	0.57±0.04 10.65±1.47	4.09±1.62 8.63±2.48	87.5±2.1 **0.00**±0.06	9.4±2.0 3.1±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 13.3±3.4	52.06±0.48
			V-C	**0.64**±0.07 **0.17**±0.02	**0.56**±0.04 10.53±1.46	4.56±1.89 9.49±2.93	**87.6**±2.2 **0.00**±0.06	9.5±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 14.1±3.5	**51.96**±0.54

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.20	0.10	200	V-U	0.94±0.02 **0.17**±0.02	0.76±0.03 7.50±1.30	**0.57**±0.49 **1.30**±0.82	24.5±2.9 **0.00**±0.00	**9.1**±2.0 66.5±2.3	**0.02**±0.00 **0.21**±0.01	4.93±1.09 8.9±3.8	54.03±0.32
			V-Δ	0.94±0.02 **0.17**±0.02	0.76±0.03 7.49±1.29	**0.57**±0.49 1.31±0.82	24.5±2.9 **0.00**±0.00	**9.1**±2.0 66.4±2.3	**0.02**±0.00 **0.21**±0.01	4.93±1.09 8.9±3.8	54.03±0.33
			V-L	0.72±0.06 **0.17**±0.02	0.59±0.04 7.20±1.21	3.35±1.59 6.95±2.35	75.7±3.3 **0.00**±0.06	9.4±2.0 15.0±2.6	0.01±0.00 **0.21**±0.01	4.93±1.09 13.0±3.5	52.38±0.48
			V-C	**0.64**±0.07 **0.17**±0.02	**0.57**±0.04 7.04±1.18	4.52±1.91 9.43±2.95	**87.6**±2.2 **0.00**±0.06	9.5±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 14.0±3.5	**51.96**±0.54
		∞	V-U	0.96±0.02 **0.17**±0.02	0.78±0.02 4.67±0.71	**0.36**±0.43 **0.82**±0.67	13.3±1.9 **0.00**±0.00	**9.1**±2.0 77.6±0.3	**0.03**±0.00 **0.21**±0.01	4.93±1.09 7.7±4.3	54.21±0.34
			V-Δ	0.96±0.02 **0.17**±0.02	0.78±0.02 4.67±0.71	**0.36**±0.43 **0.82**±0.67	13.3±1.9 **0.00**±0.00	**9.1**±2.0 77.6±0.3	**0.03**±0.00 **0.21**±0.01	4.93±1.09 7.7±4.3	54.21±0.34
			V-L	0.74±0.06 **0.17**±0.02	0.61±0.04 4.67±0.71	3.19±1.57 6.49±2.38	65.3±2.2 **0.00**±0.06	9.4±2.0 25.3±1.0	0.01±0.00 **0.21**±0.01	4.93±1.09 12.8±3.5	52.51±0.50
			V-C	**0.64**±0.07 **0.17**±0.02	**0.57**±0.04 4.66±0.72	4.52±1.91 9.42±2.95	**87.6**±2.2 **0.00**±0.06	9.5±2.0 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.93±1.09 13.9±3.5	**51.96**±0.54
	0.15	10	V-U	0.80±0.04 **0.17**±0.02	0.67±0.03 68.61±10.03	**1.54**±0.78 **3.90**±1.09	83.0±7.0 **0.00**±0.00	**8.1**±2.0 8.9±6.8	**0.02**±0.00 **0.21**±0.01	4.30±1.08 10.5±3.7	53.10±0.40
			V-Δ	0.80±0.04 **0.17**±0.02	0.67±0.03 68.77±10.10	1.57±0.73 3.91±1.07	83.4±6.6 **0.00**±0.00	**8.1**±2.0 8.5±6.4	**0.02**±0.00 **0.21**±0.01	4.30±1.08 10.4±3.7	53.09±0.41
			V-L	0.64±0.07 **0.17**±0.02	0.56±0.04 66.45±9.06	4.63±1.90 9.46±2.73	88.6±2.2 **0.00**±0.06	8.2±2.1 3.1±0.8	0.01±0.00 **0.21**±0.01	4.30±1.08 14.0±3.5	51.91±0.55
			V-C	**0.62**±0.07 **0.17**±0.02	**0.55**±0.04 66.06±9.15	5.12±2.18 10.22±3.21	**88.8**±2.1 **0.00**±0.06	8.3±2.1 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.30±1.08 14.9±3.5	**51.82**±0.57
		50	V-U	0.84±0.04 **0.17**±0.02	0.69±0.03 18.43±2.46	**1.38**±0.81 **3.32**±1.15	64.7±5.7 **0.00**±0.00	**8.0**±2.0 27.2±5.3	**0.02**±0.00 **0.21**±0.01	4.31±1.08 11.0±3.8	53.28±0.35
			V-Δ	0.84±0.04 **0.17**±0.02	0.68±0.03 18.51±2.51	**1.38**±0.80 3.35±1.15	66.1±5.7 **0.00**±0.00	**8.0**±2.0 25.8±5.3	**0.02**±0.00 **0.21**±0.01	4.31±1.08 11.0±3.8	53.29±0.37
			V-L	0.63±0.07 **0.17**±0.02	0.56±0.04 17.84±2.27	4.68±1.90 9.54±2.83	88.6±2.2 **0.00**±0.06	8.3±2.1 3.1±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 14.2±3.7	51.91±0.53
			V-C	**0.62**±0.08 **0.17**±0.02	**0.55**±0.04 17.81±2.28	5.05±2.14 10.18±3.22	**88.7**±2.1 **0.00**±0.06	8.3±2.1 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 15.0±3.7	**51.83**±0.56
		100	V-U	0.90±0.03 **0.17**±0.02	0.73±0.03 11.54±1.66	0.91±0.67 **2.11**±1.04	38.7±3.6 **0.00**±0.00	**8.0**±2.0 53.3±3.1	**0.02**±0.00 **0.21**±0.01	4.31±1.08 10.6±4.0	53.66±0.35
			V-Δ	0.90±0.03 **0.17**±0.02	0.73±0.03 11.52±1.64	**0.90**±0.68 **2.11**±1.05	39.1±3.7 **0.00**±0.00	**8.0**±2.0 52.9±3.1	**0.02**±0.00 **0.21**±0.01	4.31±1.08 10.5±4.0	53.67±0.36
			V-L	0.64±0.07 **0.17**±0.02	**0.56**±0.04 11.08±1.46	4.50±1.77 9.23±2.64	88.6±2.2 **0.00**±0.06	8.3±2.1 3.1±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 14.0±3.6	51.94±0.52
			V-C	**0.62**±0.08 **0.17**±0.02	**0.56**±0.04 11.02±1.47	5.01±2.11 10.08±3.16	**88.7**±2.2 **0.00**±0.06	8.3±2.1 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 14.8±3.5	**51.84**±0.55

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.15	0.15	200	V-U	0.93±0.02 **0.17**±0.02	0.76±0.03 7.77±1.34	**0.65**±0.57 **1.54**±0.92	26.6±3.0 **0.00**±0.00	**7.9**±2.0 65.5±2.4	**0.02**±0.00 **0.21**±0.01	4.31±1.08 9.9±4.0	53.97±0.34
			V-Δ	0.93±0.02 **0.17**±0.02	0.76±0.03 7.77±1.34	**0.65**±0.57 1.55±0.92	26.6±3.0 **0.00**±0.00	**7.9**±2.0 65.4±2.4	**0.02**±0.00 **0.21**±0.01	4.31±1.08 9.9±4.0	53.97±0.34
			V-L	0.70±0.06 **0.17**±0.02	0.58±0.04 7.47±1.18	3.73±1.75 7.54±2.60	77.8±3.1 **0.00**±0.06	8.2±2.1 14.1±2.3	0.01±0.00 **0.21**±0.01	4.31±1.08 13.7±3.7	52.27±0.50
			V-C	**0.62**±0.08 **0.17**±0.02	**0.56**±0.04 7.40±1.20	4.99±2.12 10.04±3.19	**88.7**±2.1 **0.00**±0.06	8.3±2.1 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 14.8±3.5	**51.85**±0.56
		∞	V-U	0.96±0.02 **0.17**±0.02	0.78±0.02 4.81±0.71	**0.41**±0.48 **0.97**±0.76	14.3±1.9 **0.00**±0.00	**7.9**±2.0 77.8±0.4	**0.03**±0.00 **0.21**±0.01	4.31±1.08 8.4±4.6	54.18±0.34
			V-Δ	0.96±0.02 **0.17**±0.02	0.78±0.02 4.81±0.71	**0.41**±0.48 **0.97**±0.76	14.3±1.9 **0.00**±0.00	**7.9**±2.0 77.8±0.4	**0.03**±0.00 **0.21**±0.01	4.31±1.08 8.4±4.6	54.18±0.34
			V-L	0.73±0.06 **0.17**±0.02	0.60±0.04 4.82±0.72	3.53±1.71 7.00±2.58	66.6±2.4 **0.00**±0.06	8.2±2.1 25.2±1.0	0.01±0.00 **0.21**±0.01	4.31±1.08 13.5±3.7	52.41±0.52
			V-C	**0.62**±0.08 **0.17**±0.02	**0.56**±0.04 4.83±0.72	4.99±2.12 10.04±3.19	**88.7**±2.1 **0.00**±0.06	8.3±2.1 **3.0**±0.8	0.01±0.00 **0.21**±0.01	4.31±1.08 14.8±3.5	**51.85**±0.56
0.20	0.20	10	V-U	0.79±0.04 **0.17**±0.02	0.66±0.03 70.05±9.82	**1.59**±0.71 **4.21**±1.16	85.1±5.7 **0.00**±0.00	**7.6**±1.8 7.4±5.5	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.1±3.8	52.99±0.44
			V-Δ	0.79±0.04 **0.17**±0.02	0.66±0.04 70.15±9.47	1.62±0.71 4.26±1.19	85.1±6.0 **0.00**±0.00	**7.6**±1.8 7.3±5.7	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.1±3.8	52.99±0.43
			V-L	0.62±0.07 **0.17**±0.02	0.56±0.04 67.48±8.99	4.88±1.86 9.85±2.79	**89.2**±2.0 **0.00**±0.06	7.7±1.9 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 14.2±3.6	51.87±0.54
			V-C	**0.61**±0.08 **0.17**±0.02	**0.55**±0.04 67.50±9.03	5.31±2.18 10.56±3.29	**89.2**±2.0 **0.00**±0.06	7.7±1.8 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 15.1±3.7	**51.79**±0.57
		50	V-U	0.82±0.04 **0.17**±0.02	0.68±0.03 18.91±2.53	**1.49**±0.82 **3.70**±1.23	67.8±5.6 **0.00**±0.00	**7.5**±1.8 24.7±5.3	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.7±3.8	53.19±0.38
			V-Δ	0.82±0.04 **0.17**±0.02	0.68±0.03 18.83±2.57	1.54±0.84 3.76±1.23	69.4±5.5 **0.00**±0.00	**7.5**±1.8 23.1±5.1	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.6±3.9	53.17±0.37
			V-L	0.62±0.07 **0.17**±0.02	**0.55**±0.04 18.19±2.35	5.04±1.93 9.98±2.87	**89.1**±2.0 **0.00**±0.06	7.7±1.9 3.2±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 14.5±3.8	51.86±0.54
			V-C	**0.61**±0.08 **0.17**±0.02	**0.55**±0.04 18.26±2.33	5.38±2.19 10.63±3.31	**89.1**±1.9 **0.00**±0.06	7.7±1.8 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 15.3±3.8	**51.79**±0.58
		100	V-U	0.89±0.03 **0.17**±0.02	0.72±0.03 11.80±1.60	**1.05**±0.72 **2.36**±1.18	40.3±3.4 **0.00**±0.00	**7.5**±1.8 52.2±3.0	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.2±3.8	53.62±0.36
			V-Δ	0.89±0.03 **0.17**±0.02	0.72±0.03 11.86±1.66	1.07±0.72 2.38±1.16	40.9±3.3 **0.00**±0.00	**7.5**±1.8 51.6±2.9	**0.02**±0.00 **0.21**±0.01	4.01±0.97 11.2±3.9	53.62±0.36
			V-L	0.63±0.07 **0.17**±0.02	0.56±0.04 11.26±1.62	4.83±1.81 9.71±2.76	**89.1**±2.0 **0.00**±0.06	7.7±1.8 3.2±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 14.4±3.8	51.88±0.53
			V-C	**0.61**±0.08 **0.17**±0.02	**0.55**±0.04 11.31±1.60	5.31±2.17 10.54±3.33	**89.1**±2.0 **0.00**±0.06	7.7±1.8 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 15.2±3.8	**51.79**±0.58

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.20$ $\delta = 0.20$	200	V-U	0.92±0.02 **0.17**±0.02	0.76±0.03 8.04±1.30	**0.71**±0.60 **1.73**±1.03	27.6±2.6 **0.00**±0.00	**7.4**±1.8 65.0±2.1	**0.02**±0.00 **0.21**±0.01	4.01±0.97 10.8±4.1	53.97±0.36
		V-Δ	0.92±0.02 **0.17**±0.02	0.76±0.03 8.04±1.30	0.72±0.60 **1.73**±1.03	27.6±2.6 **0.00**±0.00	**7.4**±1.8 65.0±2.1	**0.02**±0.00 **0.21**±0.01	4.01±0.97 10.8±4.1	53.97±0.36
		V-L	0.69±0.07 **0.17**±0.02	0.58±0.04 7.68±1.19	4.01±1.75 7.98±2.75	78.8±2.6 **0.00**±0.06	7.7±1.8 13.5±2.1	0.01±0.00 **0.21**±0.01	4.01±0.97 14.0±3.7	52.20±0.52
		V-C	**0.61**±0.08 **0.17**±0.02	**0.55**±0.04 7.57±1.21	5.26±2.17 10.45±3.32	**89.2**±2.0 **0.00**±0.06	7.7±1.8 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 15.0±3.7	**51.81**±0.58
	∞	V-U	0.95±0.02 **0.17**±0.02	0.78±0.03 4.91±0.72	**0.44**±0.48 **1.08**±0.82	14.8±1.7 **0.00**±0.00	**7.4**±1.8 77.8±0.4	**0.03**±0.00 **0.21**±0.01	4.01±0.97 9.4±4.5	54.18±0.34
		V-Δ	0.95±0.02 **0.17**±0.02	0.78±0.03 4.91±0.72	**0.44**±0.48 **1.08**±0.82	14.8±1.7 **0.00**±0.00	**7.4**±1.8 77.8±0.4	**0.03**±0.00 **0.21**±0.01	4.01±0.97 9.4±4.5	54.18±0.34
		V-L	0.72±0.07 **0.17**±0.02	0.60±0.04 4.86±0.77	3.78±1.73 7.36±2.65	67.1±2.1 **0.00**±0.06	7.7±1.8 25.2±1.0	0.01±0.00 **0.21**±0.01	4.01±0.97 13.8±3.7	52.35±0.53
		V-C	**0.61**±0.08 **0.17**±0.02	**0.55**±0.04 4.87±0.75	5.26±2.17 10.44±3.32	**89.2**±2.0 **0.00**±0.06	7.7±1.8 **3.1**±0.9	0.01±0.00 **0.21**±0.01	4.01±0.97 14.9±3.7	**51.81**±0.58
$\alpha = 0.25$ $\delta = 0.025$	10	V-U	1.00±0.00 **0.17**±0.02	0.81±0.02 83.68±11.09	**0.00**±0.00 **0.04**±0.12	1.1±0.7 **0.00**±0.00	**10.0**±1.6 88.9±1.0	**0.03**±0.00 **0.22**±0.01	5.00±0.00 3.6±3.2	54.44±0.32
		V-Δ	1.00±0.00 **0.17**±0.02	0.81±0.02 83.68±11.09	**0.00**±0.00 **0.04**±0.13	1.3±0.8 **0.00**±0.00	**10.0**±1.6 88.7±1.0	**0.03**±0.00 **0.22**±0.01	5.00±0.00 3.6±3.2	54.44±0.32
		V-L	0.71±0.05 **0.17**±0.02	0.60±0.03 79.66±9.60	3.23±1.06 6.76±1.39	**86.6**±1.7 **0.00**±0.00	10.4±1.6 3.0±0.7	0.01±0.00 0.21±0.01	5.45±0.85 13.0±3.4	52.37±0.40
		V-C	**0.60**±0.06 **0.17**±0.02	**0.54**±0.04 77.50±9.16	5.73±1.88 11.21±2.53	**86.6**±1.8 **0.00**±0.04	10.5±1.6 **2.9**±0.8	0.01±0.00 0.21±0.01	5.45±0.85 17.1±3.2	**51.67**±0.47
	50	V-U	1.00±0.00 **0.17**±0.02	0.81±0.02 20.98±2.56	**0.00**±0.03 **0.04**±0.12	1.3±0.7 **0.00**±0.00	**10.0**±1.6 88.7±1.0	**0.03**±0.00 **0.21**±0.01	6.56±3.89 3.9±3.2	54.44±0.31
		V-Δ	1.00±0.00 **0.17**±0.02	0.81±0.02 21.00±2.61	**0.00**±0.04 0.05±0.15	1.7±0.9 **0.00**±0.00	**10.0**±1.6 88.3±1.0	**0.03**±0.00 **0.21**±0.01	7.09±4.38 3.9±3.1	54.46±0.31
		V-L	0.72±0.04 **0.17**±0.02	0.60±0.03 20.63±2.34	3.13±1.08 6.56±1.47	86.2±2.0 **0.00**±0.00	10.4±1.6 3.4±1.3	0.01±0.00 **0.21**±0.01	5.45±0.85 13.5±3.4	52.40±0.43
		V-C	**0.60**±0.06 **0.17**±0.02	**0.54**±0.04 20.19±2.36	5.77±1.86 11.20±2.48	**86.6**±1.8 **0.00**±0.04	10.5±1.6 **2.9**±0.8	0.01±0.00 **0.21**±0.01	5.45±0.85 17.2±3.2	**51.67**±0.48
	100	V-U	1.00±0.00 **0.17**±0.01	0.81±0.02 12.86±1.58	**0.00**±0.00 **0.04**±0.12	1.3±0.7 **0.00**±0.00	**10.0**±1.6 88.7±1.0	**0.03**±0.00 **0.22**±0.01	7.85±7.25 3.9±3.1	54.43±0.32
		V-Δ	1.00±0.00 **0.17**±0.01	0.81±0.02 12.86±1.58	**0.00**±0.00 0.05±0.13	1.4±0.8 **0.00**±0.00	**10.0**±1.6 88.5±1.0	**0.03**±0.00 **0.22**±0.01	7.85±7.25 3.9±3.1	54.43±0.32
		V-L	0.81±0.04 **0.17**±0.02	0.65±0.03 12.74±1.59	2.05±0.95 4.27±1.47	58.5±3.1 **0.00**±0.00	10.3±1.6 31.2±2.7	0.02±0.00 0.21±0.01	5.45±0.85 13.2±3.5	53.01±0.40
		V-C	**0.60**±0.06 **0.17**±0.02	**0.54**±0.04 12.20±1.44	5.75±1.88 11.17±2.50	**86.6**±1.8 0.01±0.05	10.5±1.6 **2.9**±0.8	0.01±0.00 0.21±0.01	5.45±0.85 17.2±3.2	**51.67**±0.47

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.25	0.025	200	V-U	1.00±0.00 **0.17**±0.02	0.81±0.02 8.55±1.22	**0.00**±0.03 **0.04**±0.12	1.3±0.7 **0.00**±0.00	**10.0**±1.6 88.7±1.0	**0.03**±0.00 **0.22**±0.01	9.02±12.26 3.9±3.3	54.44±0.32
			V-Δ	1.00±0.00 **0.17**±0.02	0.81±0.02 8.55±1.22	**0.00**±0.03 **0.04**±0.13	1.5±0.8 **0.00**±0.00	**10.0**±1.6 88.5±1.0	**0.03**±0.00 **0.22**±0.01	10.33±13.87 3.9±3.3	54.44±0.32
			V-L	0.88±0.03 **0.17**±0.02	0.72±0.03 8.51±1.27	1.38±0.79 2.84±1.17	37.2±2.3 **0.00**±0.00	10.2±1.6 52.7±1.8	0.02±0.01 0.21±0.01	5.45±0.85 12.2±3.7	53.58±0.36
			V-C	**0.60**±0.06 **0.17**±0.02	**0.54**±0.04 8.12±1.13	5.75±1.88 11.14±2.52	**86.6**±1.8 **0.00**±0.04	10.5±1.6 **2.9**±0.8	0.01±0.00 0.21±0.01	5.45±0.85 17.1±3.2	**51.67**±0.47
		∞	V-U	1.00±0.00 **0.17**±0.02	0.81±0.02 5.29±0.59	**0.00**±0.00 **0.04**±0.12	1.1±0.7 **0.00**±0.00	**10.0**±1.6 88.9±1.0	**0.03**±0.00 **0.22**±0.01	5.00±0.00 3.6±3.2	54.44±0.32
			V-Δ	1.00±0.00 **0.17**±0.02	0.81±0.02 5.29±0.59	**0.00**±0.00 **0.04**±0.13	1.3±0.8 **0.00**±0.00	**10.0**±1.6 88.7±1.0	**0.03**±0.00 **0.22**±0.01	5.00±0.00 3.6±3.2	54.44±0.32
			V-L	0.91±0.03 **0.17**±0.02	0.74±0.03 5.33±0.66	1.14±0.72 2.21±1.04	25.1±1.6 **0.00**±0.00	10.1±1.6 64.8±0.5	0.02±0.01 0.21±0.01	5.45±0.85 11.5±3.9	53.74±0.36
			V-C	**0.60**±0.06 **0.17**±0.02	**0.54**±0.04 5.31±0.64	5.74±1.89 11.14±2.52	**86.6**±1.8 **0.00**±0.04	10.5±1.6 **2.9**±0.8	0.01±0.00 0.21±0.01	5.45±0.85 17.1±3.2	**51.67**±0.47
	0.05	10	V-U	0.99±0.00 **0.17**±0.02	0.81±0.02 86.66±11.26	**0.00**±0.03 **0.13**±0.22	2.6±1.1 **0.00**±0.00	**8.4**±1.5 89.0±0.7	**0.03**±0.00 **0.22**±0.01	4.46±0.57 5.7±4.3	54.42±0.32
			V-Δ	0.99±0.00 **0.17**±0.02	0.81±0.02 86.66±11.26	**0.00**±0.03 **0.13**±0.22	2.8±1.2 **0.00**±0.00	**8.4**±1.5 88.8±0.7	**0.03**±0.00 **0.22**±0.01	4.46±0.57 5.7±4.3	54.42±0.32
			V-L	0.68±0.05 **0.17**±0.02	0.59±0.03 82.00±9.39	3.63±1.03 7.56±1.52	**88.2**±1.7 **0.00**±0.00	8.7±1.5 3.1±0.8	0.01±0.00 0.21±0.01	4.56±0.79 13.9±3.5	52.23±0.43
			V-C	**0.58**±0.06 **0.17**±0.02	**0.53**±0.04 80.70±9.27	6.27±1.93 12.09±2.71	**88.2**±1.6 0.01±0.06	8.8±1.5 **3.0**±0.7	0.01±0.00 0.21±0.01	4.56±0.79 17.7±3.2	**51.58**±0.50
		50	V-U	0.99±0.00 **0.17**±0.02	0.81±0.02 21.76±2.52	**0.02**±0.08 **0.17**±0.25	3.5±1.3 **0.00**±0.00	**8.4**±1.5 88.1±0.9	**0.03**±0.00 **0.21**±0.01	4.99±2.92 6.3±4.1	54.43±0.31
			V-Δ	0.99±0.01 **0.17**±0.02	0.81±0.02 21.76±2.54	**0.02**±0.09 0.18±0.27	4.0±1.4 **0.00**±0.00	**8.4**±1.5 87.7±1.0	**0.03**±0.00 **0.21**±0.01	4.89±2.50 6.4±4.1	54.42±0.32
			V-L	0.69±0.05 **0.17**±0.02	0.59±0.03 21.20±2.37	3.64±1.13 7.56±1.51	88.1±1.7 **0.00**±0.03	8.7±1.5 3.2±1.0	0.01±0.00 **0.21**±0.01	4.56±0.79 14.6±3.5	52.24±0.40
			V-C	**0.58**±0.06 **0.17**±0.02	**0.53**±0.04 20.93±2.32	6.26±1.92 12.11±2.68	**88.2**±1.7 0.01±0.06	8.8±1.5 **3.0**±0.7	0.01±0.00 **0.21**±0.01	4.56±0.79 17.8±3.3	**51.57**±0.49
		100	V-U	0.99±0.00 **0.17**±0.02	0.81±0.02 13.28±1.68	**0.01**±0.06 **0.16**±0.25	3.3±1.2 **0.00**±0.00	**8.4**±1.5 88.3±0.8	**0.03**±0.00 **0.22**±0.01	5.19±3.96 6.4±4.1	54.42±0.33
			V-Δ	0.99±0.00 **0.17**±0.02	0.81±0.02 13.23±1.61	**0.01**±0.07 **0.16**±0.25	3.7±1.3 **0.00**±0.00	**8.4**±1.5 87.9±0.9	**0.03**±0.00 **0.22**±0.01	5.29±4.69 6.4±4.1	54.42±0.33
			V-L	0.79±0.04 **0.17**±0.02	0.64±0.03 13.14±1.67	2.45±1.08 5.10±1.70	62.4±2.8 **0.00**±0.00	8.6±1.5 29.0±2.3	0.01±0.00 0.21±0.01	4.56±0.79 14.3±3.4	52.84±0.37
			V-C	**0.58**±0.06 **0.17**±0.02	**0.53**±0.04 12.70±1.59	6.24±1.93 12.05±2.76	**88.2**±1.7 0.01±0.06	8.8±1.5 **3.0**±0.7	0.01±0.00 0.21±0.01	4.56±0.79 17.8±3.3	**51.58**±0.49

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
0.25	0.05	200	V-U	0.99±0.00 **0.17**±0.02	0.81±0.02 8.87±1.25	**0.01**±0.06 **0.15**±0.23	3.2±1.2 **0.00**±0.00	**8.4**±1.5 88.4±0.9	**0.03**±0.00 **0.22**±0.01	5.69±7.11 6.3±4.1	54.43±0.32
			V-Δ	0.99±0.00 **0.17**±0.02	0.81±0.02 8.88±1.26	**0.01**±0.07 **0.15**±0.23	3.5±1.2 **0.00**±0.00	**8.4**±1.5 88.1±0.9	**0.03**±0.00 **0.22**±0.01	5.69±7.09 6.2±4.1	54.43±0.32
			V-L	0.86±0.03 **0.17**±0.02	0.71±0.03 8.91±1.30	1.63±0.84 3.44±1.35	40.0±2.0 **0.00**±0.00	8.5±1.5 51.5±1.5	0.02±0.00 0.21±0.01	4.56±0.79 13.2±3.9	53.49±0.36
			V-C	**0.58**±0.06 **0.17**±0.02	**0.53**±0.04 8.39±1.16	6.22±1.94 12.02±2.76	**88.2**±1.7 0.01±0.06	8.8±1.5 **3.0**±0.7	0.01±0.00 0.21±0.01	4.56±0.79 17.7±3.3	**51.58**±0.49
		∞	V-U	0.99±0.00 **0.17**±0.02	0.81±0.02 5.38±0.63	**0.00**±0.00 **0.12**±0.22	2.5±1.1 **0.00**±0.00	**8.4**±1.5 89.2±0.7	**0.03**±0.00 **0.22**±0.01	4.44±0.52 5.6±4.2	54.44±0.32
			V-Δ	0.99±0.00 **0.17**±0.02	0.81±0.02 5.38±0.63	**0.00**±0.00 0.13±0.22	2.6±1.1 **0.00**±0.00	**8.4**±1.5 89.0±0.6	**0.03**±0.00 **0.22**±0.01	4.44±0.53 5.6±4.2	54.44±0.32
			V-L	0.90±0.03 **0.17**±0.02	0.73±0.02 5.48±0.63	1.29±0.75 2.60±1.19	26.7±1.6 **0.00**±0.00	8.5±1.5 64.8±0.5	0.02±0.01 0.21±0.01	4.56±0.79 12.4±3.9	53.70±0.36
			V-C	**0.58**±0.06 **0.17**±0.02	**0.53**±0.04 5.43±0.65	6.22±1.94 12.01±2.77	**88.2**±1.7 0.01±0.06	8.8±1.5 **3.0**±0.7	0.01±0.00 0.21±0.01	4.56±0.79 17.6±3.3	**51.58**±0.49
	0.10	10	V-U	0.99±0.01 **0.17**±0.02	0.80±0.02 90.06±11.54	**0.03**±0.12 **0.29**±0.33	5.4±1.7 **0.00**±0.00	**7.2**±1.4 87.4±1.0	**0.03**±0.00 **0.21**±0.01	3.96±1.03 7.4±4.6	54.43±0.34
			V-Δ	0.99±0.01 **0.17**±0.02	0.80±0.02 90.06±11.54	**0.03**±0.12 **0.29**±0.33	5.6±1.7 **0.00**±0.00	**7.2**±1.4 87.2±1.0	**0.03**±0.00 **0.21**±0.01	3.96±1.03 7.4±4.6	54.43±0.34
			V-L	0.66±0.05 **0.17**±0.02	0.57±0.03 85.18±9.56	4.20±1.20 8.42±1.51	**89.3**±1.6 **0.00**±0.05	7.5±1.5 3.2±0.8	0.01±0.00 **0.21**±0.01	3.93±0.77 15.0±3.7	52.11±0.45
			V-C	**0.55**±0.07 **0.17**±0.02	**0.52**±0.04 83.58±9.71	6.76±2.17 13.03±3.13	**89.3**±1.7 0.01±0.09	7.6±1.5 **3.0**±0.8	0.01±0.00 **0.21**±0.01	3.93±0.77 18.5±3.3	**51.43**±0.52
		50	V-U	0.98±0.01 **0.17**±0.02	0.80±0.02 22.49±2.81	**0.07**±0.18 **0.36**±0.41	6.6±1.7 **0.00**±0.00	**7.2**±1.4 86.2±1.4	**0.03**±0.00 **0.22**±0.01	4.41±3.34 8.2±4.7	54.38±0.30
			V-Δ	0.98±0.01 **0.17**±0.02	0.80±0.02 22.51±2.76	0.08±0.19 0.38±0.43	7.3±1.8 **0.00**±0.00	**7.2**±1.4 85.4±1.5	**0.03**±0.00 0.21±0.01	4.31±2.61 8.3±4.6	54.37±0.31
			V-L	0.66±0.05 **0.17**±0.02	0.57±0.03 22.06±2.56	4.25±1.32 8.52±1.73	**89.3**±1.6 0.01±0.06	7.5±1.5 **3.1**±0.9	0.01±0.00 0.21±0.01	3.93±0.77 15.6±3.8	52.08±0.45
			V-C	**0.55**±0.07 **0.17**±0.02	**0.52**±0.04 21.78±2.47	6.78±2.18 13.05±3.18	**89.3**±1.6 0.01±0.09	7.6±1.5 **3.1**±0.8	0.01±0.00 0.21±0.01	3.93±0.77 18.7±3.5	**51.44**±0.54
		100	V-U	0.98±0.01 **0.17**±0.01	0.80±0.02 13.86±1.72	**0.03**±0.11 **0.29**±0.34	6.1±1.6 **0.00**±0.00	**7.2**±1.4 86.7±1.3	**0.03**±0.00 **0.22**±0.01	4.65±4.66 7.7±4.5	54.39±0.33
			V-Δ	0.98±0.01 **0.17**±0.02	0.80±0.02 13.84±1.77	**0.03**±0.10 0.30±0.34	6.5±1.7 **0.00**±0.00	**7.2**±1.4 86.2±1.4	**0.03**±0.00 **0.22**±0.01	4.65±4.66 7.7±4.5	54.41±0.34
			V-L	0.76±0.04 **0.17**±0.02	0.62±0.03 13.67±1.61	2.93±1.17 5.90±1.82	66.0±2.7 **0.00**±0.05	7.5±1.5 26.6±2.3	0.01±0.00 0.21±0.01	3.93±0.77 15.6±3.8	52.71±0.40
			V-C	**0.55**±0.07 **0.17**±0.02	**0.52**±0.04 13.27±1.57	6.74±2.17 12.94±3.20	**89.3**±1.6 0.01±0.09	7.6±1.5 **3.0**±0.8	0.01±0.00 0.21±0.01	3.93±0.77 18.6±3.4	**51.46**±0.54

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.25	0.10	200	V-U	0.99±0.01 **0.17**±0.02	0.81±0.02 9.33±1.29	**0.01**±0.06 **0.28**±0.34	5.3±1.6 **0.00**±0.00	**7.2**±1.4 87.5±1.3	**0.03**±0.00 **0.22**±0.01	5.26±7.63 7.4±4.6	54.42±0.33
			V-Δ	0.99±0.01 **0.17**±0.02	0.81±0.02 9.32±1.29	**0.01**±0.07 0.29±0.34	5.6±1.6 **0.00**±0.00	**7.2**±1.4 87.2±1.3	**0.03**±0.00 **0.22**±0.01	5.26±7.63 7.4±4.6	54.42±0.33
			V-L	0.84±0.03 **0.17**±0.02	0.70±0.03 9.33±1.24	1.96±0.93 4.05±1.51	42.4±1.9 **0.00**±0.05	7.4±1.4 50.2±1.6	0.02±0.01 0.21±0.01	3.93±0.77 14.6±4.1	53.42±0.37
			V-C	**0.56**±0.07 **0.17**±0.02	**0.52**±0.04 8.82±1.15	6.71±2.18 12.88±3.20	**89.3**±1.6 0.01±0.09	7.6±1.5 **3.0**±0.8	0.01±0.00 0.21±0.01	3.93±0.77 18.4±3.4	**51.47**±0.54
		∞	V-U	0.99±0.01 **0.17**±0.02	0.81±0.02 5.57±0.65	**0.00**±0.00 **0.21**±0.29	3.4±1.1 **0.00**±0.00	**7.2**±1.4 89.3±0.5	**0.03**±0.00 **0.22**±0.01	3.81±0.50 6.4±4.9	54.44±0.32
			V-Δ	0.99±0.01 **0.17**±0.02	0.81±0.02 5.57±0.65	**0.00**±0.00 **0.21**±0.30	3.6±1.2 **0.00**±0.00	**7.2**±1.4 89.1±0.5	**0.03**±0.00 **0.22**±0.01	3.81±0.50 6.4±4.9	54.44±0.32
			V-L	0.88±0.03 **0.17**±0.02	0.73±0.03 5.58±0.66	1.53±0.84 3.02±1.33	27.8±1.5 **0.00**±0.05	7.3±1.4 64.9±0.6	0.02±0.01 0.21±0.01	3.93±0.77 13.7±4.3	53.66±0.38
			V-C	**0.56**±0.07 **0.17**±0.02	**0.52**±0.04 5.58±0.67	6.71±2.18 12.88±3.20	**89.3**±1.6 0.01±0.09	7.6±1.5 **3.0**±0.8	0.01±0.00 0.21±0.01	3.93±0.77 18.4±3.4	**51.47**±0.54
	0.15	10	V-U	0.98±0.01 **0.17**±0.02	0.80±0.02 92.34±11.68	**0.05**±0.16 **0.44**±0.38	8.3±2.1 **0.00**±0.00	**6.3**±1.1 85.4±1.6	**0.03**±0.00 **0.21**±0.01	3.54±1.11 8.6±5.2	54.39±0.33
			V-Δ	0.98±0.01 **0.17**±0.02	0.80±0.02 92.31±11.80	**0.05**±0.16 0.45±0.38	8.6±2.1 **0.00**±0.00	**6.3**±1.1 85.1±1.7	**0.03**±0.00 **0.21**±0.01	3.46±0.63 8.6±5.2	54.40±0.33
			V-L	0.64±0.05 **0.17**±0.02	0.56±0.03 87.22±9.35	4.46±1.35 9.20±1.72	**90.3**±1.4 **0.00**±0.03	6.6±1.2 3.0±0.8	0.01±0.00 **0.21**±0.01	3.45±0.62 16.0±3.8	51.97±0.43
			V-C	**0.53**±0.07 **0.17**±0.02	**0.51**±0.04 86.17±10.05	7.22±2.34 13.83±3.42	**90.3**±1.4 0.01±0.07	6.7±1.2 **2.9**±0.8	0.01±0.00 **0.21**±0.01	3.45±0.62 19.1±3.6	**51.34**±0.54
		50	V-U	0.98±0.01 **0.17**±0.02	0.80±0.02 23.00±2.66	**0.13**±0.25 **0.54**±0.46	9.4±1.9 **0.00**±0.00	**6.4**±1.1 84.3±1.5	**0.03**±0.00 **0.21**±0.01	3.72±1.99 9.6±4.9	54.36±0.34
			V-Δ	0.97±0.01 **0.17**±0.02	0.80±0.02 23.05±2.68	**0.13**±0.25 0.56±0.47	10.2±2.0 **0.00**±0.00	**6.4**±1.1 83.5±1.7	**0.03**±0.00 **0.21**±0.01	3.50±0.98 9.7±4.9	54.36±0.32
			V-L	0.64±0.05 **0.17**±0.02	0.56±0.03 22.56±2.38	4.59±1.37 9.25±1.84	**90.3**±1.5 **0.00**±0.04	6.6±1.2 3.1±0.8	0.01±0.00 **0.21**±0.01	3.45±0.62 16.7±3.9	51.93±0.45
			V-C	**0.53**±0.08 **0.17**±0.02	**0.51**±0.04 22.37±2.50	7.28±2.37 13.89±3.46	**90.3**±1.5 0.01±0.07	6.7±1.2 **2.9**±0.8	0.01±0.00 **0.21**±0.01	3.45±0.62 19.2±3.7	**51.33**±0.53
		100	V-U	0.98±0.01 **0.17**±0.02	0.81±0.02 14.28±1.74	**0.05**±0.15 **0.42**±0.44	8.1±1.8 **0.00**±0.00	**6.4**±1.1 85.6±1.5	**0.03**±0.00 **0.22**±0.01	3.92±3.38 9.2±5.0	54.37±0.34
			V-Δ	0.98±0.01 **0.17**±0.02	0.81±0.02 14.31±1.72	**0.05**±0.16 0.44±0.44	8.6±1.8 **0.00**±0.00	**6.4**±1.1 85.1±1.6	**0.03**±0.00 **0.22**±0.01	3.55±1.60 9.4±4.9	54.38±0.35
			V-L	0.74±0.05 **0.17**±0.02	0.61±0.03 13.99±1.63	3.26±1.37 6.63±2.11	68.5±2.5 **0.00**±0.04	6.6±1.2 25.0±2.2	0.01±0.00 0.21±0.01	3.45±0.62 16.6±4.0	52.56±0.43
			V-C	**0.53**±0.08 **0.17**±0.02	**0.51**±0.04 13.73±1.73	7.21±2.33 13.77±3.45	**90.3**±1.5 0.01±0.08	6.7±1.2 **2.9**±0.8	0.01±0.00 0.21±0.01	3.45±0.62 19.3±3.7	**51.33**±0.54

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$							
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\text{m},\ T_s = 0.2\,\text{s},\ N_{\text{MC}} = 200$									
α	δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\text{nom}}\|^2}\downarrow$ $\widehat{m}_{\text{cert}}$	$\widehat{q}^+_{\text{CBF}}\downarrow(\%)$ $\widehat{q}^+_Y\downarrow(\%)$	$\widehat{q}^+_{\text{use}}\uparrow(\%)$ $\widehat{q}^+_{\text{unsafe}}\downarrow(\%)$	$\widehat{q}^+_{\text{nocert}}\downarrow(\%)$ $\widehat{q}^+_{\text{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\text{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\text{m})$	$\widehat{m}_{\tau^{\text{use}}}\downarrow(\text{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\text{mis}}\downarrow(\text{s})$
0.25	0.15	200	V-U	0.98±0.01 **0.17**±0.02	0.81±0.02 9.57±1.28	**0.02**±0.09 **0.39**±0.41	6.8±1.7 **0.00**±0.00	**6.3**±1.1 86.9±1.3	**0.03**±0.00 **0.22**±0.01	4.62±7.08 8.8±5.1	54.41±0.32
			V-Δ	0.98±0.01 **0.17**±0.02	0.81±0.02 9.56±1.27	**0.02**±0.09 0.40±0.41	7.1±1.7 **0.00**±0.00	**6.3**±1.1 86.6±1.3	**0.03**±0.00 **0.22**±0.01	3.65±2.96 9.0±5.0	54.42±0.32
			V-L	0.83±0.04 **0.17**±0.02	0.70±0.03 9.56±1.18	2.15±1.02 4.48±1.74	43.9±1.9 **0.00**±0.03	6.5±1.2 49.7±1.6	0.01±0.01 0.21±0.01	3.45±0.62 15.5±4.3	53.34±0.36
			V-C	**0.53**±0.08 **0.17**±0.02	**0.51**±0.04 9.22±1.25	7.15±2.36 13.69±3.46	**90.4**±1.5 0.01±0.07	6.7±1.2 **2.9**±0.8	0.01±0.00 0.21±0.01	3.45±0.62 19.0±3.6	**51.35**±0.53
		∞	V-U	0.99±0.01 **0.17**±0.02	0.81±0.02 5.71±0.68	**0.00**±0.00 **0.28**±0.34	4.2±1.1 **0.00**±0.00	**6.3**±1.1 89.4±0.2	**0.03**±0.00 **0.22**±0.01	3.38±0.47 7.3±5.4	54.44±0.32
			V-Δ	0.99±0.01 **0.17**±0.02	0.81±0.02 5.71±0.68	**0.00**±0.00 0.29±0.34	4.4±1.1 **0.00**±0.00	**6.3**±1.1 89.2±0.2	**0.03**±0.00 **0.22**±0.01	3.44±0.60 7.4±5.4	54.44±0.32
			V-L	0.87±0.03 **0.17**±0.02	0.73±0.03 5.74±0.67	1.65±0.91 3.39±1.53	28.8±1.3 **0.00**±0.03	6.4±1.2 64.8±0.6	0.02±0.01 0.21±0.01	3.45±0.62 14.7±4.6	53.59±0.37
			V-C	**0.53**±0.08 **0.17**±0.02	**0.51**±0.04 5.68±0.72	7.15±2.36 13.68±3.47	**90.4**±1.5 0.01±0.07	6.7±1.2 **2.9**±0.8	0.01±0.00 0.21±0.01	3.45±0.62 18.9±3.6	**51.35**±0.53
	0.20	10	V-U	0.97±0.01 **0.17**±0.02	0.80±0.02 93.89±11.74	**0.09**±0.20 **0.62**±0.50	10.5±2.3 **0.00**±0.00	**5.6**±1.2 83.9±1.9	**0.03**±0.00 **0.21**±0.01	3.04±0.63 10.3±5.4	54.39±0.31
			V-Δ	0.97±0.01 **0.17**±0.02	0.80±0.02 94.07±11.89	**0.09**±0.21 **0.62**±0.50	10.9±2.4 **0.00**±0.00	**5.6**±1.2 83.5±2.0	**0.03**±0.00 **0.21**±0.01	3.04±0.63 10.4±5.4	54.36±0.30
			V-L	0.62±0.05 **0.17**±0.02	0.56±0.03 88.90±9.88	4.67±1.20 9.54±1.75	91.0±1.4 **0.00**±0.03	5.9±1.2 3.1±0.8	0.01±0.00 **0.21**±0.01	3.04±0.63 16.4±3.8	51.90±0.43
			V-C	**0.52**±0.07 **0.17**±0.02	**0.51**±0.04 87.29±10.22	7.38±2.26 14.39±3.53	**91.1**±1.4 0.01±0.06	5.9±1.2 **3.0**±0.8	0.01±0.00 **0.21**±0.01	3.04±0.63 19.4±3.5	**51.26**±0.52
		50	V-U	0.97±0.01 **0.17**±0.02	0.80±0.02 23.50±2.71	**0.15**±0.25 **0.68**±0.53	11.3±2.0 **0.00**±0.00	**5.6**±1.2 83.1±1.6	**0.03**±0.00 **0.21**±0.01	3.04±0.63 11.1±5.3	54.36±0.32
			V-Δ	0.97±0.01 **0.17**±0.02	0.80±0.02 23.44±2.67	**0.15**±0.26 0.71±0.54	12.2±2.1 **0.00**±0.00	**5.6**±1.2 82.2±1.8	**0.03**±0.00 **0.21**±0.01	3.04±0.63 11.1±5.2	54.34±0.32
			V-L	0.62±0.05 **0.17**±0.02	0.56±0.03 22.91±2.42	4.77±1.42 9.74±1.93	91.0±1.4 **0.00**±0.04	5.9±1.2 3.2±0.8	0.01±0.00 **0.21**±0.01	3.04±0.63 17.2±3.8	51.88±0.42
			V-C	**0.52**±0.07 **0.17**±0.02	**0.51**±0.04 22.83±2.54	7.40±2.27 14.52±3.57	**91.1**±1.4 0.01±0.06	5.9±1.2 **2.9**±0.8	0.01±0.00 **0.21**±0.01	3.04±0.63 19.7±3.5	**51.25**±0.52
		100	V-U	0.97±0.01 **0.17**±0.02	0.80±0.02 14.58±1.67	**0.08**±0.17 **0.57**±0.51	9.5±1.9 **0.00**±0.00	**5.6**±1.2 84.9±1.5	**0.03**±0.00 0.21±0.01	3.04±0.63 11.0±5.1	54.38±0.36
			V-Δ	0.97±0.01 **0.17**±0.02	0.80±0.02 14.57±1.69	0.09±0.18 0.59±0.53	10.0±1.9 **0.00**±0.00	**5.6**±1.2 84.4±1.5	**0.03**±0.00 **0.22**±0.01	3.04±0.63 11.0±5.1	54.37±0.34
			V-L	0.72±0.05 **0.17**±0.02	0.60±0.03 14.17±1.59	3.37±1.39 7.16±2.17	70.3±2.4 0.01±0.05	5.8±1.2 24.0±2.0	0.01±0.00 0.21±0.01	3.04±0.63 17.1±3.9	52.50±0.45
			V-C	**0.52**±0.07 **0.17**±0.02	**0.51**±0.04 14.03±1.74	7.42±2.27 14.45±3.56	**91.1**±1.4 0.01±0.06	5.9±1.2 **3.0**±0.8	0.01±0.00 0.21±0.01	3.04±0.63 19.8±3.6	**51.25**±0.52

Table IV (continued).

Maze **Baselines** $\alpha_0 = 0.30,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$								
Method	$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
Unfiltered	N-A 0.17±0.02	0.00±0.00	55.68±1.58 N-A	N-A 53.27±0.95	−0.08±0.00 0.21±0.01	N-A N-A	42.16±0.14
$c = 0$	0.00±0.00 0.17±0.02	0.32±0.02	24.93±2.57 49.79±2.97	100.0±0.0 3.86±2.16	−0.01±0.00 0.21±0.01	0.00±0.00 66.8±12.7	48.67±0.27
$c_{\max}$	1.00±0.00 0.17±0.02	0.81±0.02	0.00±0.00 0.00±0.00	0.0±0.0 0.00±0.00	0.03±0.00 0.22±0.01	N-A N-A	54.44±0.32
ACI	0.22±0.04 0.17±0.02	0.39±0.02	15.84±1.93 29.86±1.85	98.8±0.0 0.48±0.65	−0.00±0.00 0.21±0.01	0.60±0.00 47.1±8.6	49.71±0.32
QT	0.50±0.03 0.17±0.02	0.50±0.02	8.04±1.31 15.68±1.10	99.6±0.0 0.05±0.17	0.01±0.00 0.21±0.01	0.20±0.00 33.8±7.6	51.18±0.32

Maze $\alpha_0 = 0.30,\ \phi = 1,\ L_\mu = 0.0005,\ c_{\max} = 0.01\,\mathrm{m},\ T_s = 0.2\,\mathrm{s},\ N_{\mathrm{MC}} = 200$									
α δ	K		$\widehat{m}_c/c_{\max}\downarrow$ $\widehat{m}_Y/c_{\max}\downarrow$	$\widehat{m}_{\|u-u^{\mathrm{nom}}\|^2}\downarrow$ $\widehat{m}_{\mathrm{cert}}$	$\widehat{q}^{+}_{\mathrm{CBF}}\downarrow(\%)$ $\widehat{q}^{+}_{Y}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{use}}\uparrow(\%)$ $\widehat{q}^{+}_{\mathrm{unsafe}}\downarrow(\%)$	$\widehat{q}^{+}_{\mathrm{nocert}}\downarrow(\%)$ $\widehat{q}^{+}_{\mathrm{nodep}}\downarrow(\%)$	$\widehat{m}_{h^{\min}}\uparrow(\mathrm{m})$ $\widehat{m}_{\psi^{\min}}\uparrow(\mathrm{m})$	$\widehat{m}_{\tau^{\mathrm{use}}}\downarrow(\mathrm{s})$ $\widehat{m}_{p^{\max}}\downarrow(\%)$	$\widehat{T}_{\mathrm{mis}}\downarrow(\mathrm{s})$
$\alpha = 0.25$ $\delta = 0.20$	200	V-U	0.98±0.01 **0.17**±0.02	0.81±0.02 9.75±1.36	**0.06**±0.17 **0.54**±0.50	8.0±1.6 **0.00**±0.00	**5.6**±1.2 86.4±1.2	**0.03**±0.00 **0.22**±0.01	3.04±0.63 10.7±5.4	54.41±0.31
		V-Δ	0.98±0.01 **0.17**±0.02	0.81±0.02 9.75±1.36	**0.06**±0.17 0.56±0.51	8.3±1.6 **0.00**±0.00	**5.6**±1.2 86.1±1.2	**0.03**±0.00 **0.22**±0.01	3.04±0.63 10.8±5.4	54.40±0.32
		V-L	0.81±0.04 **0.17**±0.02	0.70±0.03 9.75±1.28	2.22±1.09 4.83±1.80	45.1±1.7 **0.00**±0.03	5.7±1.2 49.2±1.4	0.01±0.01 0.21±0.01	3.04±0.63 16.0±4.3	53.30±0.34
		V-C	**0.52**±0.08 **0.17**±0.02	**0.51**±0.04 9.35±1.29	7.32±2.30 14.31±3.61	**91.1**±1.4 0.01±0.06	5.9±1.2 **2.9**±0.8	0.01±0.00 0.21±0.01	3.04±0.63 19.4±3.5	**51.27**±0.52
	∞	V-U	0.98±0.01 **0.17**±0.02	0.81±0.02 5.78±0.69	**0.01**±0.07 **0.37**±0.42	5.0±1.1 **0.00**±0.00	**5.6**±1.2 89.4±0.2	**0.03**±0.00 **0.22**±0.01	3.04±0.63 9.2±5.9	54.43±0.33
		V-Δ	0.98±0.01 **0.17**±0.02	0.81±0.02 5.78±0.69	**0.01**±0.07 0.38±0.42	5.2±1.1 **0.00**±0.00	**5.6**±1.2 89.2±0.2	**0.03**±0.00 **0.22**±0.01	3.04±0.63 9.3±5.8	54.43±0.33
		V-L	0.86±0.04 **0.17**±0.02	0.72±0.02 5.78±0.68	1.70±0.93 3.65±1.60	29.6±1.2 **0.00**±0.03	5.7±1.2 64.7±0.5	0.01±0.01 0.21±0.01	3.04±0.63 15.1±4.5	53.57±0.34
		V-C	**0.52**±0.08 **0.17**±0.02	**0.51**±0.04 5.77±0.69	7.32±2.30 14.31±3.62	**91.1**±1.4 0.01±0.06	5.9±1.2 **2.9**±0.8	0.01±0.00 0.21±0.01	3.04±0.63 19.4±3.5	**51.27**±0.52